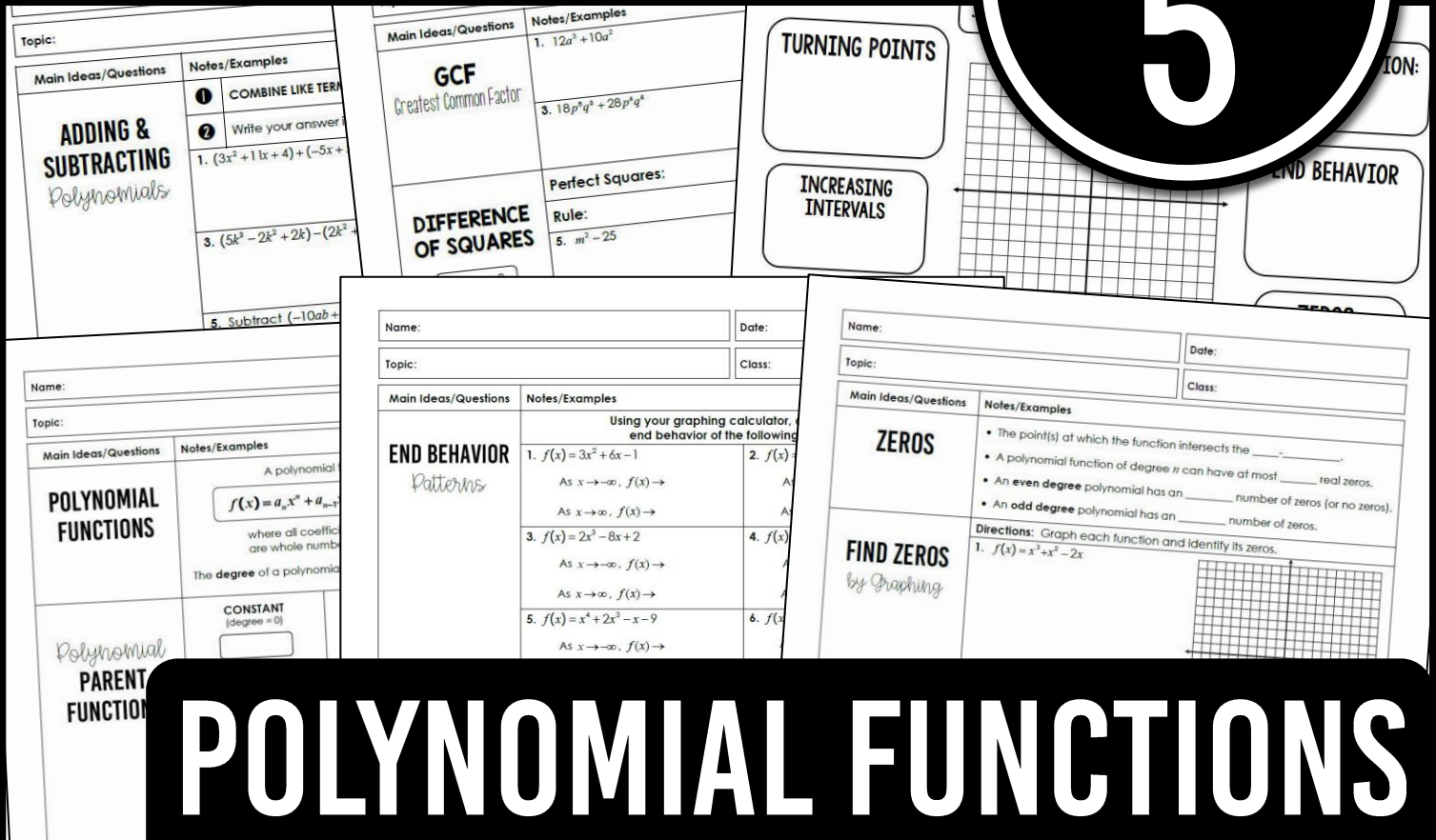


ALGEBRA 2

Unit 5



POLYNOMIAL FUNCTIONS

NOTES • HOMEWORK • QUIZZES • TEST

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Gina



Unit 5 – Polynomial Functions: Sample Unit Outline

	TOPIC	HOMEWORK
DAY 1	Monomials & Polynomials Review (Exponent Rules, Classifying Polynomials, Operations with Polynomials)	HW #1
DAY 2	Factoring Polynomials (Day 1 – GCF, Difference of Squares, Sum of Cubes, & Difference of Cubes)	HW #2
DAY 3	Factoring Polynomials (Day 2 – Trinomials & Four Term Polynomials)	↓
DAY 4	More Practice Factoring Polynomials	Brochure Project
DAY 5	Quiz 5-1	None
DAY 6	Graphing Polynomial Functions (End Behavior, Turning Points, Increasing/Decreasing Intervals}	HW #3
DAY 7	Roots/Zeros of a Polynomial Function, Multiplicity	HW #4
DAY 8	Solving Polynomial Equations by Factoring	HW #5
DAY 9	Quiz 5-2	None
DAY 10	Dividing Polynomials (Day 1 – Long Division)	HW #6
DAY 11	Dividing Polynomials (Day 2 – Synthetic Division)	HW #7
DAY 12	Remainder Theorem	HW #8
DAY 13	Function Operations & Compositions of Functions	HW #9
DAY 14	Quiz 5-3	None
DAY 15	*Polynomial Regression	None
DAY 16	Unit 5 Review	Study for Test
DAY 17	Unit 5 Test	None
*Note: A quick overview of using cubic and quartic models to find the curve of best fit is included in which the model to use is given. Once exponential and logarithmic models are introduced in a later unit, a lesson will be included in which students must select the most appropriate model.		

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples	
MONOMIALS	- A monomial is a number, variable, or a product of numbers and variables. - Examples: _____ - Use the EXPONENT RULES to simplify monomial expressions:	
	NAME	RULE
	Product Rule	
	Power Rule	
	Quotient Rule	
	Negative Exponent Rule	
	Zero Exponent Rule	
	When ADDING OR SUBTRACTING monomials, COMBINE LIKE TERMS!	
EXAMPLES	1. $5x^2 \cdot -7x^6$	2. $(-2a^3b)^2 \cdot 8ab^9$
	3. $\frac{54m^6n^4}{3m^2n} - 10m^4n^3$	4. $2k^4 \cdot 10k^{-7}$
	5. $\left(\frac{2}{3}r^2s^7\right)^2 \cdot \left(\frac{1}{6}r^3s\right)$	6. $\left(\frac{14w^{12}}{7w^3}\right)^{-1}$
	7. $\frac{15x^{10}y^4}{24x^{12}y^3}$	8. $\left(\frac{c}{c^2}\right)^4 \cdot (-3c)^4$

	9. Give an example of two monomials with a quotient of $\frac{-3n^2}{m}$.																								
POLYNOMIALS	- A polynomial is the sum or difference of many monomials. - The highest exponent of a polynomial is called the _____. Standard Form: _____ Write the polynomials below in standard form: 10. $-k^5 - 1 + 8k - 3k^3 + \frac{1}{4}k^2$ _____ 11. $18a^2b^2 + 7ab - b^2 + 4a^3$ _____ 12. $5xy^2 - x^2 + 9x^3y - y^4 + 2$ _____																								
CLASSIFYING POLYNOMIALS <table border="1" style="display: inline-table; border-collapse: collapse; text-align: center;"> <thead> <tr><th colspan="2">Degree</th></tr> </thead> <tbody> <tr><td>0</td><td></td></tr> <tr><td>1</td><td></td></tr> <tr><td>2</td><td></td></tr> <tr><td>3</td><td></td></tr> <tr><td>4</td><td></td></tr> <tr><td>5</td><td></td></tr> </tbody> </table> <table border="1" style="display: inline-table; border-collapse: collapse; text-align: center;"> <thead> <tr><th colspan="2">Number of Terms</th></tr> </thead> <tbody> <tr><td>1</td><td></td></tr> <tr><td>2</td><td></td></tr> <tr><td>3</td><td></td></tr> <tr><td>4+</td><td></td></tr> </tbody> </table>	Degree		0		1		2		3		4		5		Number of Terms		1		2		3		4+		Polynomials are classified by degree (highest exponent) and number of terms. Use the charts to the left to classify each polynomial below. 13. $-3x + 1$ _____ 14. $9x^5 - x^4 + 2x$ _____ 15. 24 _____ 16. $\frac{1}{2}x^3 - 2x^2 + 4x + 15$ _____ 17. $-x^2 - 18x + 31$ _____ 18. $-\frac{3}{2}x^4$ _____ 19. Give an example of a cubic binomial. _____ 20. Give an example of a linear monomial. _____
Degree																									
0																									
1																									
2																									
3																									
4																									
5																									
Number of Terms																									
1																									
2																									
3																									
4+																									

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples	
ADDING & SUBTRACTING <i>Polynomials</i>	①	COMBINE LIKE TERMS! (Watch out for subtraction problems!)
	②	Write your answer in STANDARD FORM .
	1. $(3x^2 + 11x + 4) + (-5x + x^2 - 13)$	2. $(9n^3 - 4n^2 + 2n - 10) + (-2n^2 + n + 7)$
	3. $(5k^3 - 2k^2 + 2k) - (2k^2 + 2k + 17)$	4. $(y + 4y^2 - 3) - (1 + 2y^2 - 5y - y^3)$
	5. Subtract $(-10ab + 7a^2 - b^2)$ from $(8b^2 + ab - 2a^2)$.	
MULTIPLYING <i>Polynomials</i>	①	DISTRIBUTE to expand and remove parentheses.
	②	COMBINE LIKE TERMS!
	③	Write your answer in STANDARD FORM .
	6. $(w - 7)(w^2 + 2w + 1)$	7. $(2x + 1)(4 - 9x) - 2x(3x + 11)$
	8. $-3(a + 5)(a - 2)(a + 8)$	9. $(2c + 5d)^3$

	10. $(3m^2 - 4m + 1)(2m^2 + 5m - 9)$	
	11. $(x^2 + 6x - 7)(-x^2 - 9x - 4)$	
	12. The height of a trapezoid can be expressed as $(2x + 9)$ while the bases can be expressed as $(3x - 7)$ and $(x + 5)$. Write an expression to represent the area of the trapezoid.	
DIVIDING Polynomials (by a Monomial)	1	DIVIDE each term in the numerator by the monomial in the denominator.
	2	Write your answer in STANDARD FORM . Rewrite all terms with negative exponents.
	13. $\frac{9m^3 - 24m^2}{3m}$	14. $\frac{20x^3 - 4x^2 - 8x}{4x}$
	15. $(14a^4b^3 - 20a^2b^2 + 2ab) \cdot (2ab)^{-1}$	16. $\frac{30c^8d^3 - 36c^6d^2 + 6c^3d}{12c^2d}$
	17. $\frac{30x^3 - 12x^2 + 9x}{3x^2}$	18. $\frac{40m^3n^4 - 24m^2n^7}{8m^2n^5}$

Name: _____

Unit 5: Polynomial Functions



Date: _____ Per: _____

Homework 1: Monomials & Polynomials

**** This is a 2-page document! ****
Simplify the monomials below. Final answers should contain positive exponents only.

1. Subtract $-3n^2$ from $-7n^2$

2. $(-8x^4y^3) \cdot (2x^5y^2) + 7x^9y^5$

3. $(-4a^3b^2)^2 \cdot (3a^2b)$

4. $5c^{-7}d^2 \cdot (-cd^2)^4$

5. $\frac{18w^4x^9}{14w^5x^5}$

6. $\frac{15x^2y \cdot -6x^7y}{(3xy)^2}$

7. Write two monomials with a product of $-84a^6b^{10}$.

8. The side length of a cube can be represented by the expression $2x^5$. If the side length is doubled, write an expression to represent the new volume of the cube.

Write in standard form if needed, then classify the polynomial.

9. $25 - 9m^2 - m + 4m^3$

10. $-\frac{1}{2}k^3 + \frac{3}{4}k^5$

11. $14y^2 - x^2 - 3xy$

12. $9x$

Simplify. Final answers should be written in standard form.

13. $(8x^3 - 6x^4 + 3) - (3x^3 - 3 + 8x^4)$

14. $(2a^2b - 5b^3 + 4a^4b^2) - (7b^3 + 8a^4b^2 - 7a^2b)$

15. $-3x(5x + 1)(8 - 2x)$	16. $(a + b)(3a - b)(2a + 7b)$
17. $(1 - 2n)^3 - 7n(n^2 - 2)$	18. $4(2 - 3w)(w^2 - 2w + 10)$
19. $\frac{-42x^{10}y^5 + 12x^8y^3 - 6x^2y}{6x^2y}$	20. $\frac{16a^4 - 40a^2 + 24a}{12a^3}$
21. The length, width, and height of a rectangular prism can be represented by the expressions $(x + 3)$, $(x + 7)$ and $(x - 1)$. Write an expression to represent the surface area of the prism.	
22. The area of a triangle can be represented by the expression $14x^5 + 63x^2$. If the base is $7x^2$, write an expression to represent its height.	

Name:		Date:	
Topic:		Class:	

Main Ideas/Questions	Notes/Examples	
GCF Greatest Common Factor	1. $12a^3 + 10a^2$	2. $14m^8n^5 - 7m^2n^2$
	3. $18p^8q^3 + 28p^6q^5$	4. $16x^3y - 12x^2y^2 + 32xy^3$
DIFFERENCE OF SQUARES $a^2 - b^2$	Perfect Squares:	
	Rule:	
	5. $m^2 - 25$	6. $9x^2 - 49y^2$
	7. $5r^2 - 80s^2$	8. $72c^4 - 2$
	9. $7m^7n - 28mn^3$	10. $10w^5 - 10w$
SUM OF CUBES $a^3 + b^3$	Perfect Cubes:	
	Rule:	
	11. $x^3 + 27$	12. $8w^3 + 1$
	13. $m^3 + 216n^3$	14. $343p^3 + 64q^3$
	15. $500a^3 + 4$	16. $24c^4d + 375cd^4$

DIFFERENCE OF CUBES $a^3 - b^3$	Rule:	
	17. $x^3 - 1$	18. $64 - w^3$
	19. $216k^3 - 125$	20. $m^3 - 343n^3$
	21. $4 - 32h^3$	22. $2x^3 - 54y^3$
TRINOMIALS $ax^2 + bx + c$	23. $k^2 + 15k + 56$	24. $n^2 + 7n - 30$
	25. $p^2 - 16p + 64$	26. $r^2 - 11r - 42$
	27. $w^3 + w^2 - 20w$	28. $2a^3 + 38a^2 + 68a$
	29. $n^4 + 2n^2 - 24$	30. $x^5 - 24x^3 - 25x$
	31. $3x^2 - 10x + 8$	32. $8x^2 - 6x - 5$

	33. $3x^4 + 14x^2 - 5$	34. $12m^4 - 11m^2 + 2$
FOUR TERMS	Rule: Use grouping!	
	Steps	Example
	① GROUP the first two terms together and the last two terms.	$x^3 + x^2 - 4x - 4$
	② FACTOR the GCF for each group.	
	③ FACTOR the common binomial.	
	35. $x^3 - x^2 + 9x - 9$	36. $k^3 + 5k^2 - k - 5$
	37. $4v^3 - 5v^2 - 16v + 20$	38. $2x^5 - 18x^4 + 7x - 63$
	39. $c^4 + 9c^3 - c - 9$	40. $2g^4 - 3g^3 + 16g - 24$

Name: _____

Unit 5: Polynomial Functions

Date: _____ Per: _____

Homework 2: Factoring Polynomials

**Directions:** Complete the following rules.

1. Difference of Squares

$a^2 - b^2 =$

2. Sum of Cubes

$a^3 + b^3 =$

3. Difference of Cubes

$a^3 - b^3 =$

Directions: Factor each polynomial **completely**. Make sure to check for a **GCF** first.

4. $x^4 - 36$

5. $64c^3 + 1$

6. $k^3 - 27$

7. $54x^3 + 250y^3$

8. $3m^4 - 48n^2$

9. $a^7b^2 - ab^2$

10. $x^3y^2 - 343y^5$

11. $9y^7 - 144y$

12. $w^2 - 13w + 36$

13. $p^3 + 5p^2 - 84p$

14. $x^4 - 12x^2 + 36$	15. $w^4 - 14w^2 - 32$
16. $k^3 + 7k^2 - 44k$	17. $2a^3 + 28a^2 + 96a$
18. $-x^3 + 4x^2 + 21x$	19. $m^6 - 7m^4 - 18m^2$
20. $9y^6 + 6y^4 + y^2$	21. $8c^4 + 10c^2 - 3$
22. $x^3 - 7x^2 + x - 7$	23. $4r^3 - 3r^2 - 4r + 3$
24. $3p^3 + 5p^2 - 12p - 20$	25. $15n^3 - 6n^2 - 25n + 10$

Check for a **GCF** first!

FACTORING GUIDE

2
terms

**DIFFERENCE
OF SQUARES**

$$a^2 - b^2$$

SUM OF CUBES

$$a^3 + b^3$$

**DIFFERENCE
OF CUBES**

$$a^3 - b^3$$

3
terms

TRINOMIALS ($a = 1$)

TRINOMIALS ($a > 1$)

4
terms

TRY GROUPING!

Group Members:

Per: _____



REVIEW: *Factoring Polynomials*

Directions: Work together to factor each polynomial **completely**!
Work together to complete each problem. Do not divide up the work!
Each person should be participating.

2 TERMS

1. $8a^3 + 1$

2. $x^4 - 100y^2$

3. $m^3 - 64$

4. $27w^5 - 75w$

5. $125c^4 + cd^3$

6. $p^6q^4 - q^7$

7. $5a^4 - 245$

8. $10x^8y^2 + 80x^2y^5$

3 TERMS

9. $x^4 + 2x^2 - 63$

10. $a^4 - 3a^2 + 2$

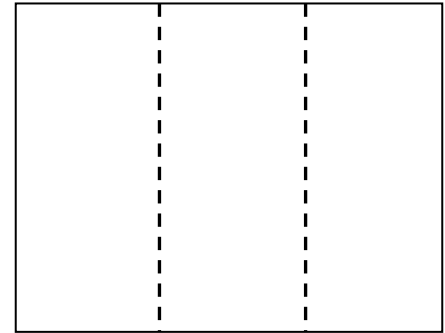
11. $x^3 - 12x^2 + 36x$	12. $2m^4 - 10m^2 - 28$
13. $5x^4 - 10x^2 - 315$	14. $3u^5 - 9u^3 - 84u$
15. $3a^4 + 16a^2 + 20$	16. $8n^4 - 2n^2 - 1$

4 TERMS	
17. $x^3 - 2x^2 + x - 2$	18. $6b^3 - 3b^2 + 4b - 2$
19. $2w^3 + 3w^2 - 6w - 9$	20. $x^5 + 5x^2 - 4x^3 - 20$

FACTORING BROCHURE

Your Role: To create a brochure to serve as a guide to factoring polynomials.

Directions: Fold a piece of paper into thirds.



- ☐ Design a **cover** on the front section.
- ☐ On each remaining section, make a **tutorial** for the following topics:
 - **Difference of Squares** ($a^2 - b^2$)
 - **Sum of Cubes** ($a^3 + b^3$)
 - **Difference of Cubes** ($a^3 - b^3$)
 - **Trinomials** ($ax^2 + bx + c$)
 - **Four Terms**
- ☐ For each topic you must include the following:
 - A **TITLE** for the factoring technique.
 - An **EXPLANATION** of the factoring rule or technique.
 - **THREE EXAMPLES** with **various levels of difficulty**.
 - ****Two of the examples must include a GCF first!****

★ Final product must be neat, organized,
accurate, and attractive to the reader!

Name: _____

Algebra 2

Date: _____ Per: _____

Unit 5: Polynomial Functions**Quiz 5-1: Polynomial Operations & Factoring Polynomials****Write each polynomial in standard form. Then classify it by degree and number of terms.**

1. $-2x^2 + x^4 + 3$

2. $-7 + 5m^3 + 9m - 3m^2$

a. Standard form: _____

a. Standard form: _____

b. Classify by degree: _____

b. Classify by degree: _____

c. Classify by terms: _____

c. Classify by terms: _____

Simplify. All answers should be in standard form and contain only positive exponents.

3. $(-6a^4b^3)^2 \cdot -2ab^5$

4. $\frac{8x^{-3}y \cdot 5x^5y}{10x^6y^2}$

3. _____

4. _____

5. _____

6. _____

7. _____

8. _____

9. _____

10. _____

5. $(6x^2 + 7x) + (5 - 2x^2 + 9x)$

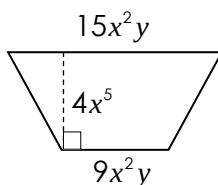
6. $(4h^3 + 6h^2 + 1) - (5h^2 - 8h^4 + 6)$

7. $(x + 2)(2x^2 - x - 9)$

8. $3x(x - 7)(x - 1)^2$

9. $\frac{21c^{10}d^3 + 56c^6d^2 - 7c^2d}{7c^2d}$

10. Write an expression in simplified form to represent the area of the trapezoid.



Factor each polynomial completely.

11. $27p^3 - 1$

12. $10x^5y - 40xy^3$

13. $a^5b - 64ab$

14. $250m^3 + 54$

15. $r^2 + 16r + 63$

16. $k^4 - 10k^2 - 39$

17. $6x^2 + 7x - 20$

18. $4w^4 + 16w^2 + 15$

19. $3w^3 - 2w^2 + 27w - 18$

20. $2n^3 + 7n^2 - 2n - 7$

11. _____

12. _____

13. _____

14. _____

15. _____

16. _____

17. _____

18. _____

19. _____

20. _____

HOW TO FIND

Turning Points

Find the turning points of the function: $f(x) = x^3 - 6x^2 + 7x + 2$

Step 1: Graph the function. (Enter the function in $y =$, then hit **GRAPH**)

Step 2: Use the **CALC** menu to find the minimum and maximum values.

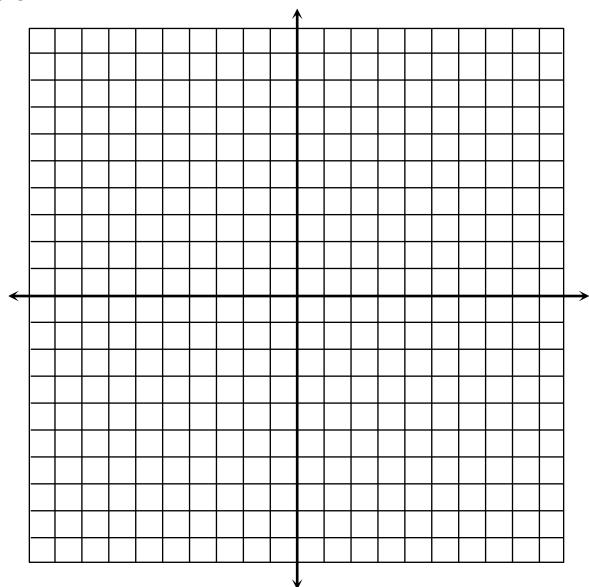
Step 3: Move the cursor to the left bound of the turning point.
Hit **ENTER**, then move the cursor to the right bound of the turning point. Hit **ENTER** twice.

Rel. Maximum(s): _____ Rel. Minimum(s): _____

Inc. Intervals: _____ Dec. Intervals: _____

Directions: Graph each polynomial function using a table. Then give the domain, range, turning points, increasing intervals, decreasing intervals, and end behavior.

1. $f(x) = x^3 - 3x^2 - 3$



Domain: _____

Range: _____

Rel. Maximum(s): _____

Rel. Minimum(s): _____

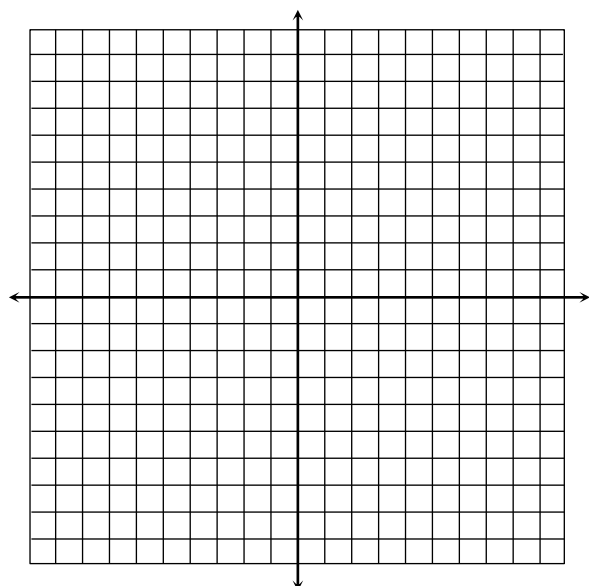
Inc. Intervals: _____

Dec. Intervals: _____

End Behavior: As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

2. $f(x) = x^4 + x^3 - 4x^2 + 5$



Domain: _____

Range: _____

Rel. Maximum(s): _____

Rel. Minimum(s): _____

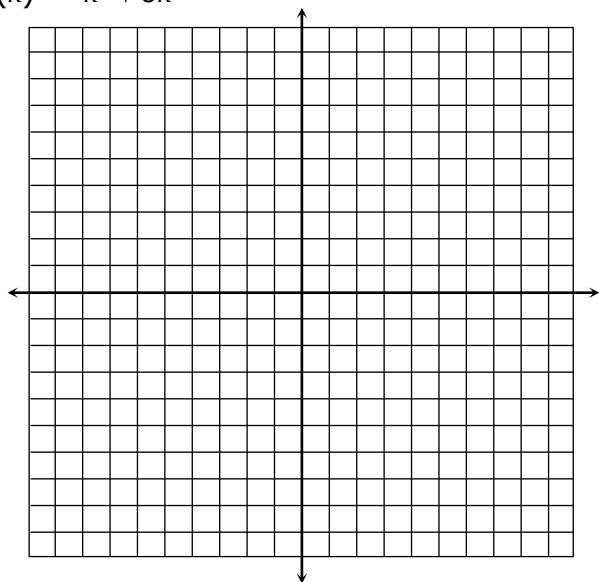
Inc. Intervals: _____

Dec. Intervals: _____

End Behavior: As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

3. $f(x) = -x^3 + 3x^2$



Domain: _____

Range: _____

Rel. Maximum(s): _____

Rel. Minimum(s): _____

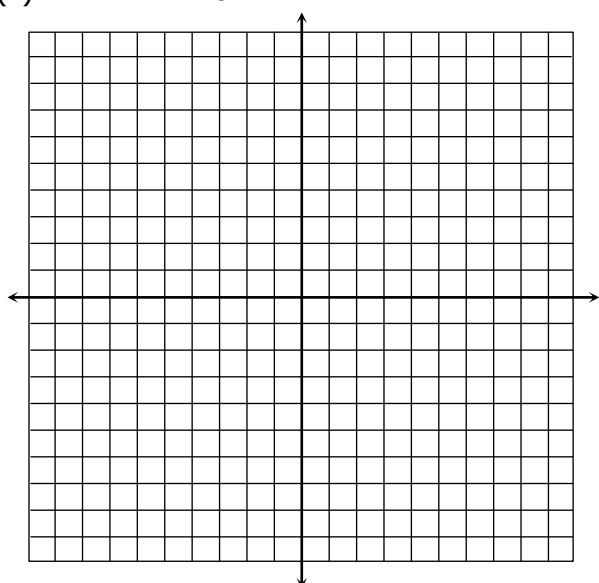
Inc. Intervals: _____

Dec. Intervals: _____

End Behavior: As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

4. $f(x) = -x^4 + 4x^2 - 3x - 2$



Domain: _____

Range: _____

Rel. Maximum(s): _____

Rel. Minimum(s): _____

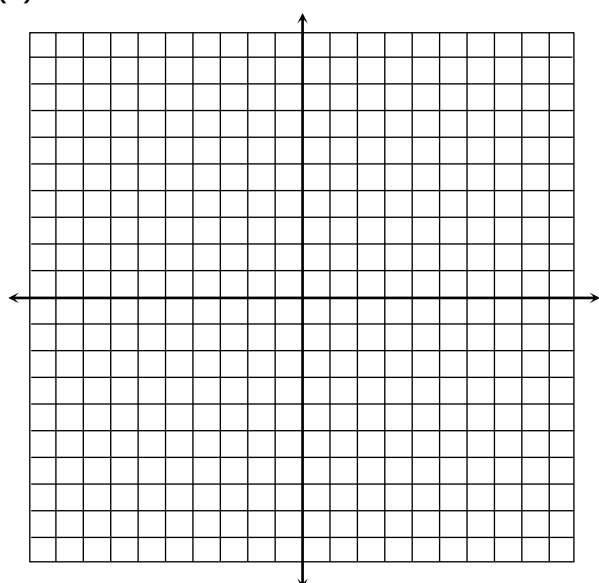
Inc. Intervals: _____

Dec. Intervals: _____

End Behavior: As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

5. $f(x) = 2x^3 - 4x^2 - x$



Domain: _____

Range: _____

Rel. Maximum(s): _____

Rel. Minimum(s): _____

Inc. Intervals: _____

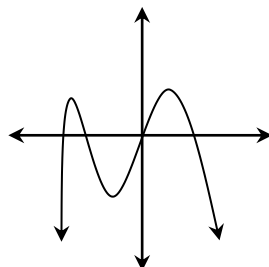
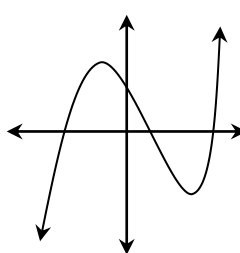
Dec. Intervals: _____

End Behavior: As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples															
END BEHAVIOR Patterns	Using your graphing calculator, determine the end behavior of the following functions.															
	1. $f(x) = 3x^2 + 6x - 1$ As $x \rightarrow -\infty$, $f(x) \rightarrow$ As $x \rightarrow \infty$, $f(x) \rightarrow$		2. $f(x) = -x^2 + x$ As $x \rightarrow -\infty$, $f(x) \rightarrow$ As $x \rightarrow \infty$, $f(x) \rightarrow$													
	3. $f(x) = 2x^3 - 8x + 2$ As $x \rightarrow -\infty$, $f(x) \rightarrow$ As $x \rightarrow \infty$, $f(x) \rightarrow$		4. $f(x) = -x^3 + 5x + 1$ As $x \rightarrow -\infty$, $f(x) \rightarrow$ As $x \rightarrow \infty$, $f(x) \rightarrow$													
	5. $f(x) = x^4 + 2x^3 - x - 9$ As $x \rightarrow -\infty$, $f(x) \rightarrow$ As $x \rightarrow \infty$, $f(x) \rightarrow$		6. $f(x) = -x^4 + 3x^3 - x^2$ As $x \rightarrow -\infty$, $f(x) \rightarrow$ As $x \rightarrow \infty$, $f(x) \rightarrow$													
	When written in standard form, the leading coefficient and degree of the polynomial function determines the end behavior of the function.															
	<table><tr><td colspan="2" rowspan="2"></td><th colspan="2">LEADING COEFFICIENT</th></tr><tr><th>POSITIVE</th><th>NEGATIVE</th></tr><tr><th rowspan="2">DEGREE</th><th>EVEN</th><td></td><td></td></tr><tr><th>ODD</th><td></td><td></td></tr></table>						LEADING COEFFICIENT		POSITIVE	NEGATIVE	DEGREE	EVEN			ODD	
		LEADING COEFFICIENT														
		POSITIVE	NEGATIVE													
DEGREE	EVEN															
	ODD															
Draw sketches to help remember the end behavior patterns!																
EXAMPLES	Identify the degree as even or odd, the leading coefficient as positive or negative, then sketch the end behavior with arrows.															
	FUNCTION	DEGREE	LEADING COEFFICIENT	END BEHAVIOR												
	7. $f(x) = -5x + 1$															
	8. $f(x) = 12x^2 - 11x + 31$															
	9. $f(x) = -x^3 - x^2 + 8x + 1$															

	10. $f(x) = -3x^4 + x^2 - 18$			
	11. $f(x) = 9x^5 + 2x^4 - x^3 + 9x^2 - 2$			
	12. $f(x) = 5x^4 + 2x^3 - 17x + 1$			
	13. $f(x) = -x^7 - 5x^4 + 2x^2$			
	14. $f(x) = -3x^2 - 14x + 28$			
	Determine the sign of the leading coefficient and whether the function has an even or odd degree.			
15.		16.		
IDENTIFYING FUNCTIONS	Without graphing, check all functions with the described end behavior.			
	17. As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$ As $x \rightarrow \infty$, $f(x) \rightarrow -\infty$ ☐ $f(x) = -x^3 + 10x^2 - 6x + 8$ ☐ $f(x) = 3x^2 - 17x - 11$ ☐ $f(x) = -4x^4 - x^3 + 8x^2 - 2x$ ☐ $f(x) = 2x^5 + 8x^4 - 10x^3$ ☐ $f(x) = -x^7 - 3x^2 + 9x - 14$ ☐ $f(x) = -x^2 + 8x - 15$		18. As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$ As $x \rightarrow \infty$, $f(x) \rightarrow \infty$ ☐ $f(x) = -x^2 - 14x + 32$ ☐ $f(x) = x^3 + x^2 - 13x$ ☐ $f(x) = 4x^7 - 8x^4 + 10x^2 + 2$ ☐ $f(x) = -x^5 + 5x^2 - 13x - 4$ ☐ $f(x) = -3x^4 + 2x^3 - x^2 + x$ ☐ $f(x) = -x^3 - 12x^2 + 9x - 18$	
	19. As $x \rightarrow -\infty$, $f(x) \rightarrow \infty$ As $x \rightarrow \infty$, $f(x) \rightarrow -\infty$ ☐ $f(x) = -2x^3 + 10x^2 - 7x$ ☐ $f(x) = 5x^2 + x - 14$ ☐ $f(x) = -x^6 + 4x^2 - 11x - 17$ ☐ $f(x) = -x^5 - 4x^4 - x^3 + 2x - 1$ ☐ $f(x) = -2x^4 - 7x^3 + x^2 - 6x$ ☐ $f(x) = -x^3 - 12x + 20$		20. As $x \rightarrow -\infty$, $f(x) \rightarrow \infty$ As $x \rightarrow \infty$, $f(x) \rightarrow \infty$ ☐ $f(x) = 4x^3 - 8x^2 - 2$ ☐ $f(x) = 2x^6 + 5x^3 - 14x$ ☐ $f(x) = -x^4 + x^3 - 6x^2 + x$ ☐ $f(x) = x^2 - 9x + 36$ ☐ $f(x) = 3x^4 + 7x^3 - 16x^2 + 3$ ☐ $f(x) = -x^2 + 10x - 2$	

Name: _____

Unit 5: Polynomial Functions

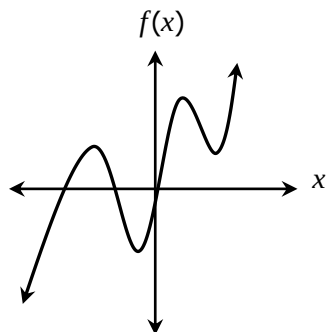
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Homework 3: Graphing Polynomial Functions

**** This is a 2-page document! ****

Directions: For each graph, **(a)** Describe the end behavior, **(b)** Determine whether it's the graph of an even or odd degree function, and **(c)** Determine the sign of the leading coefficient.

1.

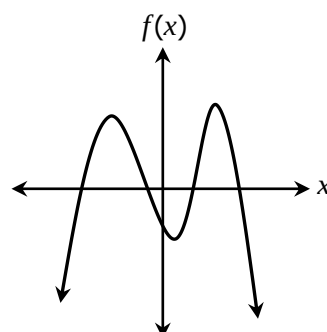


a) _____

b) _____

c) _____

2.



a) _____

b) _____

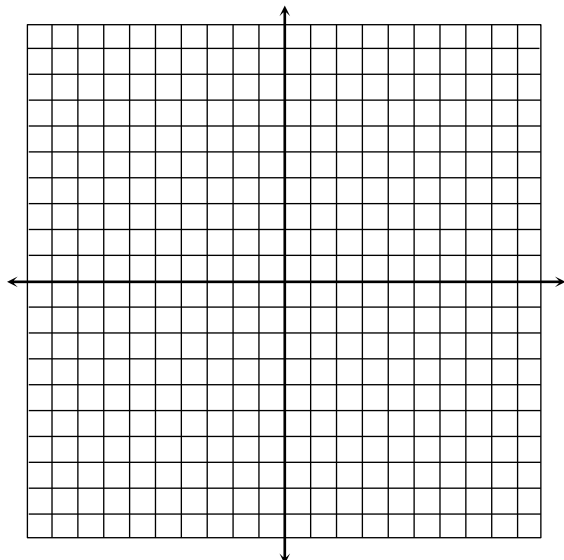
c) _____

3. Using proper notation, describe the end behavior of a 14th degree polynomial with a positive leading coefficient.

4. Using proper notation, describe the end behavior of a 9th degree polynomial with a negative leading coefficient.

Directions: Graph each function and give its key characteristics. Use a graphing calculator for the turning points and round to the nearest hundredth.

5. $f(x) = -x^3 + 6x^2 - 10x + 4$



Domain: _____

Range: _____

Rel. Maximum(s): _____

Rel. Minimum(s): _____

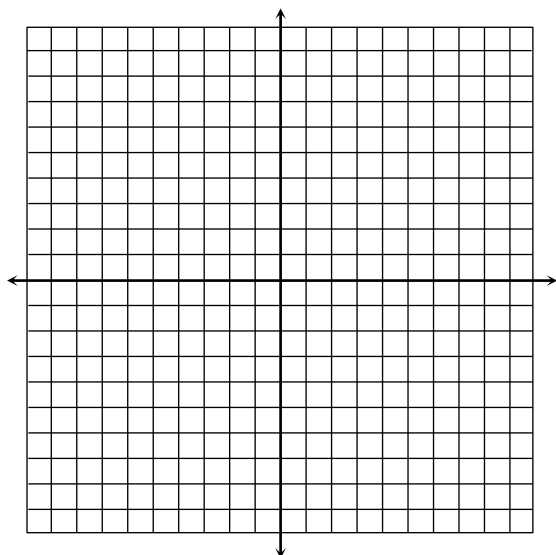
End Behavior: As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

Inc. Intervals: _____

Dec. Intervals: _____

6. $f(x) = 2x^3 - 7x^2 + 6$



Domain: _____

Range: _____

Rel. Maximum(s): _____

Rel. Minimum(s): _____

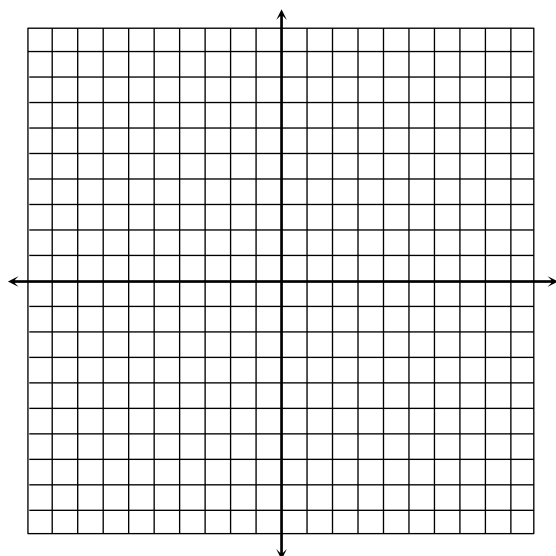
End Behavior: As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

Inc. Intervals: _____

Dec. Intervals: _____

7. $f(x) = x^4 - 4x^3 + 3x^2 + 4x - 5$



Domain: _____

Range: _____

Rel. Maximum(s): _____

Rel. Minimum(s): _____

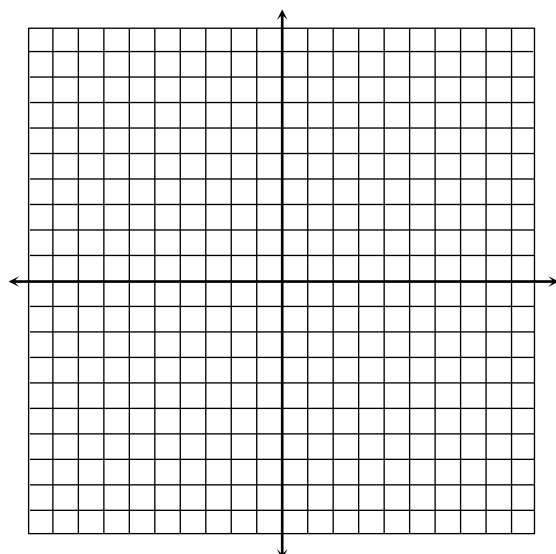
End Behavior: As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

Inc. Intervals: _____

Dec. Intervals: _____

8. $f(x) = -x^4 + 2x^2 - x - 2$



Domain: _____

Range: _____

Rel. Maximum(s): _____

Rel. Minimum(s): _____

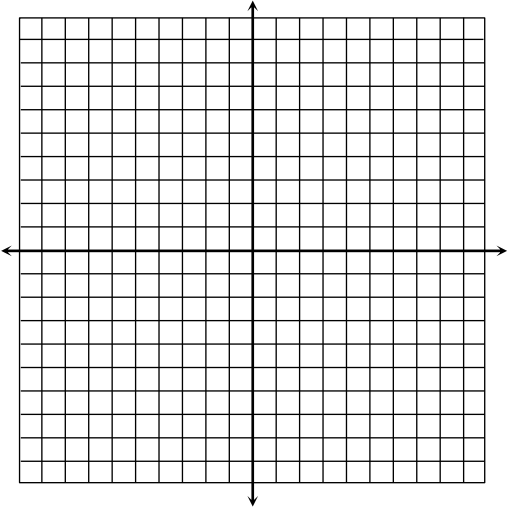
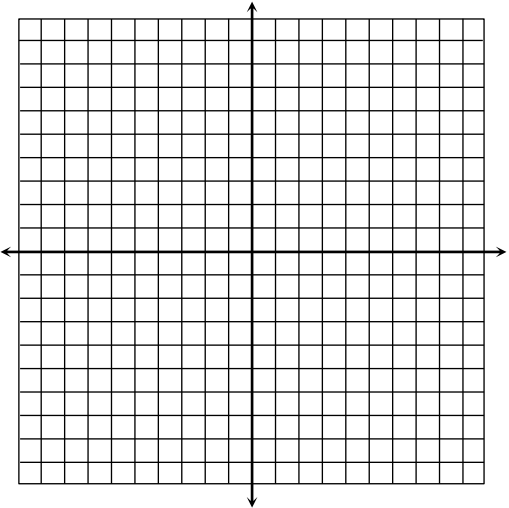
End Behavior: As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

Inc. Intervals: _____

Dec. Intervals: _____

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples
ZEROS	- The point(s) at which the function intersects the ____-_____. - A polynomial function of degree n can have at most _____ real zeros. - An even degree polynomial has an _____ number of zeros (or no zeros). - An odd degree polynomial has an _____ number of zeros.
FIND ZEROS <i>by Graphing</i>	Directions: Graph each function and identify its zeros.
	1. $f(x) = x^3 + x^2 - 2x$ 
	2. $f(x) = -x^4 + 5x^2 - 4$ 
LINEAR FACTORS	The linear factors of a polynomial reveal its zeros. Linear factors can be found by writing the polynomial in factored form. If $(x - a)$ is a factor of $f(x)$, then a is a zero of $f(x)$. 3. Write $f(x) = x^3 - 2x^2 - 15x$ in factored form and identify the zeros. Factored Form: _____ Zeros: _____

MULTIPLICITY

The **multiplicity** of a zero is equal to the number of times its corresponding linear factor occurs.

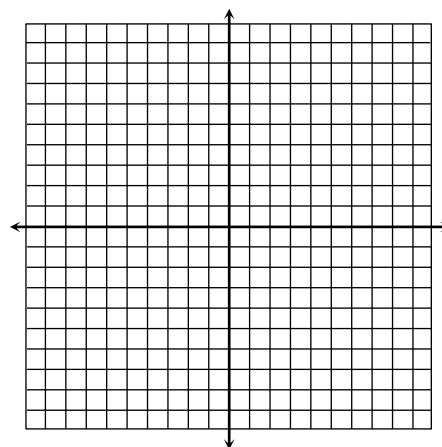
4. Write $f(x) = x^3 - 6x^2 + 9x$ in factored form and identify the zeros and their multiplicity.

Factored Form: _____

Zeros and Multiplicity: _____

Effect of MULTIPLICITY ON A GRAPH

5. Graph $f(x) = x^3 - 6x^2 + 9x$



- If the **multiplicity** is **ODD**, then the graph _____ at that zero.
- If the **multiplicity** is **EVEN**, then the graph _____ at that zero.

EXAMPLES

Identify the zeros and state their multiplicities. Describe the effect on the graph.

6. $f(x) = (x + 7)^2(2x + 1)(x - 4)^3$

Zero	Multiplicity	Effect

7. $f(x) = -2x^4 + 4x^3 + 16x^2$

Zero	Multiplicity	Effect

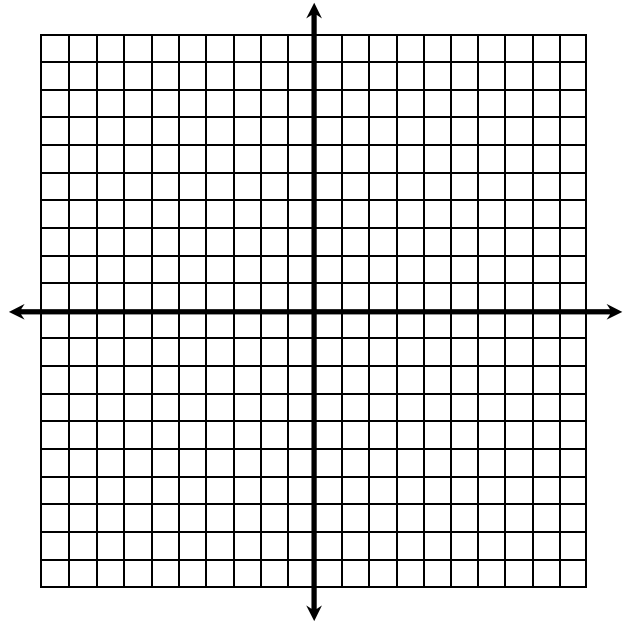
8. A polynomial function has a zero at -6 (multiplicity 2) and 2 (multiplicity 1). Write a polynomial function in standard form that could represent this function.

Analyzing Functions & Graphs

DIRECTIONS: Graph the function below. Then, write a written description of the graph that includes all key characteristics below.

CHARACTERISTICS: Type of polynomial function, domain, range, end behavior, turning points (including whether it's a relative maximum or relative minimum), zeros and their multiplicity, and the effect of the multiplicity on the graph.

$$f(x) = x^3 - 10x^2 + 28x - 24$$

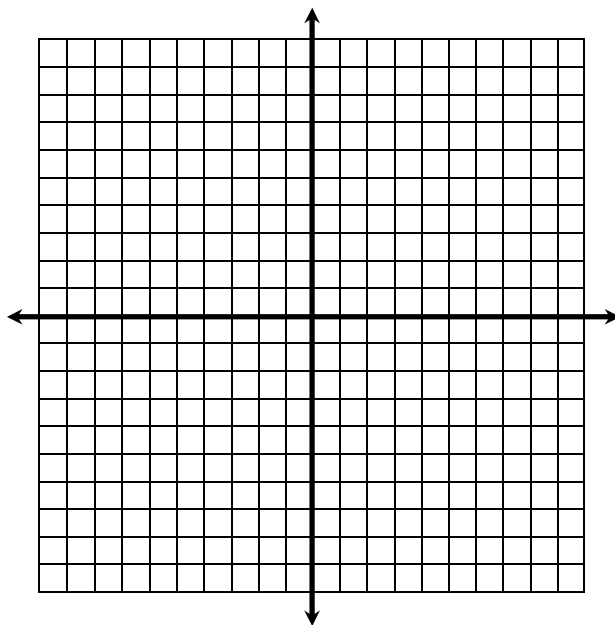
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Analyzing Functions & Graphs

DIRECTIONS: Graph the function below. Then, write a written description of the graph that includes all key characteristics below.

CHARACTERISTICS: Type of polynomial function, domain, range, end behavior, turning points (including whether it's a relative maximum or relative minimum), zeros and their multiplicity, and the effect of the multiplicity on the graph.

$$f(x) = -x^4 - 9x^3 - 24x^2 - 16x$$

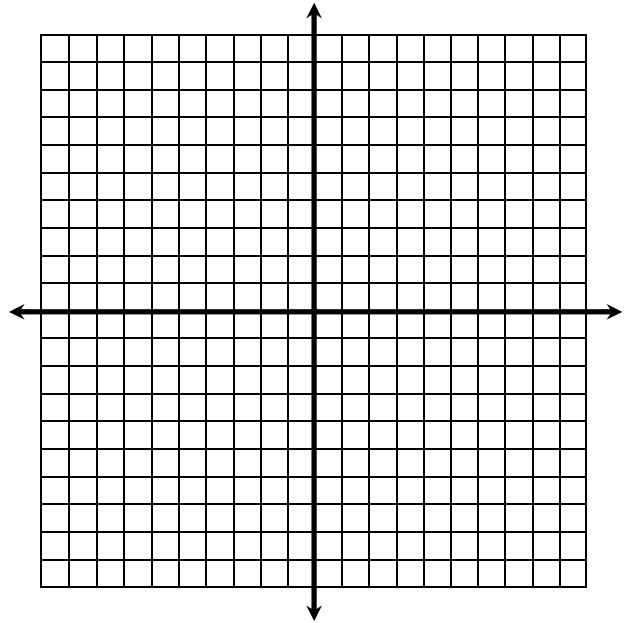
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Analyzing Functions & Graphs

DIRECTIONS: Graph the function below. Then, write a written description of the graph that includes all key characteristics below.

CHARACTERISTICS: Type of polynomial function, domain, range, end behavior, turning points (including whether it's a relative maximum or relative minimum), zeros and their multiplicity, and the effect of the multiplicity on the graph.

$$f(x) = -x^3 + 15x^2 - 75x + 125$$

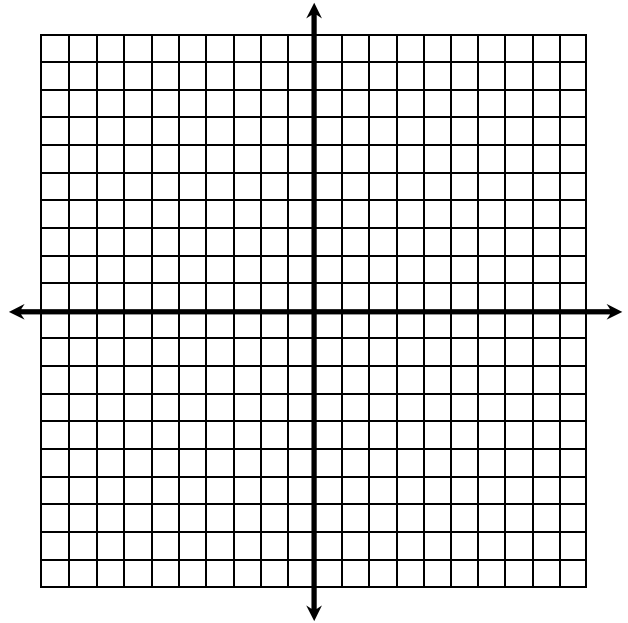
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Analyzing Functions & Graphs

DIRECTIONS: Graph the function below. Then, write a written description of the graph that includes all key characteristics below.

CHARACTERISTICS: Type of polynomial function, domain, range, end behavior, turning points (including whether it's a relative maximum or relative minimum), zeros and their multiplicity, and the effect of the multiplicity on the graph.

$$f(x) = x^5 - 6x^4 + 8x^3 + 6x^2 - 9x$$

[illegible]

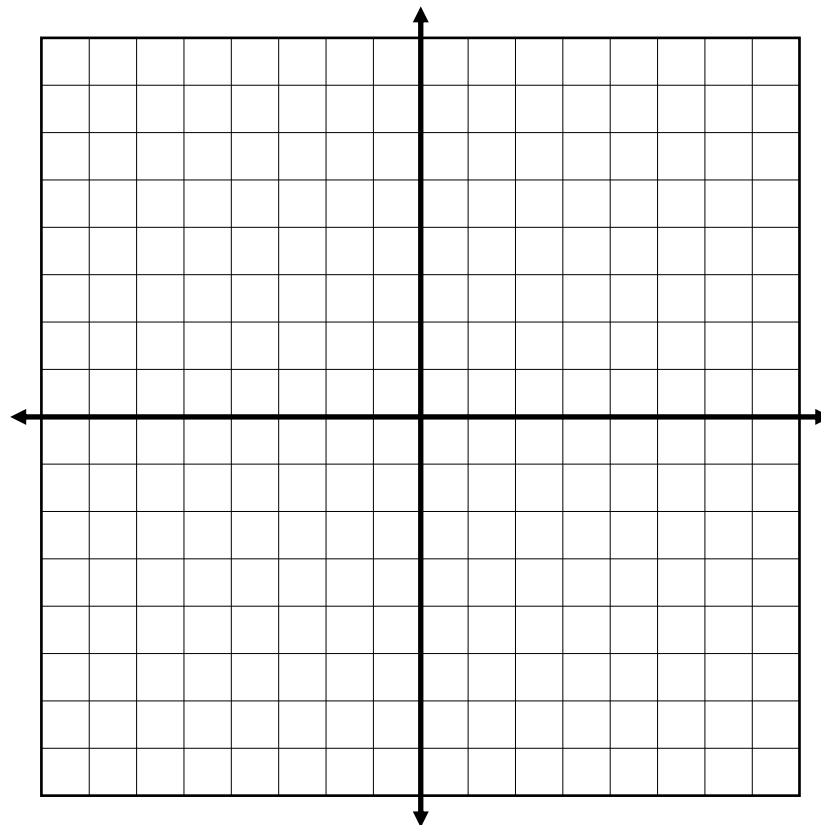
CHARACTERISTICS *of a* POLYNOMIAL FUNCTION

$$f(x) = x^3 + 6x^2 + 8x$$

TURNING POINTS

**INCREASING
INTERVALS**

**DECREASING
INTERVALS**



DOMAIN

RANGE

PARENT FUNCTION

END BEHAVIOR

ZEROS

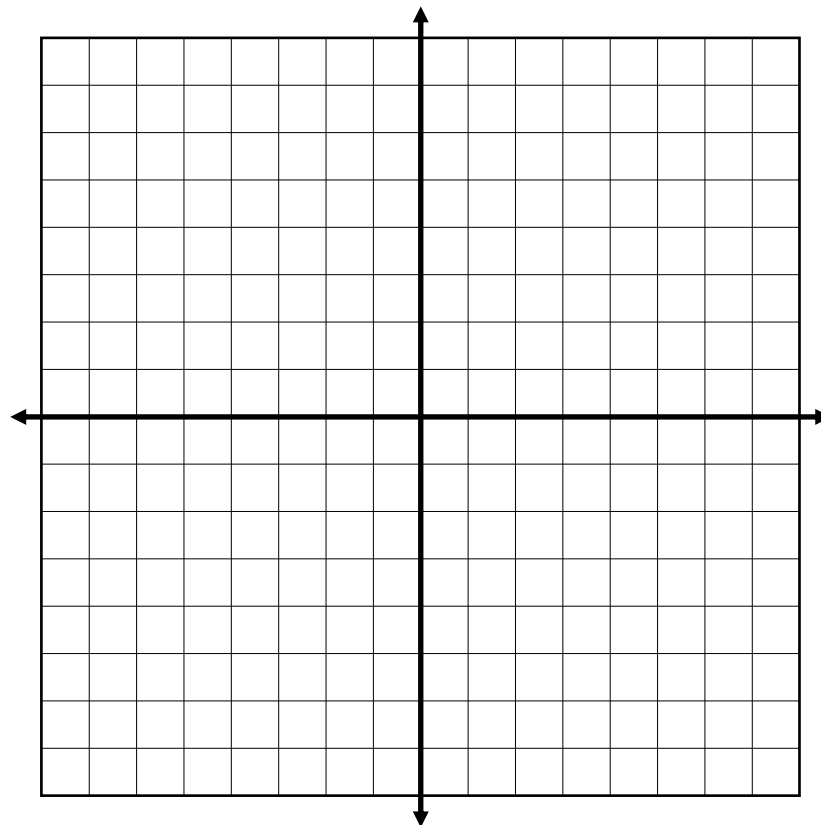
CHARACTERISTICS of a POLYNOMIAL FUNCTION

$$f(x) = x^4 - 3x^2 - 2x$$

TURNING POINTS

**INCREASING
INTERVALS**

**DECREASING
INTERVALS**



DOMAIN

RANGE

PARENT FUNCTION

END BEHAVIOR

ZEROS

Name: _____

Unit 5: Polynomial Functions

Date: _____ Per: _____

Homework 4: Zeros and Multiplicity



**** This is a 2-page document! ****

Directions: Identify the zeros, their multiplicity, and the effect of the multiplicity on the graph.

1. $f(x) = x^2(x-1)^4(x+5)$

Zero	Multiplicity	Effect

2. $f(x) = -2x(3x+1)^2(x+7)^7$

Zero	Multiplicity	Effect

3. $f(x) = x^3 - 12x^2 + 35x$

Zero	Multiplicity	Effect

4. $f(x) = 3x^3 - 21x^2 - 54x$

Zero	Multiplicity	Effect

5. $f(x) = x^4 - 16x^3 + 64x^2$

Zero	Multiplicity	Effect

6. $f(x) = -8x^3 - 20x^2$

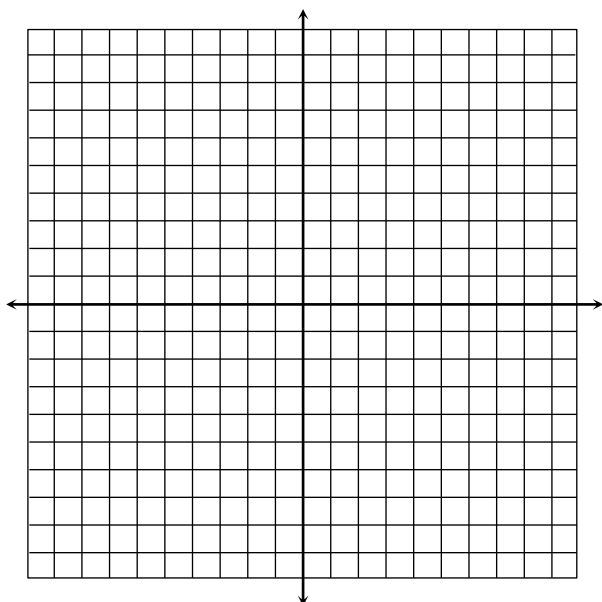
Zero	Multiplicity	Effect

7. A polynomial function has zeros at -1, 2, and 7 (all multiplicity 1). Write a function in standard form that could represent this function.

8. A polynomial function has zeros at $\frac{5}{2}$ (multiplicity 2), 3 (multiplicity 1), and 0 (multiplicity 4). Write a function in standard form that could represent this function.

Directions: Graph each function and identify its key characteristics.

9. $f(x) = x^3 - 10x^2 + 27x - 18$



Domain: _____

Range: _____

Rel. Maximum(s): _____

Rel. Minimum(s): _____

End Behavior: As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

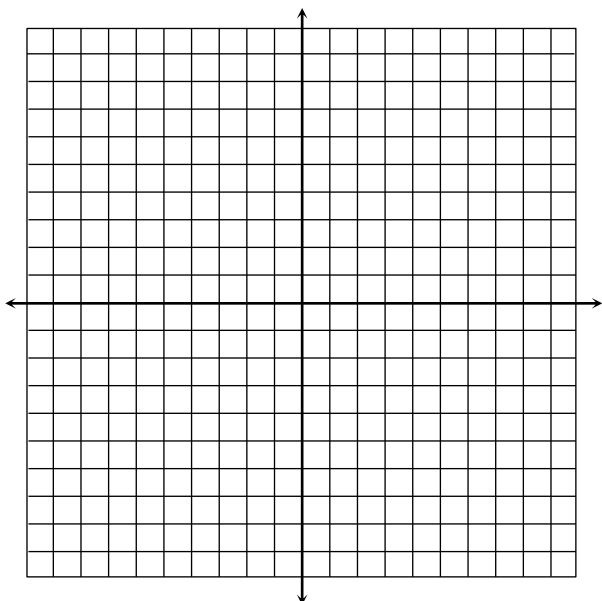
As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

Inc. Intervals: _____

Dec. Intervals: _____

Zeros: _____

10. $f(x) = -x^4 - 7x^3 - 12x^2$



Domain: _____

Range: _____

Rel. Maximum(s): _____

Rel. Minimum(s): _____

End Behavior: As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

Inc. Intervals: _____

Dec. Intervals: _____

Zeros: _____

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples
SOLVING POLYNOMIAL EQUATIONS <i>By Factoring</i>	① MOVE ALL TERMS to one side and set the equation EQUAL TO 0 .
	② FACTOR the polynomial completely!
	③ SET EACH FACTOR EQUAL TO 0 and SOLVE for each x-value.
	➤ For quadratic equations, solve by square roots, completing the square , or the quadratic formula . ➤ SIMPLIFY all irrational and complex solutions!

Directions: Solve each equation by factoring.	
1. $x^3 + x = 0$	2. $4x^3 - 7x^2 = 3x^2$
3. $x^4 - 64 = 0$	4. $5x^5 - 80x = 0$

5. $x^3 + 125 = 0$

6. $8x^5 + 5x^2 = 4x^2$

7. $x^3 - 216 = 0$

8. $16x^3 = 54$

9. $x^4 + x^2 - 42 = 0$

10. $x^4 + 3 = 13 - 9x^4$

11. $x^3 - 8x^2 + 16x = 0$

12. $2x^6 - 10x^4 - 48x^2 = 0$

13. $3x^4 + 2x^2 = 8$

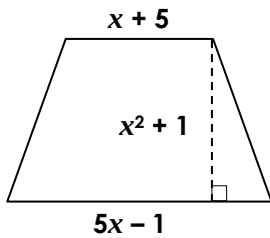
14. $5x^3 - 3x^2 + 32x = 2x + 18$

15. $x^3 + 2x^2 + 5x + 10 = 0$

16. $2x^3 - 3x^2 - 32x + 48 = 0$

17. A rectangular prism with dimensions of $(x - 3)$, $(x - 1)$, and $(x - 8)$ has a volume of $33x$ cubic units. What are the dimensions of the prism?

18. If the area of the trapezoid below is $(30x + 20)$ square feet, find the area of the trapezoid.



19. The revenue R (in thousands of dollars) for a small business can be modeled by $R(t) = 2t^3 - 12t^2 + 3t + 57$ where t is the number of years since 1995. In what year did the revenue reach \$75,000?

20. The population of a species is modeled by the equation $p(t) = -t^4 + 72t^2 + 225$, where t is the number of years. Approximate the number of years until the species is extinct.

Name: _____

Unit 5: Polynomial Functions

Date: _____ Per: _____

Homework 5: Solving Polynomial Equations

**** This is a 2-page document! ****

Directions: Solve each equation by **factoring**. Simplify all irrational and complex solutions.

1. $12x^3 - 3x^2 = 0$

2. $x^4 - 64 = 0$

3. $2x^4 = 24x^2$

4. $x^3 + 512 = 0$

5. $8x^3 - 125 = 0$

6. $81x^4 = 3x$

7. $2x^3 - 16x^2 - 40x = 0$

8. $x^4 - 16x^2 = x^2 + 18$

9. $4x^4 + 35x^2 - 9 = 0$

10. $x^3 + 3x^2 = 24x + 72$

11. $2x^3 - 5x^2 + 40x - 100 = 0$

12. The population of a species is modeled by the equation $p(t) = -t^4 + 56t^2 + 240$, where t is the number of years. Find the approximate number of years until the species is extinct.

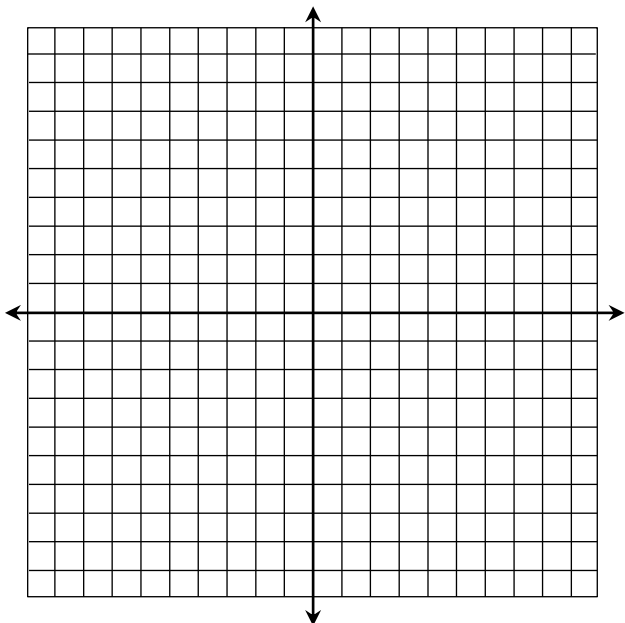
Name: _____

Algebra 2

Date: _____ Per: _____

Unit 5: Polynomial Functions**Quiz 5-2: Graphing Polynomial Functions & Solving Polynomial Equations****Graph the function below and identify its key characteristics.**

1. $f(x) = -x^3 + 8x^2 - 15x$



Domain: _____

Range: _____

Rel. Maximum(s): _____

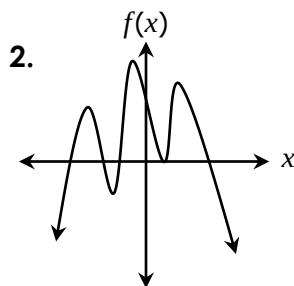
Rel. Minimum(s): _____

End Behavior: As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

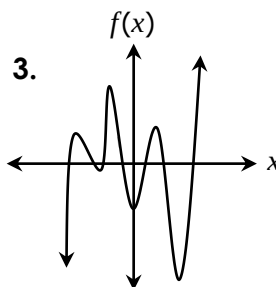
Inc. Intervals: _____

Dec. Intervals: _____

Zeros: _____

Given the graph, determine whether the degree is even or odd and the leading coefficient is positive or negative.

Degree	Leading Coefficient
☐ Even	☐ Positive
☐ Odd	☐ Negative



Degree	Leading Coefficient
☐ Even	☐ Positive
☐ Odd	☐ Negative

Identify the zeros, their multiplicity, and their effect on the graph of the function.

4 $f(x) = -x(3x - 2)^2(x + 9)^5$

5 $f(x) = x^3 + 10x^2 + 25x$

Zero	Multiplicity	Effect

Zero	Multiplicity	Effect

6. A polynomial function has a zero at 4 (multiplicity 3) and 0 (multiplicity 1). Write a function in standard form that could represent this function. Box your final answer.

Solve each equation by factoring. Simplify all irrational and complex solutions.

7. $2x^4 = 50$

8. $x^3 + 216 = 0$

7. _____

8. _____

9. _____

10. _____

11. _____

12. _____

9. $x^3 = x^2 + 20x$

10. $x^4 + 13x^2 + 40 = 0$

11. $x^4 - x^2 = x^2 + 8$

12. $4x^3 + 5x^2 - 16x - 20 = 0$

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples	
DIVIDING by a Monomial	1. $\frac{6x^4 - 15x^3 + 3x^2}{3x^2}$	2. $\frac{21a^5b^3 - 14a^3b^2 + 63a^2b}{7ab}$
	Use factoring to divide the polynomials below.	
DIVIDING by a Binomial	3. $\frac{w^2 - 7w + 10}{w - 5}$	4. $(4h^2 - 7h - 2) \div (h - 2)$
	5. $(x^3 + 9x^2 - 4x - 36) \div (x + 9)$	6. $(2n^3 + 3n^2 - 16n - 24) \div (2n + 3)$

Dividing Polynomials by
LONG DIVISION

When factoring is not possible, we can use long division to divide polynomials.

7. $(r^3 - 4r^2 - 26r - 15) \div (r + 3)$

8. $(k^3 - 8k^2 - 3k + 8) \div (k - 1)$

9. $(6n^3 - 20n^2 - 42n - 47) \div (n - 5)$

10. $(3a^3 - 7a^2 - 30a - 21) \div (3a + 2)$

MISSING POWERS

NOTE: USE 0 as the coefficient when there is a missing power in the dividend.

11. $(k^3 - 9k - 2) \div (k - 3)$

12. $(a^4 + 8a^3 - 3a - 16) \div (a + 8)$

13. $(3a^4 - 20a^2 - 6a + 14) \div (a + 2)$

14. $(36w^3 - 27w^2 - 6) \div (4w - 3)$

Name: _____

Unit 5: Polynomial Functions

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Homework 6: Long Division with Polynomials

**** This is a 2-page document! ****

Find each quotient using long division.

1. $(k^3 - 10k^2 + k + 1) \div (k - 1)$

2. $(x^4 + 4x^3 - 28x^2 - 45x + 26) \div (x + 7)$

3. $(20c^3 + 22c^2 - 7c + 7) \div (5c - 2)$

4. $(3n^4 + 6n^3 - 15n^2 + 32n - 25) \div (n + 4)$

5. $(p^3 - 17p - 13) \div (p + 4)$

6. $(y^4 - y^2 + 8y + 5) \div (y + 1)$

7. $(4v^3 - 12v^2 + 6) \div (v - 3)$

8. $(3a^4 + 4a^3 + 15a + 10) \div (3a + 4)$

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples	
Dividing Polynomials by SYNTHETIC DIVISION	Synthetic division is a shortcut method to divide polynomials. *This method only works when dividing by a binomial with a coefficient of 1.*	
	1. $(x^2 + 3x - 43) \div (x + 8)$	2. $(2n^2 - 16n + 19) \div (n - 7)$
	3. $(y^3 + 5y^2 - 7y + 2) \div (y - 2)$	4. $(c^4 + 9c^3 + 13c^2 - 33c - 16) \div (c + 4)$
MISSING POWERS NOTE: USE 0 as the coefficient when there is a missing power in the dividend	5. $(k^3 - 17k + 32) \div (k + 5)$	6. $(8x^4 - 64x^3 + 3x - 28) \div (x - 8)$

LONG DIVISION *vs.* SYNTHETIC DIVISION

Find each quotient using long and synthetic division. Compare answers to check.

LONG DIVISION

SYNTHETIC DIVISION

7. $(n^3 - 3n^2 - 11n - 3) \div (n + 1)$

8. $(2k^4 + 10k^3 - 40k^2 + 61k - 24) \div (k + 8)$

9. $(w^4 - 4w^3 - 3w + 13) \div (w - 4)$

Name: _____

Unit 5: Polynomial Functions

Date: _____ Per: _____

Homework 7: Synthetic Division with Polynomials

**** This is a 2-page document! ****

Find each quotient using synthetic division.

1. $(n^2 + 11n + 25) \div (n + 5)$

2. $(3k^2 - k - 20) \div (k + 2)$

3. $(x^3 + 2x^2 - 22x - 45) \div (x + 5)$

4. $(w^3 - 2w^2 - 2w + 1) \div (w - 1)$

5. $(2c^3 + 13c^2 + 24c + 8) \div (c + 3)$

6. $(p^4 - 7p^2 - 32p - 15) \div (p - 4)$

Find each quotient using long and synthetic division. Compare answers to check.

LONG DIVISION

SYNTHETIC DIVISION

7. $(m^3 - 12m^2 + 33m) \div (m - 5)$

8. $(2x^3 - 14x + 10) \div (x + 3)$

9. $(x^4 - 2x^3 - 29x^2 - 43x + 8) \div (x - 7)$

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples	
WARM-UP	1. Consider the polynomial function $f(x) = x^3 + 9x^2 + 22x - 1$	
	a) Find $f(-1)$	b) Divide the function by $(x + 1)$.
	2. Consider the polynomial function $f(x) = 2x^4 - 30x^2 + 2x - 17$.	
	a) Find $f(4)$	b) Divide the function by $(x - 4)$.
REMAINDER <i>Theorem</i>	Remainder Theorem: If a polynomial function $f(x)$ is divided by _____, then the _____ is _____. - Using synthetic division to evaluate a function is called _____.	
EXAMPLES	Use The Remainder Theorem (Synthetic Substitution) to find the given value.	
	3. $f(x) = x^3 - 13x^2 + 39x - 4$; $f(5)$	4. $f(x) = 5x^3 + 11x^2 - 22x - 28$; $f(-2)$

	5. $f(x) = x^4 + 7x^3 - 12x^2 - 34x - 10$; $f(-8)$	6. $f(x) = x^3 - 10x^2 - 4$; $f(10)$
	7. $f(x) = x^4 + x^3 + 7x - 1$; $f(-3)$	8. $f(x) = 2x^4 - 18x^2 + 7$; $f(2)$
APPLICATIONS	9. A bicycle manufacturer determines that the profit per bicycle when producing x bicycles per day is $P(x) = -x^4 + 7x^3 - 5x^2 + x$. Use the Remainder Theorem to find the profit per bicycle when four bicycles are produced each day.	
	10. The value V (in dollars) of a signed baseball card can be modeled by the function $V(t) = 2t^3 - 48t^2 + 300t$ where t is the number of years since 1980. Use the Remainder Theorem to find the value of the card in 2025.	

Name: _____

Unit 5: Polynomial Functions

Date: _____ Per: _____

Homework 8: Remainder Theorem

**Evaluate each function using the Remainder Theorem (Synthetic Substitution).**

1. $f(x) = x^3 - 11x^2 + 9x + 8$; $f(2)$

2. $f(x) = 4x^3 + 33x^2 + 25x - 68$; $f(-7)$

3. $f(x) = x^4 + 6x^3 + 6x^2 - 2x + 4$; $f(-1)$

4. $f(x) = 6x^4 - 26x^3 - 23x^2 + 17x - 4$; $f(5)$

5. $f(x) = -x^3 - 8x^2 + 3$; $f(-4)$

6. $f(x) = 2x^4 - 7x^3 - 2x + 19$; $f(3)$

7. $f(x) = x^4 - 7x^2 + 23$; $f(-2)$

8. The height h , in feet, of a rollercoaster x seconds into a ride can be modeled by the function $h(x) = -x^4 + 26x^3 - 212x^2 + 590x + 22$. Use the Remainder Theorem to find the height of the rollercoaster 4 seconds into the ride.

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples	
<i>Function</i> OPERATIONS	Functions can added, subtracted, multiplied, and divided to create new functions.	
	Example: If $f(x) = 3x - 7$ and $g(x) = 13 - 2x$, find $f(x) + g(x)$:	
	This new function is denoted as	
OPERATIONS <i>Rules</i>	SUM	$(f + g)(x) =$
	DIFFERENCE	$(f - g)(x) =$
	PRODUCT	$(f \cdot g)(x) =$
	*QUOTIENT	$\left(\frac{f}{g}\right)(x) =$
Functions: $f(x) = x^2 - 8x + 4$ $g(x) = 4x - 3$ $h(x) = x + 2$	Find each function using the functions given to the left. Indicate any restrictions in the domain.	
	1. $(f + g)(x)$	2. $(f - h)(x)$
	3. $(h \cdot g)(x)$	4. $\left(\frac{f}{h}\right)(x)$
	5. $\left(\frac{h}{g}\right)(x)$	6. $(f \cdot g)(x)$

Functions: $f(x) = 2x^2 - x - 12$ $g(x) = x + 7$	Evaluate each function using the functions given to the left.	
	7. $(f + g)(-2)$	8. $(f - g)(8)$
	9. $(f \cdot g)(-1)$	10. $\left(\frac{f}{g}\right)(5)$
COMPOSITIONS <i>of Functions</i>	Another method to combine functions is called a composition . Given $f(x)$ and $g(x)$, the composite function $(f \circ g)(x)$ is defined as: 	
Functions: $f(x) = 5x - 3$ $g(x) = x - 1$ $h(x) = x^3 + 8$	Find each function value using the functions given to the left.	
	11. $(f \circ g)(x)$	12. $(g \circ h)(x)$
	13. $(h \circ g)(x)$	14. $(g \circ f)(x)$
	Using the same set of functions, evaluate each function.	
	15. $(g \circ f)(2)$	16. $(f \circ h)(-6)$

Name: _____

Unit 5: Polynomial Functions

Date: _____ Per: _____

Homework 9: Operations on Functions
& Compositions of Functions

Directions: Given $f(x) = 2x^2 - 9x + 2$, $g(x) = 1 - 6x$, and $h(x) = x^2 - 4$, find each function.
Indicate any restrictions in the domain.

1. $(f - g)(x)$

2. $\left(\frac{f}{h}\right)(x)$

3. $(g + f)(x)$

4. $(f \cdot h)(x)$

5. $\left(\frac{h}{g}\right)(x)$

6. $(g \cdot h)(x)$

7. $(f \circ g)(x)$

8. $(h \circ f)(x)$

Directions: Using the functions above, find each function value.

9. $(h - g)(-7)$

10. $(f \circ h)(2)$

Name: _____

Algebra 2

Date: _____ Per: _____

Unit 5: Polynomial Functions

Quiz 5-3: Dividing Polynomials, Remainder Theorem, Functions Operations

Find each quotient.

1. $(x^2 - 16x + 63) \div (x - 9)$

2. $(y^2 - 2y - 29) \div (y - 6)$

1. _____

2. _____

3. _____

4. _____

5. _____

6. _____

3. $(a^3 + 8a^2 + 11a + 20) \div (a + 1)$

4. $(12v^3 + 31v^2 + 17v - 1) \div (3v + 4)$

5. $(x^3 - 16x + 9) \div (x - 4)$

6. $(k^4 + 4k^3 - 20k^2 - 39) \div (k + 7)$

7. If $f(x) = 3x^4 - 8x^3 + 14x - 29$, use the Remainder Theorem to find $f(2)$.

7. _____

8. _____

8. The population P of an endangered species can be modeled by the function $P(t) = -x^3 + 32x^2 - 50x + 450$, where t is the years since 2010. Use the Remainder Theorem to find the population of the species in 2024.

Given $f(x) = x + 5$, $g(x) = x^2 - 9$, and $h(x) = x^2 - 10x + 7$, find each function.
State any restrictions in the domain.

9. $(g - h)(x)$

10. $(f \cdot g)(x)$

9. _____

10. _____

11. _____

12. _____

13. _____

14. _____

11. $\left(\frac{h}{f}\right)(x)$

12. $(g \circ f)(x)$

Find each function value.

13. $(h - f)(-3)$

14. $(h \circ g)(5)$

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples														
REVIEW: Linear & Quadratic Regression	1. The data below shows the average student loan debt accumulated by college graduates during certain years since 1996. Write an equation for the line of best fit, then estimate the average student loan debt for a student graduating in 2020. <table><tr><th>Year</th><th>Loan Debt</th></tr><tr><td>1996</td><td>\$12,750</td></tr><tr><td>2000</td><td>\$17,350</td></tr><tr><td>2004</td><td>\$18,650</td></tr><tr><td>2008</td><td>\$23,200</td></tr><tr><td>2009</td><td>\$24,000</td></tr></table>	Year	Loan Debt	1996	\$12,750	2000	\$17,350	2004	\$18,650	2008	\$23,200	2009	\$24,000		
	Year	Loan Debt													
1996	\$12,750														
2000	\$17,350														
2004	\$18,650														
2008	\$23,200														
2009	\$24,000														
	2. Dan hit a golf ball from the top of a hill. The height of the ball after each second is given in the table below. Using a quadratic model, write an equation for the curve of best fit, then estimate the height of the ball after 5 seconds. <table><tr><th>Time (s)</th><th>Height (ft)</th></tr><tr><td>0</td><td>60</td></tr><tr><td>1</td><td>83</td></tr><tr><td>2</td><td>95</td></tr><tr><td>3</td><td>81</td></tr><tr><td>4</td><td>59</td></tr></table>	Time (s)	Height (ft)	0	60	1	83	2	95	3	81	4	59		
Time (s)	Height (ft)														
0	60														
1	83														
2	95														
3	81														
4	59														
NEW: Modeling with Cubic & Quartic Functions	3. The Millennium Force at Cedar Point is one of the world's most popular roller coasters. The table below shows the average number of people in line at each hour after the park opens. Use a cubic model to find an equation for the curve of best fit, then estimate the number of people in line 6 hours after the park opened. <table><tr><th>Hour</th><th>People</th></tr><tr><td>0</td><td>0</td></tr><tr><td>1</td><td>120</td></tr><tr><td>2</td><td>324</td></tr><tr><td>3</td><td>375</td></tr><tr><td>4</td><td>394</td></tr><tr><td>5</td><td>312</td></tr></table>	Hour	People	0	0	1	120	2	324	3	375	4	394	5	312
Hour	People														
0	0														
1	120														
2	324														
3	375														
4	394														
5	312														

4. On the first day back to school after winter break, 20 students were absent from school due to illness. The table below shows the number of students absent from school on certain days since returning from winter break due to illness. Use a **quartic model** to find an equation for the curve of best fit, then estimate the number of students absent 9 days after returning from break.

Day	# Absent
0	20
2	28
4	35
6	42
8	43

5. The table below shows the average minimum wage during certain years since 1955. Use a **cubic model** to find an equation for the curve of best fit, then estimate the minimum wage in 2025.

Year	Min. Wage
1955	\$0.75
1961	\$1.15
1967	\$1.40
1974	\$2.00
1978	\$2.65
1980	\$3.10
1990	\$3.80

6. The table below shows the opening value of a stock index on the first day of trading during certain years since 1988. Use a **quartic model** to find an equation for the curve of best fit, then estimate the value of the stock index in 2010.

Year	Price
1988	\$1,800
1990	\$2,100
1993	\$5,500
1995	\$9,800
1999	\$7,000
2005	\$16,000

Unit 5 Test Study Guide

(Polynomial Functions)

Name: _____

Date: _____ Per: _____

Topic 1: Classifying Polynomials & Polynomial Operations

Simplify each expression. Final answers should be written in standard form.		
1. $(-4m^2n)^4 \cdot \frac{1}{6}m^{-10}n^{-4}$	2. $(8a^2 - 6 - 8a) + (1 - 6a - 7a^2)$	3. $(6x - 7x^2 + 7) - (5x^2 + 2x - 2x^3 - 1)$
4. $(y + 4)^3 - 2y(y - 1)$	5. $(3k - 6)(k^2 - k + 7)$	6. $\frac{-8c^6d^4 + 56c^4d^2 - 24c^2d}{8c^2d}$

Topic 2: Factoring Polynomials

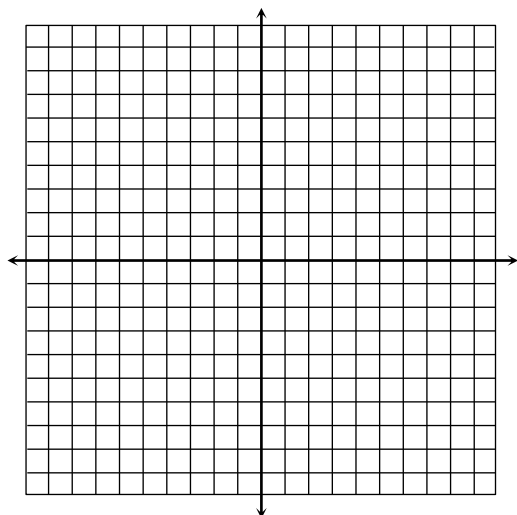
Differences of Squares $a^2 - b^2 =$	Sum of Cubes $a^3 + b^3 =$	Differences of Cubes $a^3 - b^3 =$
Factor each polynomial below completely.		
7. $9x^3 + 21x^2$	8. $3n^4 - 147$	
9. $64a^3 - 343b^3$	10. $648w + 1029w^4$	
11. $32c^5d - 162cd^3$	12. $216pq - p^7q$	

13. $2c^5 - 2c^3 - 60c$	14. $9y^4 - 7y^2 - 16$
15. $n^3 + 2n^2 - 36n - 72$	16. $8x^3 - 10x^2 + 28x - 35$

Topic 3: Graphing Polynomial Functions

Graph each function and identify its key characteristics.

17. $f(x) = x^3 + 8x^2 + 16x + 7$



Domain: _____

Range: _____

Rel. Maximum(s): _____

Rel. Minimum(s): _____

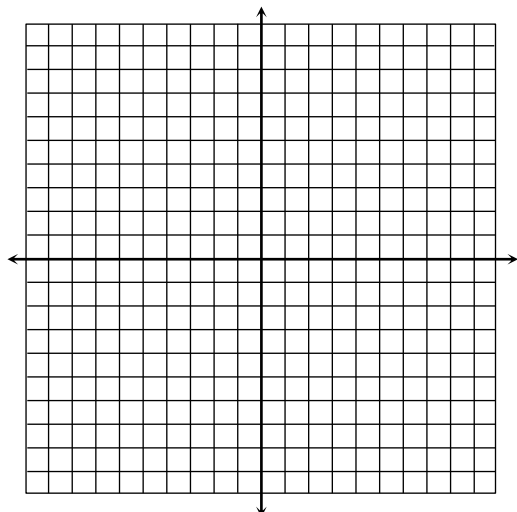
Inc. Intervals: _____

Dec. Intervals: _____

End Behavior: As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

18. $f(x) = -x^4 + 3x^2 + 2x$



Domain: _____

Range: _____

Rel. Maximum(s): _____

Rel. Minimum(s): _____

Inc. Intervals: _____

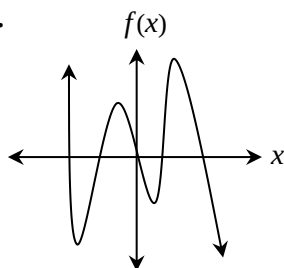
Dec. Intervals: _____

End Behavior: As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

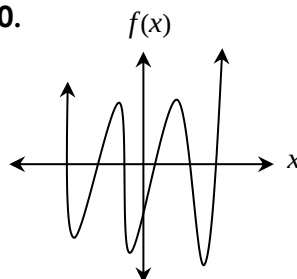
As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

Determine the end behavior, whether the function is an even or odd degree, and the sign of the leading coefficient given each graph below.

19.



20.



Identify the zeros (and their multiplicity) of each function below and the effect on the graph.

21. $f(x) = 3x^3(x-1)^2(x-8)$

Zero	Multiplicity	Effect

22.. $f(x) = -x(5x+3)^3(x-2)^8$

Zero	Multiplicity	Effect

23. The graph of a polynomial function has zeros of 0 (multiplicity 2), 1 (multiplicity 2), and $5/2$ (multiplicity 2). Write a function in standard form that could represent this function.

Topic 4: Solving Polynomial Equations

Solve each equation. Simplify all irrational and complex solutions.

24. $2x^4 - 48x^2 = 0$

25. $25x^3 = 64x$

26. $108x^3 + 37 = 5$

27. $9x^5 - 72x^2 = 0$

28. $x^4 + 19x^2 - 20 = 0$

29. $x^5 = 18x^3 - 81x$

30. $3x^4 - 14x^2 = 5$

31. $2x^3 + 7x^2 - 16x - 56 = 0$

Topic 5: Dividing Polynomials & The Remainder Theorem**Find each quotient.**

32. $(x^3 + x^2 - 71x + 9) \div (x + 9)$

33. $(8n^3 - 36n^2 - 15n - 16) \div (n - 5)$

34. $(12a^3 + 2a^2 - 6a - 30) \div (3a - 4)$

35. $(y^4 + 6y^3 - 4y - 31) \div (y + 6)$

36. Using the Remainder Theorem, find $f(-2)$ when $f(x) = 3x^3 - 28x^2 + 70$.**37.** The profit P of a small business (in thousands of dollars) since it was founded can be modeled by the function below, where t is the years since 1990. Use the Remainder Theorem to find the company's profit in 2017.

$$P(t) = 0.5t^4 - 3t^3 + t^2 + 25$$

Topic 6: Operations & Compositions of Functions

Given $f(x) = x^2 + 4x - 12$, $g(x) = 5x^2 - 2$, and $h(x) = x + 7$, find each function. Indicate any restrictions in the domain.

38. $(f - g)(x)$

39. $(h \cdot g)(x)$

40. $\left(\frac{h}{f}\right)(x)$

41. $(g \circ h)(x)$

Use the same functions above, evaluate each function.

42. $(g + h)(-4)$

43. $(h \circ f)(2)$

Topic 7: Regression

44. The population present in a bacteria culture over 5 days is given in the table below. Write a **cubic function** to represent the data.

Time (days)	0	1	2	3	4	5
Population	28	135	219	332	520	834

45. Use a **cubic function** to estimate the value of y when x is -8. How does the estimate change when a quartic function is used instead?

x	-4	0	4	8	12
y	975	128	-9	-160	-893

Name: _____

Date: _____ Per: _____

Unit 5 Test

Polynomial Functions

1. Simplify the expression given below.

$$\frac{(-6x^2y^3)^4}{8x^9y^7 \cdot 3xy^5}$$

A. $-x^2y$

C. $-\frac{1}{x^2}$

B. $-54x^2y$

D. $\frac{54}{x^2}$

2. Simplify the expression below. Give your answer in standard form.

$$(18 - 2x^3 + 9x) - (4 - x^2 - 3x^3 + 11x)$$

3. Simplify the expression below. Give your answer in standard form.

$$(2w - 5)^3$$

4. Simplify the expression below. Give your answer in standard form.

$$\frac{-18a^7b^3 + 9a^4b^2 - 63a^2b}{9ab}$$

Factor each polynomial completely.

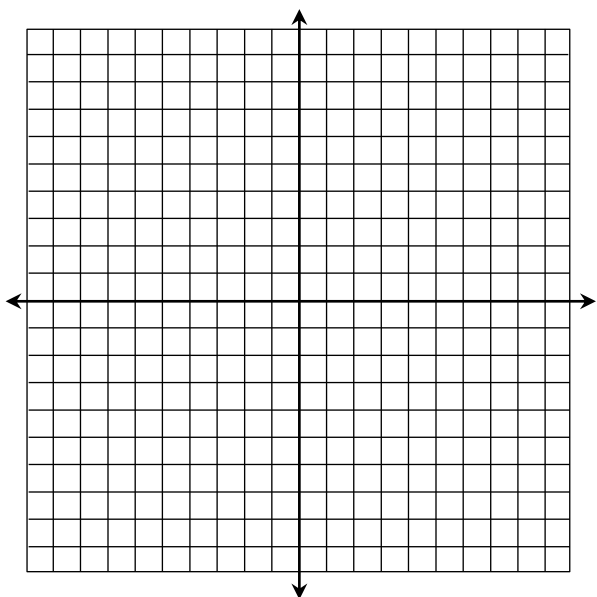
5. $50x^5 - 18x$

6. $x^4 + 216x$

7. $14x^3 - 49x^2 + 21x$

8. $x^3 - 5x^2 - x + 5$

9. Graph the function and identify all key characteristics: $f(x) = x^3 - 12x^2 + 45x - 54$



Domain: _____

Range: _____

Rel. Maximum(s): _____

Rel. Minimum(s): _____

Inc. Intervals: _____

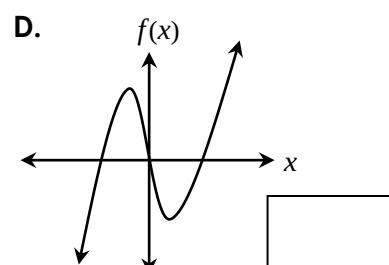
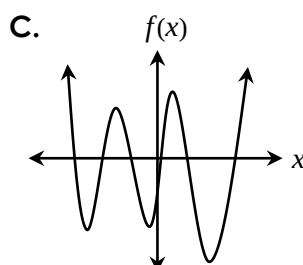
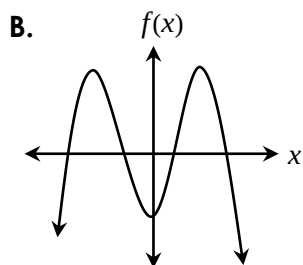
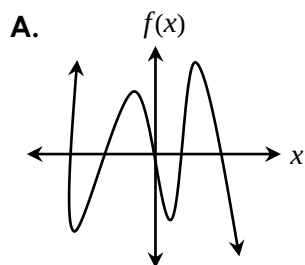
Dec. Intervals: _____

Zeros: _____

End Behavior: As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

10. If an even-degree polynomial function has a positive leading coefficient, which graph could represent this function?



11. Which of the following describes the end behavior of the function below? Check all that apply.

$$f(x) = -x^9 + bx^4 + c$$

☐ As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$

☐ As $x \rightarrow \infty$, $f(x) \rightarrow -\infty$

☐ As $x \rightarrow -\infty$, $f(x) \rightarrow \infty$

☐ As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

12. Identify the zeros, their multiplicity, and the effect (bounces or intersects) on the graph of the function given below.

$$f(x) = 2x(x + 4)^6(3x - 2)^3$$

Zero	Multiplicity	Effect

13. The graph of a polynomial function has zeros of 3 (multiplicity 2), and -7 (multiplicity 1). Write a function in standard form that could represent this function.

Solve each equation by factoring. Simplify all irrational and complex solutions.

14. $8x^3 - 98x = 0$

15. $80x^4 = 10x$

16. $x^4 - 5x^2 = 3x^2 + 48$

17. $3x^3 + 4x^2 = 12x + 16$

18. Which of the following represents the zeros of the function $f(x) = x^3 + 512$? Check all that apply.

☐ 8	☐ -8
☐ $4 + 4\sqrt{3}$	☐ $4 - 4\sqrt{3}$
☐ $4 + 4i\sqrt{3}$	☐ $4 - 4i\sqrt{3}$

Find the quotient.

19. $(x^3 + 3x^2 - 15x + 18) \div (x + 6)$

20. $(2a^3 - 10a^2 + 17a - 7) \div (a - 1)$

21. $(16w^3 + 58w^2 + 39w - 17) \div (2w + 5)$

22. $(y^4 - 64y^2 + 2y - 9) \div (y - 8)$

23. Kyle ran for 30 minutes. His elevation f (in feet) x minutes into his run can be represented by the function $f(x) = 0.04x^3 - 2x^2 + 24x + 150$. Using the Remainder theorem, find his elevation halfway through his run.

Use $f(x) = x^2 - x$, $g(x) = 9 - 2x$, and $h(x) = x^2 + 2x - 63$ for questions 24-29.

24. Find $(f + h)(x)$

25. Find $(g \cdot f)(x)$

26. The domain of the function $\left(\frac{g}{h}\right)(x)$ is all real numbers except -

- A. -9 and 7
- B. -7 and 9
- C. -9, 0 and 7
- D. -7, 0 and 9

27. Find $(f \circ g)(x)$

28. Evaluate $(h - g)(4)$

29. Evaluate $(g \circ h)(-1)$

30. The data below shows the attendance at a water park over five days. Write a **quartic** function to model the data, then estimate the attendance at the park on the 8th day.

Day	Attendance
1	605
2	921
3	977
4	1,037
5	1,221

Equation:

Attendance:

Name:

Date:

Topic:

Class:

Main Ideas/Questions

Notes/Examples

MONOMIALS

- A **monomial** is a number, variable, or a product of numbers and variables.

- Examples: 4, 7x, 3x²

- Use the **EXPONENT RULES** to simplify monomial expressions:

NAME	RULE	EXAMPLE
Product Rule	$a^m \cdot a^n = a^{m+n}$	$x^2 \cdot x^7 = x^9$
Power Rule	$(a^m)^n = a^{m \cdot n}$	$(x^2)^7 = x^{14}$
Quotient Rule	$\frac{a^m}{a^n} = a^{m-n}$	$\frac{x^7}{x^2} = x^5$
Negative Exponent Rule	$a^{-m} = \frac{1}{a^m}$	$x^{-2} = \frac{1}{x^2}$
Zero Exponent Rule	$a^0 = 1$ ($0^0 = \text{undefined}$)	$(7x^2)^0 = 1$
When ADDING OR SUBTRACTING monomials, COMBINE LIKE TERMS!		

EXAMPLES

1. $\underline{5x^2} \cdot \underline{-7x^6}$

$$= \boxed{-35x^8}$$

2. $(-2a^3b)^2 \cdot 8ab^9$

$$(4a^6b^2) \cdot 8ab^9$$

$$= \boxed{32a^7b^{11}}$$

3. $\frac{54m^6n^4}{3m^2n} - 10m^4n^3$

$$18m^4n^3 - 10m^4n^3$$

$$= \boxed{8m^4n^3}$$

4. $\underline{2k^4} \cdot \underline{10k^{-7}}$

$$20k^{-3} = \boxed{\frac{20}{k^3}}$$

5. $\left(\frac{2}{3}r^2s^7\right)^2 \cdot \left(\frac{1}{6}r^3s\right)$

$$\frac{4}{9}r^4s^{14} \cdot \frac{1}{6}r^3s = \boxed{\frac{2}{27}r^7s^{15}}$$

6. $\left(\frac{14w^{12}}{7w^3}\right)^{-1} = (2w^9)^{-1}$

$$= \boxed{\frac{1}{2w^9}}$$

7. $\frac{15x^{10}y^4}{24x^{12}y^3} = \frac{5}{8}x^{-2}y$

$$= \boxed{\frac{5y}{8x^2}}$$

8. $\left(\frac{c}{c^2}\right)^4 \cdot (-3c)^4$

$$(c^{-1})^4 \cdot 81c^4 = c^{-4} \cdot 81c^4$$

$$= \boxed{81}$$

	9. Give an example of two monomials with a <u>quotient</u> of $\frac{-3n^2}{m}$. $\frac{-36m^4n^7}{12m^5n^5}$																
POLYNOMIALS	- A polynomial is the sum or difference of many monomials. - The highest exponent of a polynomial is called the <u>degree</u>. Standard Form: <u>Always write exponents in descending order.</u> Write the polynomials below in standard form: 10. $-k^5 - 1 + 8k - 3k^3 + \frac{1}{4}k^2$ <u>$-k^5 - 3k^3 + \frac{1}{4}k^2 + 8k - 1$</u> 11. $18a^2b^2 + 7ab - b^2 + 4a^3$ <u>$4a^3 + 18a^2b^2 + 7ab - b^2$</u> 12. $5xy^2 - x^2 + 9x^3y - y^4 + 2$ <u>$9x^3y - x^2 + 5xy^2 - y^4 + 2$</u>																
CLASSIFYING POLYNOMIALS	Polynomials are classified by degree (highest exponent) and number of terms. Use the charts to the left to classify each polynomial below. <table style="width: 100%;"> <tr> <td style="width: 25%;">13. $-3x + 1$</td> <td><u>linear, binomial</u></td> </tr> <tr> <td>14. $9x^5 - x^4 + 2x$</td> <td><u>quintic, trinomial</u></td> </tr> <tr> <td>15. 24</td> <td><u>constant, monomial</u></td> </tr> <tr> <td>16. $\frac{1}{2}x^3 - 2x^2 + 4x + 15$</td> <td><u>cubic, polynomial</u></td> </tr> <tr> <td>17. $-x^2 - 18x + 31$</td> <td><u>quadratic, trinomial</u></td> </tr> <tr> <td>18. $-\frac{3}{2}x^4$</td> <td><u>quartic, monomial</u></td> </tr> <tr> <td>19. Give an example of a cubic binomial.</td> <td><u>$\frac{3}{5}x^3 + \frac{7}{6}x$</u></td> </tr> <tr> <td>20. Give an example of a linear monomial.</td> <td><u>$-5x$</u></td> </tr> </table>	13. $-3x + 1$	<u>linear, binomial</u>	14. $9x^5 - x^4 + 2x$	<u>quintic, trinomial</u>	15. 24	<u>constant, monomial</u>	16. $\frac{1}{2}x^3 - 2x^2 + 4x + 15$	<u>cubic, polynomial</u>	17. $-x^2 - 18x + 31$	<u>quadratic, trinomial</u>	18. $-\frac{3}{2}x^4$	<u>quartic, monomial</u>	19. Give an example of a cubic binomial.	<u>$\frac{3}{5}x^3 + \frac{7}{6}x$</u>	20. Give an example of a linear monomial.	<u>$-5x$</u>
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Degree	
0	Constant
1	linear
2	quadratic
3	cubic
4	quartic
5	quintic

Number of Terms	
1	monomial
2	binomial
3	trinomial
4+	polynomial

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Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples	
ADDING & SUBTRACTING <i>Polynomials</i>	❶	COMBINE LIKE TERMS! (Watch out for subtraction problems!)
	❷	Write your answer in STANDARD FORM .
	1. $(\underline{3x^2} + \underline{11x} + 4) + (\underline{-5x} + \underline{x^2} - 13)$	2. $(9n^3 - \underline{4n^2} + \underline{2n} - 10) + (\underline{-2n^2} + \underline{n} + 7)$
	$4x^2 + 6x - 9$	$9n^3 - 6n^2 + 3n - 3$
	3. $(5k^3 - \underline{2k^2} + 2k) - (\underline{2k^2} + 2k + 17)$	4. $(\underline{y} + \underline{4y^2} - 3) - (1 + \underline{2y^2} - \underline{5y} - y^3)$
MULTIPLYING <i>Polynomials</i>	5. Subtract $(-10ab + 7a^2 - b^2)$ from $(8b^2 + ab - 2a^2)$.	
	$(8b^2 + \underline{ab} - \underline{2a^2}) - (\underline{-10ab} + \underline{7a^2} - b^2)$ $= \boxed{-9a^2 + 11ab + 9b^2}$	
	❶	DISTRIBUTE to expand and remove parentheses.
	❷	COMBINE LIKE TERMS!
	❸	Write your answer in STANDARD FORM .
	6. $(w-7)(w^2 + 2w + 1)$	7. $(2x+1)(4-9x) - 2x(3x+11)$
	$w^3 + \underline{2w^2} + \underline{w} - \underline{7w^2} - \underline{14w} - \underline{7}$ $= \boxed{w^3 - 5w^2 - 13w - 7}$	$\underline{8x} - \underline{18x^2} + \underline{4} - \underline{9x} - \underline{6x^2} - \underline{22x}$ $= \boxed{-24x^2 - 23x + 4}$
	8. $-3(a+5)(a-2)(a+8)$ $(-3a-15)(a^2+6a-16)$ $= -3a^3 - \underline{18a^2} + \underline{48a} - \underline{15a^2}$ $\quad - \underline{90a} + 240$ $= \boxed{-3a^3 - 33a^2 - 42a + 240}$	9. $(2c+5d)^3$ $(2c+5d)(2c+5d)(2c+5d)$ $= (2c+5d)(4c^2 + 20cd + 25d^2)$ $= 8c^3 + \underline{40c^2d} + \underline{50cd^2} + \underline{20c^2d}$ $\quad + \underline{100cd^2} + 125d^3$ $= \boxed{8c^3 + 60c^2d + 150cd^2 + 125d^3}$

	10. $(3m^2 - 4m + 1)(2m^2 + 5m - 9)$ $6m^4 + 15m^3 - 27m^2 - 8m^3 - 20m^2 + 36m + 2m^2 + 5m - 9$ $= 6m^4 + 7m^3 - 45m^2 + 41m - 9$	
	11. $(x^2 + 6x - 7)(-x^2 - 9x - 4)$ $-x^4 - 9x^3 - 4x^2 - 6x^3 - 54x^2 - 24x + 7x^2 + 63x + 28$ $= -x^4 - 15x^3 - 51x^2 + 39x + 28$	
	12. The height of a trapezoid can be expressed as $(2x + 9)$ while the bases can be expressed as $(3x - 7)$ and $(x + 5)$. Write an expression to represent the area of the trapezoid. $A = \frac{1}{2}h(b_1 + b_2)$ $A = \frac{1}{2}(2x + 9)(3x - 7 + x + 5)$ $A = \frac{1}{2}(2x + 9)(4x - 2)$ $A = (2x + 9)(2x - 1)$ $A = 4x^2 - 2x + 18x - 9$ $A = 4x^2 + 16x - 9$	
DIVIDING Polynomials (by a Monomial)	① DIVIDE each term in the numerator by the monomial in the denominator.	
	② Write your answer in STANDARD FORM. Rewrite all terms with negative exponents.	
	13. $\frac{9m^3 - 24m^2}{3m}$ $= 3m^2 - 8m$	14. $\frac{20x^3 - 4x^2 - 8x}{4x}$ $= 5x^2 - x - 2$
	15. $(14a^4b^3 - 20a^2b^2 + 2ab) \cdot (2ab)^{-1}$ $= \frac{14a^4b^3 - 20a^2b^2 + 2ab}{2ab}$ $= 7a^3b^2 - 10ab + 1$	16. $\frac{30c^8d^3 - 36c^6d^2 + 6c^3d}{12c^2d}$ $= \frac{5}{2}c^6d^2 - 3c^4d + \frac{1}{2}c$ $= \frac{5c^6d^2}{2} - 3c^4d + \frac{c}{2}$
	17. $\frac{30x^3 - 12x^2 + 9x}{3x^2}$ $= 10x - 4 + 3x^{-1}$ $= 10x - 4 + \frac{3}{x}$	18. $\frac{40m^3n^4 - 24m^2n^7}{8m^2n^5}$ $= 5mn^{-1} - 3n^2$ $= \frac{5m}{n} - 3n^2$

Name: _____

Unit 5: Polynomial Functions

Date: _____ Per: _____

Homework 1: Monomials & Polynomials

**** This is a 2-page document! ******Simplify the monomials below. Final answers should contain positive exponents only.**

1. Subtract $-3n^2$ from $-7n^2$
 $(-7n^2) - (-3n^2)$
 $= \boxed{-4n^2}$

2. $(-8x^4y^3) \cdot (2x^5y^2) + 7x^9y^5$
 $-16x^9y^5 + 7x^9y^5$
 $= \boxed{-9x^9y^5}$

3. $(-4a^3b^2)^2 \cdot (3a^2b)$
 $16a^6b^4 \cdot 3a^2b$
 $= \boxed{48a^8b^5}$

4. $5c^{-7}d^2 \cdot (-cd^2)^4$
 $5c^{-7}d^2 \cdot c^4d^8$
 $= 5c^{-3}d^{10}$
 $= \boxed{\frac{5d^{10}}{c^3}}$

5. $\frac{18w^4x^9}{14w^5x^5}$
 $\frac{9}{7}w^{-1}x^4$
 $= \boxed{\frac{9x^4}{7w}}$

6. $\frac{15x^2y \cdot -6x^7y}{(3xy)^2}$
 $\frac{-90x^9y^2}{9x^2y^2}$
 $= \boxed{-10x^7}$

7. Write two monomials with a product of $-84a^6b^{10}$.

$$(14a^{10}b^9) \cdot (-6a^{-4}b)$$

8. The side length of a cube can be represented by the expression $2x^5$. If the side length is doubled, write an expression to represent the new volume of the cube.

$$V = s^3 \quad V = (4x^5)^3$$

$$= \boxed{64x^{15}}$$

Write in standard form if needed, then classify the polynomial.

9. $25 - 9m^2 - m + 4m^3$
 $4m^3 - 9m^2 - m + 25$
 cubic, polynomial

10. $-\frac{1}{2}k^3 + \frac{3}{4}k^5$
 $\frac{3}{4}k^5 - \frac{1}{2}k^3$
 quintic, binomial

11. $14y^2 - x^2 - 3xy$
 $-x^2 - 3xy + 14y^2$
 quadratic, trinomial

12. $9x$
 linear, monomial

Simplify. Final answers should be written in standard form.

13. $(\underline{8x^3} - \underline{6x^4} + 3) - (\underline{3x^3} - 3 + \underline{8x^4})$
 $-14x^4 + 5x^3 + 6$

14. $(\underline{2a^2b} - 5b^3 + \underline{4a^4b^2}) - (\underline{7b^3} + \underline{8a^4b^2} - \underline{7a^2b})$
 $-4a^4b^2 + 9a^2b - 12b^3$

$$15. -3x(5x+1)(8-2x)$$

$$= -3x(40x - 10x^2 + 8 - 2x)$$

$$= -3x(-10x^2 + 38x + 8)$$

$$= \boxed{30x^3 - 114x^2 - 24x}$$

$$16. (a+b)(3a-b)(2a+7b)$$

$$= (a+b)(6a^2 + 21ab - 2ab - 7b^2)$$

$$= (a+b)(6a^2 + 19ab - 7b^2)$$

$$= 6a^3 + 19a^2b - 7ab^2 + 6a^2b + 19ab^2 - 7b^3$$

$$= \boxed{6a^3 + 25a^2b + 12ab^2 - 7b^3}$$

$$17. (1-2n)^3 - 7n(n^2-2)$$

$$= (1-2n)(1-2n)(1-2n) - 7n^3 + 14n$$

$$= (1-2n)(1-4n+4n^2) - 7n^3 + 14n$$

$$= 1 - 4n + 4n^2 - 2n + 8n^2 - 8n^3 - 7n^3 + 14n$$

$$= \boxed{-15n^3 + 12n^2 + 8n + 1}$$

$$18. 4(2-3w)(w^2-2w+10)$$

$$= (8-12w)(w^2-2w+10)$$

$$= 8w^2 - 16w + 80 - 12w^3 + 24w^2 - 120w$$

$$= \boxed{-12w^3 + 32w^2 - 136w + 80}$$

$$19. \frac{-42x^{10}y^5 + 12x^8y^3 - 6x^2y}{6x^2y}$$

$$= \boxed{-7x^8y^4 + 2x^6y^2 - 1}$$

$$20. \frac{16a^4 - 40a^2 + 24a}{12a^3}$$

$$= \frac{4}{3}a - \frac{10}{3}a^{-1} + 2a^{-2}$$

$$= \boxed{\frac{4a}{3} - \frac{10}{3a} + \frac{2}{a^2}}$$

21. The length, width, and height of a rectangular prism can be represented by the expressions $(x+3)$, $(x+7)$ and $(x-1)$. Write an expression to represent the surface area of the prism. $SA = 2LW + 2LH + 2WH$

$$SA = 2(x+3)(x+7) + 2(x+3)(x-1) + 2(x+7)(x-1)$$

$$SA = 2(x^2 + 10x + 21) + 2(x^2 + 2x - 3) + 2(x^2 + 6x - 7)$$

$$SA = 2x^2 + 20x + 42 + 2x^2 + 4x - 6 + 2x^2 + 12x - 14$$

$$\boxed{SA = 6x^2 + 36x + 22}$$

22. The area of a triangle can be represented by the expression $14x^5 + 63x^2$. If the base is $7x^2$, write an expression to represent its height.

$$A = \frac{1}{2}bh \rightarrow \frac{2A}{b} = h$$

$$h = \frac{2(14x^5 + 63x^2)}{7x^2}$$

$$h = \frac{28x^5 + 126x^2}{7x^2}$$

$$\boxed{h = 4x^3 + 18}$$

Name:		Date:	
Topic:		Class:	
Main Ideas/Questions		Notes/Examples	
GCF Greatest Common Factor	1. $12a^3 + 10a^2$ $2a^2(6a + 5)$	2. $14m^8n^5 - 7m^2n^2$ $7m^2n^2(2m^6n^3 - 1)$	
	3. $18p^8q^3 + 28p^6q^5$ $2p^6q^3(9p^2 + 14q^2)$	4. $16x^3y - 12x^2y^2 + 32xy^3$ $4xy(4x^2 - 3xy + 8y^2)$	
DIFFERENCE OF SQUARES $a^2 - b^2$	Perfect Squares: 1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144, ...		
	Rule: $a^2 - b^2 = (a + b)(a - b)$		
	5. $m^2 - 25$ $= (m + 5)(m - 5)$	6. $9x^2 - 49y^2$ $= (3x + 7y)(3x - 7y)$	
	7. $5r^2 - 80s^2$ $5(r^2 - 16s^2)$ $= 5(r + 4s)(r - 4s)$	8. $72c^4 - 2$ $2(36c^4 - 1)$ $= 2(6c^2 + 1)(6c^2 - 1)$	
SUM OF CUBES $a^3 + b^3$	9. $7m^7n - 28mn^3$ $7mn(m^6 - 4n^2)$ $= 7mn(m^3 + 2n)(m^3 - 2n)$	10. $10w^5 - 10w$ $10w(w^4 - 1)$ $= 10w(w^2 + 1)(w^2 - 1)$ $= 10w(w^2 + 1)(w + 1)(w - 1)$	
	Perfect Cubes: 1, 8, 27, 64, 125, 216, 343, 512, 729, ...		
	Rule: $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$		
	11. $x^3 + 27 \rightarrow (x)^3 + (3)^3$ $= (x + 3)(x^2 - 3x + 9)$	12. $8w^3 + 1 \rightarrow (2w)^3 + (1)^3$ $= (2w + 1)(4w^2 - 2w + 1)$	
13. $m^3 + 216n^3 \rightarrow (m)^3 + (6n)^3$ $= (m + 6n)(m^2 - 6mn + 36n^2)$	14. $343p^3 + 64q^3 \rightarrow (7p)^3 + (4q)^3$ $= (7p + 4q)(49p^2 - 28pq + 16q^2)$		
15. $500a^3 + 4$ $4(125a^3 + 1) \rightarrow (5a)^3 + (1)^3$ $= 4(5a + 1)(25a^2 - 5a + 1)$	16. $24c^4d + 375cd^4$ $3cd(8c^3 + 125d^3) \rightarrow (2c)^3 + (5d)^3$ $= 3cd(2c + 5d)(4c^2 - 10cd + 25d^2)$		

DIFFERENCE OF CUBES $a^3 - b^3$	Rule: $(a^3 - b^3) = (a - b)(a^2 + ab + b^2)$	
TRINOMIALS $ax^2 + bx + c$	17. $x^3 - 1 \rightarrow (x)^3 - (1)^3$ $= (x - 1)(x^2 + x + 1)$	18. $64 - w^3 \rightarrow (4)^3 - (w)^3$ $= (4 - w)(16 + 4w + w^2)$
	19. $216k^3 - 125 \rightarrow (6k)^3 - (5)^3$ $= (6k - 5)(36k^2 + 30k + 25)$	20. $m^3 - 343n^3 \rightarrow (m)^3 - (7n)^3$ $= (m - 7n)(m^2 + 7mn + 49n^2)$
	21. $4 - 32h^3$ $= 4(1 - 8h^3) \rightarrow (1)^3 - (2h)^3$ $= 4(1 - 2h)(1 + 2h + 4h^2)$	22. $2x^3 - 54y^3$ $= 2(x^3 - 27y^3) \rightarrow (x)^3 - (3y)^3$ $= 2(x - 3y)(x^2 + 3xy + 9y^2)$
	23. $k^2 + 15k + 56$ $(k + 7)(k + 8)$	24. $n^2 + 7n - 30$ $(n + 10)(n - 3)$
	25. $p^2 - 16p + 64$ $(p - 8)(p - 8)$ $= (p - 8)^2$	26. $r^2 - 11r - 42$ $(r - 14)(r + 3)$
	27. $w^3 + w^2 - 20w$ $w(w^2 + w - 20)$ $= w(w + 5)(w - 4)$	28. $2a^3 + 38a^2 + 68a$ $2a(a^2 + 19a + 34)$ $= 2a(a + 17)(a + 2)$
	29. $n^4 + 2n^2 - 24$ $(n^2 + 6)(n^2 - 4)$ $= (n^2 + 6)(n + 2)(n - 2)$	30. $x^5 - 24x^3 - 25x$ $x(x^4 - 24x^2 - 25)$ $= x(x^2 - 25)(x^2 + 1)$ $= x(x + 5)(x - 5)(x^2 + 1)$
	31. $3x^2 - 10x + 8$ $x^2 - 10x + 24$ $= (x - \frac{6}{3})(x - \frac{4}{3})$ $= (x - 2)(3x - 4)$	32. $8x^2 - 6x - 5$ $x^2 - 6x - 40$ $(x - \frac{10}{8})(x + \frac{4}{8})$ $(x - \frac{5}{4})(x + \frac{1}{2})$ $(4x - 5)(2x + 1)$

$$\begin{aligned}
 33. \quad & 3x^4 + 14x^2 - 5 \\
 & x^4 + 14x^2 - 15 \\
 & = (x^2 + \frac{15}{3})(x^2 - 1) \\
 & = \boxed{(x^2 + 5)(3x^2 - 1)}
 \end{aligned}$$

$$\begin{aligned}
 34. \quad & 12m^4 - 11m^2 + 2 \\
 & m^4 - 11m^2 + 24 \\
 & = (m^2 - \frac{8}{12})(m^2 - \frac{3}{12}) \\
 & = (m^2 - \frac{2}{3})(m^2 - \frac{1}{4}) \\
 & = (3m^2 - 2)(4m^2 - 1) \\
 & = \boxed{(3m^2 - 2)(2m + 1)(2m - 1)}
 \end{aligned}$$

FOUR TERMS

Rule: Use grouping!

Steps		Example
①	GROUP the first two terms together and the last two terms.	$x^3 + x^2 - 4x - 4$
②	FACTOR the GCF for each group.	$\underline{x^2}(x+1) - \underline{4}(x+1)$
③	FACTOR the common binomial.	$(x^2 - 4)(x+1)$ $= (x+2)(x-2)(x+1)$
35. $x^3 - x^2 + 9x - 9$ $\underline{x^2}(x-1) + \underline{9}(x-1)$ $= \boxed{(x^2 + 9)(x-1)}$		36. $k^3 + 5k^2 - k - 5$ $\underline{k^2}(k+5) - \underline{1}(k+5)$ $= (k^2 - 1)(k+5)$ $= \boxed{(k+1)(k-1)(k+5)}$
37. $4v^3 - 5v^2 - 16v + 20$ $\underline{v^2}(4v-5) - \underline{4}(4v-5)$ $= (v^2 - 4)(4v-5)$ $= \boxed{(v+2)(v-2)(4v-5)}$		38. $2x^5 - 18x^4 + 7x - 63$ $\underline{2x^4}(x-9) + \underline{7}(x-9)$ $= \boxed{(2x^4 + 7)(x-9)}$
39. $c^4 + 9c^3 - c - 9$ $\underline{c^3}(c+9) - \underline{1}(c+9)$ $= (c^3 - 1)(c+9)$ $= \boxed{(c-1)(c^2 + c + 1)(c+9)}$		40. $2g^4 - 3g^3 + 16g - 24$ $\underline{g^3}(2g-3) + \underline{8}(2g-3)$ $= (g^3 + 8)(2g-3)$ $= \boxed{(g+2)(g^2 - 2g + 4)(2g-3)}$

Name: _____

Unit 5: Polynomial Functions

Date: _____ Per: _____

Homework 2: Factoring Polynomials

Directions: Complete the following rules.

1. Difference of Squares

$$a^2 - b^2 = (a+b)(a-b)$$

2. Sum of Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2)$$

3. Difference of Cubes

$$a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Directions: Factor each polynomial **completely**. Make sure to check for a **GCF** first.

4. $x^4 - 36$

$$= (x^2 + 6)(x^2 - 6)$$

5. $64c^3 + 1$

$$= (4c+1)(16c^2 - 4c + 1)$$

6. $k^3 - 27$

$$= (k-3)(k^2 + 3k + 9)$$

7. $54x^3 + 250y^3$

$$= 2(27x^3 + 125y^3)$$

$$= 2(3x+5y)(9x^2 - 15xy + 25y^2)$$

8. $3m^4 - 48n^2$

$$3(m^4 - 16n^2)$$

$$= 3(m^2 + 4n)(m^2 - 4n)$$

9. $a^7b^2 - ab^2$

$$ab^2(a^6 - 1)$$

$$= ab^2(a^3 + 1)(a^3 - 1)$$

$$= ab^2(a+1)(a^2 - a + 1)(a-1)(a^2 + a + 1)$$

10. $x^3y^2 - 343y^5$

$$y^2(x^3 - 343y^3)$$

$$= y^2(x - 7y)(x^2 + 7xy + 49y^2)$$

11. $9y^7 - 144y$

$$9y(y^6 - 16)$$

$$= 9y(y^3 + 4)(y^3 - 4)$$

12. $w^2 - 13w + 36$

$$= (w-9)(w-4)$$

13. $p^3 + 5p^2 - 84p$

$$p(p^2 + 5p - 84)$$

$$= p(p+12)(p-7)$$

14. $x^4 - 12x^2 + 36$ $(x^2 - 6)(x^2 - 6)$ $= \boxed{(x^2 - 6)^2}$	15. $w^4 - 14w^2 - 32$ $(w^2 - 16)(w^2 + 2)$ $= \boxed{(w + 4)(w - 4)(w^2 + 2)}$
16. $k^3 + 7k^2 - 44k$ $k(k^2 + 7k - 44)$ $= \boxed{k(k + 11)(k - 4)}$	17. $2a^3 + 28a^2 + 96a$ $2a(a^2 + 14a + 48)$ $= \boxed{2a(a + 6)(a + 8)}$
18. $-x^3 + 4x^2 + 21x$ $-x(x^2 - 4x - 21)$ $= \boxed{-x(x - 7)(x + 3)}$	19. $m^6 - 7m^4 - 18m^2$ $m^2(m^4 - 7m^2 - 18)$ $= m^2(m^2 - 9)(m^2 + 2)$ $= \boxed{m^2(m + 3)(m - 3)(m^2 + 2)}$
20. $9y^6 + 6y^4 + y^2$ $y^2(9y^4 + 6y^2 + 1)$ $= y^2(3y^2 + 1)(3y^2 + 1)$ $= \boxed{y^2(3y^2 + 1)^2}$	21. $8c^4 + 10c^2 - 3$ $c^4 + 10c^2 - 24$ $= (c^2 + \frac{12}{8})(c^2 - \frac{2}{8})$ $= (c^2 + \frac{3}{2})(c^2 - \frac{1}{4})$ $= (2c^2 + 3)(4c^2 - 1)$ $= \boxed{(2c^2 + 3)(2c - 1)(2c + 1)}$
22. $x^3 - 7x^2 + x - 7$ $x^2(x - 7) + 1(x - 7)$ $= \boxed{(x^2 + 1)(x - 7)}$	23. $4r^3 - 3r^2 - 4r + 3$ $r^2(4r - 3) - 1(4r - 3)$ $= (r^2 - 1)(4r - 3)$ $= \boxed{(r + 1)(r - 1)(4r - 3)}$
24. $3p^3 + 5p^2 - 12p - 20$ $p^2(3p + 5) - 4(3p + 5)$ $= (p^2 - 4)(3p + 5)$ $= \boxed{(p + 2)(p - 2)(3p + 5)}$	25. $15n^3 - 6n^2 - 25n + 10$ $3n^2(5n - 2) - 5(5n - 2)$ $= \boxed{(3n^2 - 5)(5n - 2)}$

Check for a **GCF** first!

FACTORING GUIDE

2
terms

DIFFERENCE OF SQUARES

$$\begin{array}{c} a^2 - b^2 \\ (a+b)(a-b) \end{array}$$

$$x^2 - 9 = (x+3)(x-3)$$

$$16x^4 - 25 = (4x^2+5)(4x^2-5)$$

$$\begin{aligned} 48x^3 - 3x &= 3x(16x^2 - 1) \\ &= 3x(4x+1)(4x-1) \end{aligned}$$

SUM OF CUBES

$$\begin{array}{c} a^3 + b^3 \\ (a+b)(a^2 - ab + b^2) \end{array}$$

$$x^3 + 27 = (x+3)(x^2 - 3x + 9)$$

$$8x^6 + 1 = (2x^2+1)(4x^4 - 2x^2 + 1)$$

$$\begin{aligned} 216x^3 + 125y^3 \\ = (6x+5y)(36x^2 - 30xy + 25y^2) \end{aligned}$$

DIFFERENCE OF CUBES

$$\begin{array}{c} a^3 - b^3 \\ (a-b)(a^2 + ab + b^2) \end{array}$$

$$64 - x^3 = (4-x)(16 + 4x + x^2)$$

$$27x^3 - 8 = (3x-2)(9x^2 + 6x + 4)$$

$$\begin{aligned} 512x^3 - 8 &= 8(64x^3 - 1) \\ &= 8(4x-1)(16x^2 + 4x + 1) \end{aligned}$$

3
terms

TRINOMIALS ($a = 1$)

Multiplies to "c" and adds to "b"

$$x^2 - 6x + 5 = (x-5)(x-1)$$

$$\begin{aligned} 3x^2 + 3x - 60 &= 3(x^2 + x - 20) \\ &= 3(x+5)(x-4) \end{aligned}$$

TRINOMIALS ($a > 1$)

Multiplies to "a·c" and adds to "b"

$$2x^2 + 15x - 77 = (2x-7)(x+11)$$

$$6x^2 - 13x - 28 = (3x+4)(2x-7)$$

$$\begin{aligned} 9x^2 + 30x + 25 &= (3x+5)(3x+5) \\ &= (3x+5)^2 \end{aligned}$$

4
terms

TRY GROUPING!

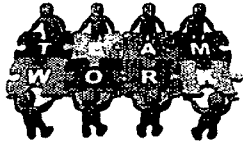
Find the GCF of each part, then factor the binomials.

$$\begin{aligned} x^3 - 2x^2 + 6x - 12 \\ = x^2(x-2) + 6(x-2) \\ = (x^2+6)(x-2) \end{aligned}$$

$$\begin{aligned} x^3 + x^2 - 9x - 9 &= x^2(x+1) - 9(x+1) \\ &= (x^2-9)(x+1) \\ &= (x+3)(x-3)(x+1) \end{aligned}$$

Group Members: _____

Per: _____



REVIEW: Factoring Polynomials

Directions: Work together to factor each polynomial **completely**!
Work together to complete each problem. Do not divide up the work!
Each person should be participating.

2 TERMS

1. $8a^3 + 1$

$$(2a+1)(4a^2-2a+1)$$

2. $x^4 - 100y^2$

$$(x^2+10y)(x^2-10y)$$

3. $m^3 - 64$

$$(m-4)(m^2+4m+16)$$

4. $27w^5 - 75w$

$$3w(9w^4 - 25)$$

$$= 3w(3w^2+5)(3w^2-5)$$

5. $125c^4 + cd^3$

$$c(125c^3 + d^3)$$

$$= c(5c+d)(25c^2-5cd+d^2)$$

6. $p^6q^4 - q^7$

$$q^4(p^6 - q^3)$$

$$q^4(p^2 - q)(p^4 + p^2q + q^2)$$

7. $5a^4 - 245$

$$5(a^4 - 49)$$

$$= 5(a^2+7)(a^2-7)$$

8. $10x^8y^2 + 80x^2y^5$

$$10x^2y^2(x^6 + 8y^3)$$

$$= 10x^2y^2(x^2+2y)(x^4-2x^2y+4y^2)$$

3 TERMS

9. $x^4 + 2x^2 - 63$

$$(x^2+9)(x^2-7)$$

10. $a^4 - 3a^2 + 2$

$$(a^2-1)(a^2-2)$$

$$= (a+1)(a-1)(a^2-2)$$

11. $x^3 - 12x^2 + 36x$ $x(x^2 - 12x + 36)$ $= x(x-6)(x-6)$ $= \boxed{x(x-6)^2}$	12. $2m^4 - 10m^2 - 28$ $2(m^4 - 5m^2 - 14)$ $= \boxed{2(m^2 - 7)(m^2 + 2)}$
13. $5x^4 - 10x^2 - 315$ $5(x^4 - 2x^2 - 63)$ $= 5(x^2 - 9)(x^2 + 7)$ $= \boxed{5(x+3)(x-3)(x^2 + 7)}$	14. $3u^5 - 9u^3 - 84u$ $3u(u^4 - 3u^2 - 28)$ $= \boxed{3u(u^2 - 7)(u^2 + 4)}$
15. $3a^4 + 16a^2 + 20$ $a^4 + 16a^2 + 60$ $(a^2 + \frac{6}{3})(a^2 + \frac{10}{3})$ $= \boxed{(a^2 + 2)(3a^2 + 10)}$	16. $8n^4 - 2n^2 - 1$ $n^4 - 2n^2 - 8$ $(n^2 - \frac{4}{8})(n^2 + \frac{2}{8})$ $(n^2 - \frac{1}{2})(n^2 + \frac{1}{4})$ $= \boxed{(2n^2 - 1)(4n^2 + 1)}$

4 TERMS

17. $x^3 - 2x^2 + x - 2$ $x^2(x-2) + 1(x-2)$ $= \boxed{(x^2 + 1)(x-2)}$	18. $6b^3 - 3b^2 + 4b - 2$ $3b^2(2b-1) + 2(2b-1)$ $= \boxed{(3b^2 + 2)(2b-1)}$
19. $2w^3 + 3w^2 - 6w - 9$ $w^2(2w+3) - 3(2w+3)$ $= \boxed{(w^2 - 3)(2w+3)}$	20. $x^5 + 5x^2 - 4x^3 - 20$ $x^2(x^3 + 5) - 4(x^3 + 5)$ $= (x^2 - 4)(x^3 + 5)$ $= \boxed{(x+2)(x-2)(x^3 + 5)}$

Name: _____ Algebra 2

Date: _____ Per: _____ Unit 5: Polynomial Functions

Quiz 5-1: Polynomial Operations & Factoring Polynomials

Write each polynomial in standard form. Then classify it by degree and number of terms.

1. $-2x^2 + x^4 + 3$

a. Standard form: $x^4 - 2x^2 + 3$ b. Classify by degree: quarticc. Classify by terms: trinomial

2. $-7 + 5m^3 + 9m - 3m^2$

a. Standard form: $5m^3 - 3m^2 + 9m - 7$ b. Classify by degree: cubicc. Classify by terms: polynomial

Simplify. All answers should be in standard form and contain only positive exponents.

3. $(-6a^4b^3)^2 \cdot -2ab^5$

$$= 36a^8b^6 \cdot -2ab^5$$

4. $\frac{8x^{-3}y \cdot 5x^5y}{10x^6y^2}$

$$= \frac{40x^2y^2}{10x^6y^2} = 4x^{-4}$$

5. $(\underline{6x^2} + \underline{7x}) + (5 - \underline{2x^2} + \underline{9x})$

$$= 4x^2 + 16x + 5$$

6. $(\underline{4h^3} + \underline{6h^2} + 1) - (\underline{5h^2} - \underline{8h^4} + 6)$

$$= 8h^4 + 4h^3 + h^2 - 5$$

7. $(x+2)(2x^2 - x - 9)$

$$= 2x^3 - x^2 - 9x + 4x^2 - 2x - 18$$

8. $3x(x-7)(x-1)^2$

$$= (3x^2 - 21x)(x^2 - 2x + 1) \\ = 3x^4 - 6x^3 + 3x^2 - 21x^3 + 42x^2 - 21x$$

3. $-72a^9b^{11}$

4. $\frac{4}{x^4}$

5. $4x^2 + 16x + 5$

6. $8h^4 + 4h^3 + h^2 - 5$

7. $2x^3 + 3x^2 - 11x - 18$

8. $3x^4 - 27x^3 + 45x^2 - 21x$

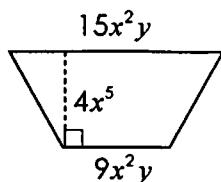
9. $3c^8d^2 + 8c^4d - 1$

10. $48x^7y$

9. $\frac{21c^{10}d^3 + 56c^6d^2 - 7c^2d}{7c^2d}$

$$3c^8d^2 + 8c^4d - 1$$

10. Write an expression in simplified form to represent the area of the trapezoid.



$$A = \frac{1}{2} (4x^5) (15x^2y + 9x^2y) \\ = 2x^5 (24x^2y) \\ = 48x^7y$$

Factor each polynomial completely.

$$11. 27p^3 - 1$$

$$= (3p-1)(9p^2+3p+1)$$

$$12. 10x^5y - 40xy^3$$

$$= 10xy(x^4 - 4y^2)$$

$$= 10xy(x^2+2y)(x^2-2y)$$

$$11. \underline{(3p-1)(9p^2+3p+1)}$$

$$12. \underline{10xy(x^2+2y)(x^2-2y)}$$

$$13. \underline{ab(a^2+8)(a^2-8)}$$

$$14. \underline{2(5m+3)(25m^2-15m+9)}$$

$$15. \underline{(r+7)(r+9)}$$

$$16. \underline{(k^2-13)(k^2+3)}$$

$$17. \underline{(2x+5)(3x-4)}$$

$$18. \underline{(2w^2+5)(2w^2+3)}$$

$$19. \underline{(w^2+9)(3w-2)}$$

$$20. \underline{(n+1)(n-1)(2n+7)}$$

$$13. a^5b - 64ab$$

$$= ab(a^4 - 64)$$

$$= ab(a^2+8)(a^2-8)$$

$$14. 250m^3 + 54$$

$$= 2(125m^3 + 27)$$

$$= 2(5m+3)(25m^2-15m+9)$$

$$15. r^2 + 16r + 63$$

$$= (r+9)(r+7)$$

$$16. k^4 - 10k^2 - 39$$

$$= (k^2-13)(k^2+3)$$

$$17. \overbrace{6x^2+7x-20}$$

$$= x^2+7x-120$$

$$= (x+\frac{15}{6})(x-\frac{8}{6})$$

$$= (x+\frac{5}{2})(x-\frac{4}{3})$$

$$= (2x+5)(3x-4)$$

$$18. \overbrace{4w^4+16w^2+15}$$

$$= w^4+16w^2+60$$

$$= (w^2+\frac{10}{4})(w^2+\frac{6}{4})$$

$$= (w^2+\frac{5}{2})(w^2+\frac{3}{2})$$

$$= (2w^2+5)(2w^2+3)$$

$$19. 3w^3 - 2w^2 + 27w - 18$$

$$= w^2(3w-2) + 9(3w-2)$$

$$= (w^2+9)(3w-2)$$

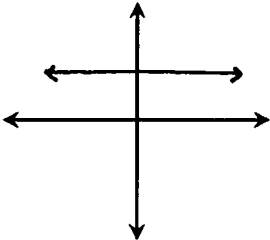
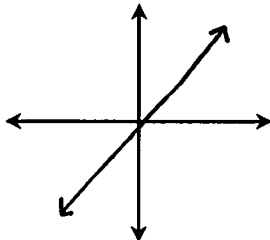
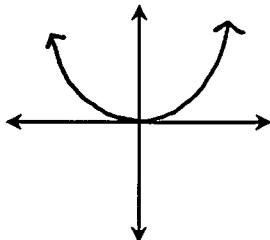
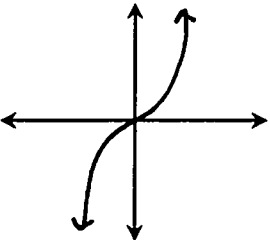
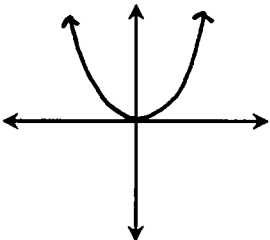
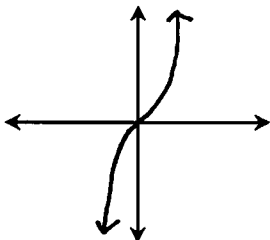
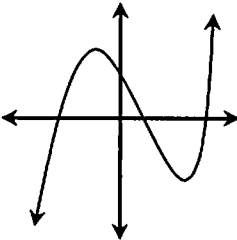
$$20. 2n^3 + 7n^2 - 2n - 7$$

$$= n^2(2n+7) - 1(2n+7)$$

$$= (n^2-1)(2n+7)$$

$$= (n+1)(n-1)(2n+7)$$

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples		
POLYNOMIAL FUNCTIONS	A polynomial function is a function of the form $f(x) = a_n x^n + a_{n-1} x^{n-1} + a_{n-2} x^{n-2} + \dots + a_2 x^2 + a_1 x + a_0$ where all coefficients are real numbers, all exponents are whole numbers, $a_n \neq 0$, and n is a positive integer. The degree of a polynomial function is the value of the greatest exponent.		
Polynomial PARENT FUNCTIONS	CONSTANT (degree = 0) $f(x) = c$ 	LINEAR (degree = 1) $f(x) = x$ 	QUADRATIC (degree = 2) $f(x) = x^2$ 
	CUBIC (degree = 3) $f(x) = x^3$ 	QUARTIC (degree = 4) $f(x) = x^4$ 	QUINTIC (degree = 5) $f(x) = x^5$ 
	TURNING POINTS  - The point(s) at which a polynomial function switches direction. - If the turning point is higher than any nearby point, it's called a <u>relative maximum</u>. - If the turning point is lower than any nearby point, it's called a <u>relative minimum</u>. - The maximum and minimum values are called <u>extrema</u>. - Turning points define where a function is increasing or decreasing.		

HOW TO FIND

Turning Points

Find the turning points of the function: $f(x) = x^3 - 6x^2 + 7x + 2$

Step 1: Graph the function. (Enter the function in $y =$, then hit **GRAPH**)

Step 2: Use the **CALC** menu to find the minimum and maximum values.

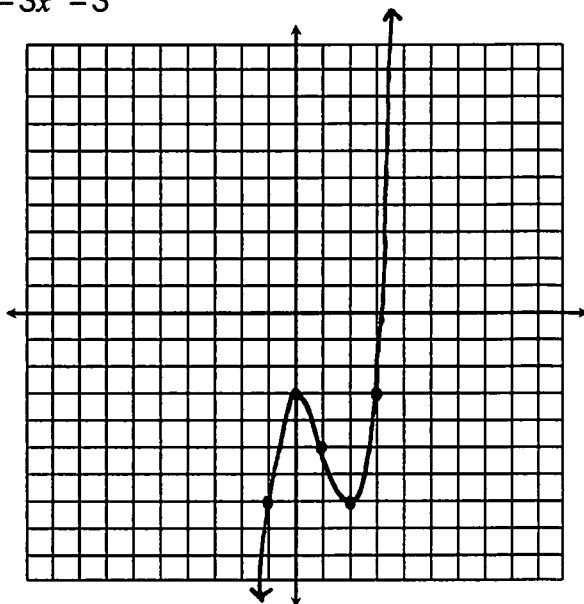
Step 3: Move the cursor to the left bound of the turning point.
Hit **ENTER**, then move the cursor to the right bound of the turning point. Hit **ENTER** twice.

Rel. Maximum(s): $(0.71, 4.30)$ Rel. Minimum(s): $(3.29, -4.30)$

Inc. Intervals: $(-\infty, 0.71), (3.29, \infty)$ Dec. Intervals: $(0.71, 3.29)$

Directions: Graph each polynomial function using a table. Then give the domain, range, turning points, increasing intervals, decreasing intervals, and end behavior.

1. $f(x) = x^3 - 3x^2 - 3$



Domain: $\{\mathbb{R}\}$

Range: $\{\mathbb{R}\}$

Rel. Maximum(s): $(0, -3)$

Rel. Minimum(s): $(2, -7)$

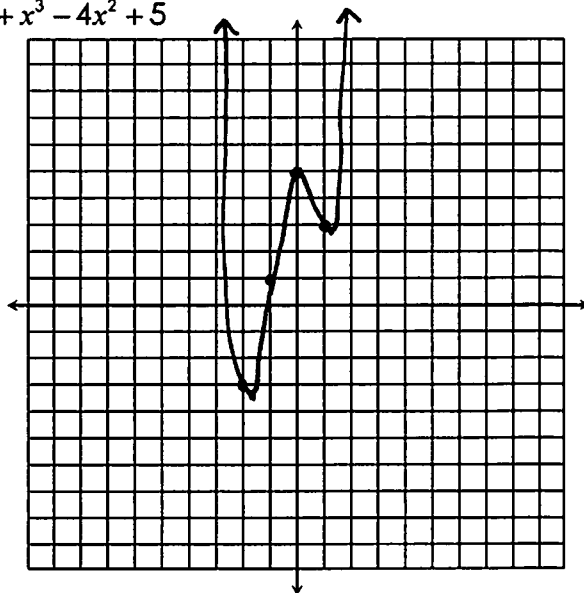
Inc. Intervals: $(-\infty, 0), (2, \infty)$

Dec. Intervals: $(0, 2)$

End Behavior: As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$

As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

2. $f(x) = x^4 + x^3 - 4x^2 + 5$



Domain: $\{\mathbb{R}\}$

Range: $\{y \mid y \geq -3.31\}$

Rel. Maximum(s): $(0, 5)$

Rel. Minimum(s): $(-1.84, -3.31), (1.09, 2.95)$

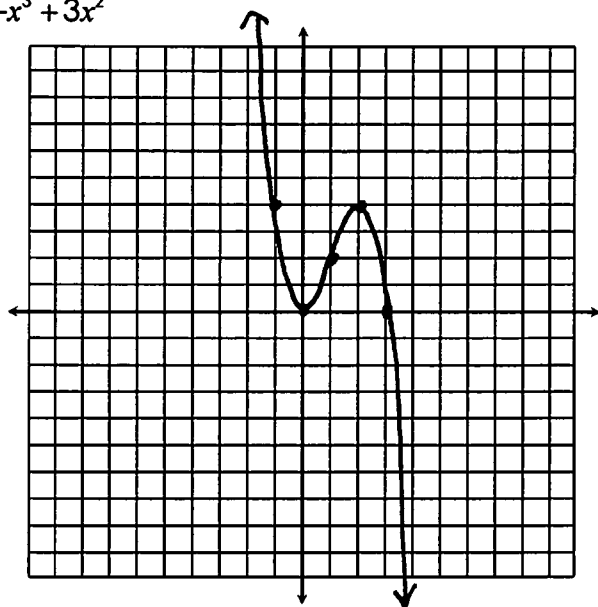
Inc. Intervals: $(-1.84, 0), (1.09, \infty)$

Dec. Intervals: $(-\infty, -1.84), (0, 1.09)$

End Behavior: As $x \rightarrow -\infty$, $f(x) \rightarrow \infty$

As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

3. $f(x) = -x^3 + 3x^2$



Domain: $\{\mathbb{R}\}$

Range: $\{\mathbb{R}\}$

Rel. Maximum(s): $(2, 4)$

Rel. Minimum(s): $(0, 0)$

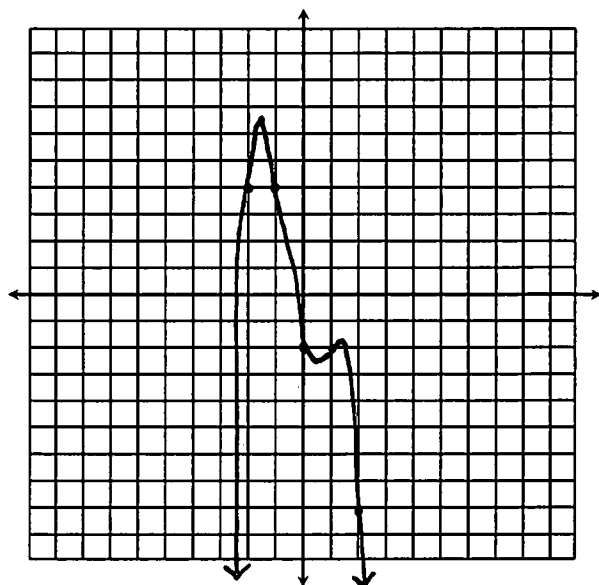
Inc. Intervals: $(0, 2)$

Dec. Intervals: $(-\infty, 0), (2, \infty)$

End Behavior: As $x \rightarrow -\infty$, $f(x) \rightarrow \infty$

As $x \rightarrow \infty$, $f(x) \rightarrow -\infty$

4. $f(x) = -x^4 + 4x^2 - 3x - 2$



Domain: $\{\mathbb{R}\}$

Range: $\{y \mid y \leq 6.5\}$

Rel. Maximum(s): $(-1.57, 6.50), (1.16, -1.91)$

Rel. Minimum(s): $(0.41, -2.59)$

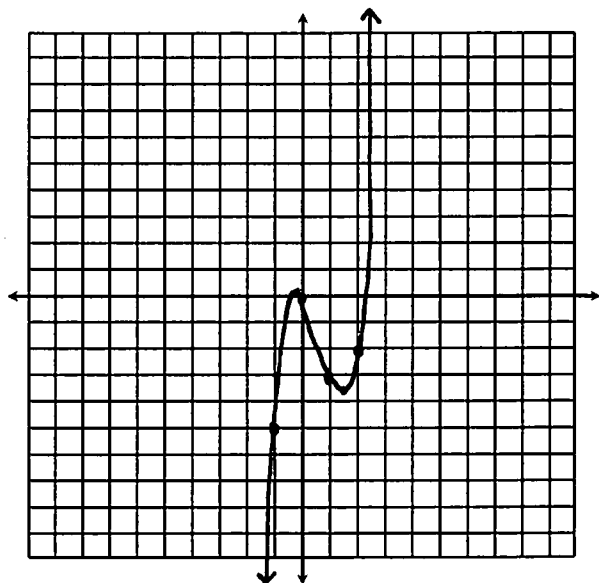
Inc. Intervals: $(-\infty, -1.57), (0.41, 1.16)$

Dec. Intervals: $(-1.57, 0.41), (1.16, \infty)$

End Behavior: As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$

As $x \rightarrow \infty$, $f(x) \rightarrow -\infty$

5. $f(x) = 2x^3 - 4x^2 - x$



Domain: $\{\mathbb{R}\}$

Range: $\{\mathbb{R}\}$

Rel. Maximum(s): $(-0.12, 0.06)$

Rel. Minimum(s): $(1.45, -3.76)$

Inc. Intervals: $(-\infty, -0.12), (1.45, \infty)$

Dec. Intervals: $(-0.12, 1.45)$

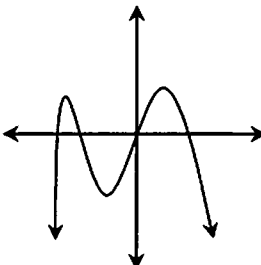
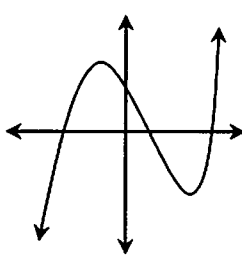
End Behavior: As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$

As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

Name:	Date:
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Topic:	Class:
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Main Ideas/Questions	Notes/Examples															
END BEHAVIOR Patterns	Using your graphing calculator, determine the end behavior of the following functions.															
	1. $f(x) = 3x^2 + 6x - 1$ As $x \rightarrow -\infty$, $f(x) \rightarrow \infty$ As $x \rightarrow \infty$, $f(x) \rightarrow \infty$		2. $f(x) = -x^2 + x$ As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$ As $x \rightarrow \infty$, $f(x) \rightarrow -\infty$													
	3. $f(x) = 2x^3 - 8x + 2$ As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$ As $x \rightarrow \infty$, $f(x) \rightarrow \infty$		4. $f(x) = -x^3 + 5x + 1$ As $x \rightarrow -\infty$, $f(x) \rightarrow \infty$ As $x \rightarrow \infty$, $f(x) \rightarrow -\infty$													
	5. $f(x) = x^4 + 2x^3 - x - 9$ As $x \rightarrow -\infty$, $f(x) \rightarrow \infty$ As $x \rightarrow \infty$, $f(x) \rightarrow \infty$		6. $f(x) = -x^4 + 3x^3 - x^2$ As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$ As $x \rightarrow \infty$, $f(x) \rightarrow -\infty$													
	When written in standard form, the leading coefficient and degree of the polynomial function determines the end behavior of the function.															
	<table><tr><th colspan="2" rowspan="2"></th><th colspan="2">LEADING COEFFICIENT</th></tr><tr><th>POSITIVE</th><th>NEGATIVE</th></tr><tr><th rowspan="2">DEGREE</th><th>EVEN</th><td>↑ ↑</td><td>↘ ↘</td></tr><tr><th>ODD</th><td>↘ ↑</td><td>↑ ↘</td></tr></table>						LEADING COEFFICIENT		POSITIVE	NEGATIVE	DEGREE	EVEN	↑ ↑	↘ ↘	ODD	↘ ↑
		LEADING COEFFICIENT														
		POSITIVE	NEGATIVE													
DEGREE	EVEN	↑ ↑	↘ ↘													
	ODD	↘ ↑	↑ ↘													
Draw sketches to help remember the end behavior patterns! 																
EXAMPLES	Identify the degree as even or odd, the leading coefficient as positive or negative, then sketch the end behavior with arrows.															
	FUNCTION	DEGREE	LEADING COEFFICIENT	END BEHAVIOR												
	7. $f(x) = -5x + 1$	odd	negative	↑ ↓												
	8. $f(x) = 12x^2 - 11x + 31$	even	positive	↑ ↑												
	9. $f(x) = -x^3 - x^2 + 8x + 1$	odd	negative	↑ ↓												

	10. $f(x) = -3x^4 + x^2 - 18$	even	negative	↓	↓
	11. $f(x) = 9x^5 + 2x^4 - x^3 + 9x^2 - 2$	odd	positive	↓	↑
	12. $f(x) = 5x^4 + 2x^3 - 17x + 1$	even	positive	↑	↑
	13. $f(x) = -x^7 - 5x^4 + 2x^2$	odd	negative	↑	↓
	14. $f(x) = -3x^2 - 14x + 28$	even	negative	↓	↓
Determine the sign of the leading coefficient and whether the function has an even or odd degree.					
	15.			16.	
	Negative ; Even degree		Positive ; odd degree		
IDENTIFYING FUNCTIONS	Without graphing, check all functions with the described end behavior.				
	17. As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$ As $x \rightarrow \infty$, $f(x) \rightarrow -\infty$ ☐ $f(x) = -x^3 + 10x^2 - 6x + 8$ ☐ $f(x) = 3x^2 - 17x - 11$ ☒ $f(x) = -4x^4 - x^3 + 8x^2 - 2x$ ☐ $f(x) = 2x^5 + 8x^4 - 10x^3$ ☐ $f(x) = -x^7 - 3x^2 + 9x - 14$ ☒ $f(x) = -x^2 + 8x - 15$		18. As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$ As $x \rightarrow \infty$, $f(x) \rightarrow \infty$ ☐ $f(x) = -x^2 - 14x + 32$ ☒ $f(x) = x^3 + x^2 - 13x$ ☒ $f(x) = 4x^7 - 8x^4 + 10x^2 + 2$ ☐ $f(x) = -x^5 + 5x^2 - 13x - 4$ ☐ $f(x) = -3x^4 + 2x^3 - x^2 + x$ ☐ $f(x) = -x^3 - 12x^2 + 9x - 18$		
	19. As $x \rightarrow -\infty$, $f(x) \rightarrow \infty$ As $x \rightarrow \infty$, $f(x) \rightarrow -\infty$ ☒ $f(x) = -2x^3 + 10x^2 - 7x$ ☐ $f(x) = 5x^2 + x - 14$ ☐ $f(x) = -x^6 + 4x^2 - 11x - 17$ ☒ $f(x) = -x^5 - 4x^4 - x^3 + 2x - 1$ ☐ $f(x) = -2x^4 - 7x^3 + x^2 - 6x$ ☒ $f(x) = -x^3 - 12x + 20$		20. As $x \rightarrow -\infty$, $f(x) \rightarrow \infty$ As $x \rightarrow \infty$, $f(x) \rightarrow \infty$ ☐ $f(x) = 4x^3 - 8x^2 - 2$ ☒ $f(x) = 2x^6 + 5x^3 - 14x$ ☐ $f(x) = -x^4 + x^3 - 6x^2 + x$ ☒ $f(x) = x^2 - 9x + 36$ ☒ $f(x) = 3x^4 + 7x^3 - 16x^2 + 3$ ☐ $f(x) = -x^2 + 10x - 2$		

Name: _____

Unit 5: Polynomial Functions

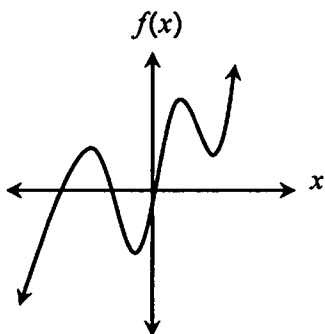
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Homework 3: Graphing Polynomial Functions

**** This is a 2-page document! ****

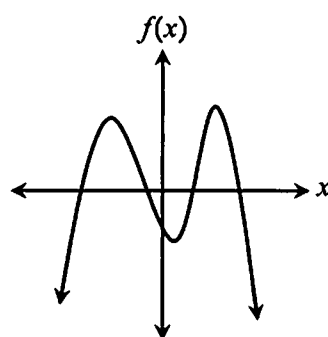
Directions: For each graph, (a) Describe the end behavior, (b) Determine whether it's the graph of an even or odd degree function, and (c) Determine the sign of the leading coefficient.

1.



- a) As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$
As $x \rightarrow \infty$, $f(x) \rightarrow \infty$
 b) Odd degree function
 c) positive

2.



- a) As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$
As $x \rightarrow \infty$, $f(x) \rightarrow -\infty$
 b) Even degree function
 c) Negative

3. Using proper notation, describe the end behavior of a 14th degree polynomial with a positive leading coefficient.

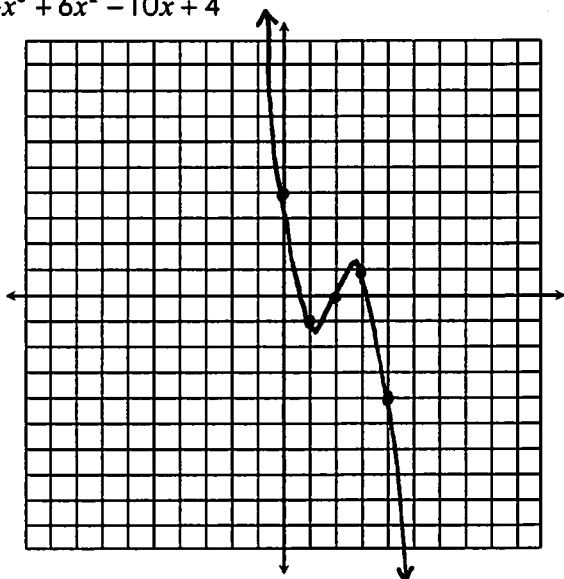
As $x \rightarrow -\infty$, $f(x) \rightarrow \infty$
As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

4. Using proper notation, describe the end behavior of a 9th degree polynomial with a negative leading coefficient.

As $x \rightarrow -\infty$, $f(x) \rightarrow \infty$
As $x \rightarrow \infty$, $f(x) \rightarrow -\infty$

Directions: Graph each function and give its key characteristics. Use a graphing calculator for the turning points and round to the nearest hundredth.

5. $f(x) = -x^3 + 6x^2 - 10x + 4$



Domain: $\{ \mathbb{R} \}$

Range: $\{ \mathbb{R} \}$

Rel. Maximum(s): (2.82, 1.09)

Rel. Minimum(s): (1.18, -1.09)

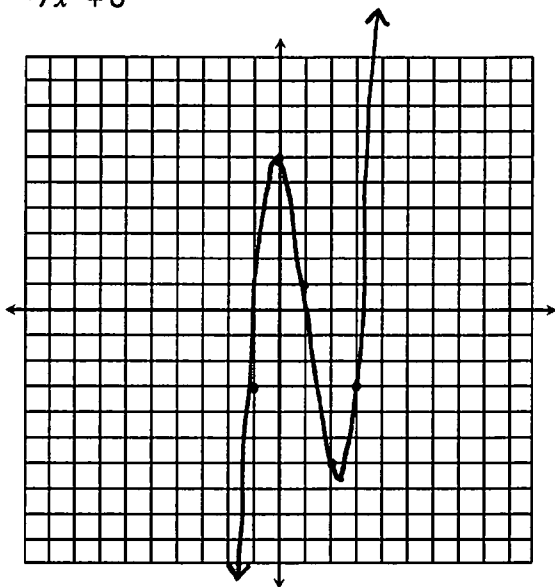
End Behavior: As $x \rightarrow -\infty$, $f(x) \rightarrow \infty$

As $x \rightarrow \infty$, $f(x) \rightarrow -\infty$

Inc. Intervals: (1.18, 2.82)

Dec. Intervals: $(-\infty, 1.18)$, $(2.82, \infty)$

6. $f(x) = 2x^3 - 7x^2 + 6$



Domain: $\mathbb{R}$

Range: $\mathbb{R}$

Rel. Maximum(s): $(0, 6)$

Rel. Minimum(s): $(2.33, -6.70)$

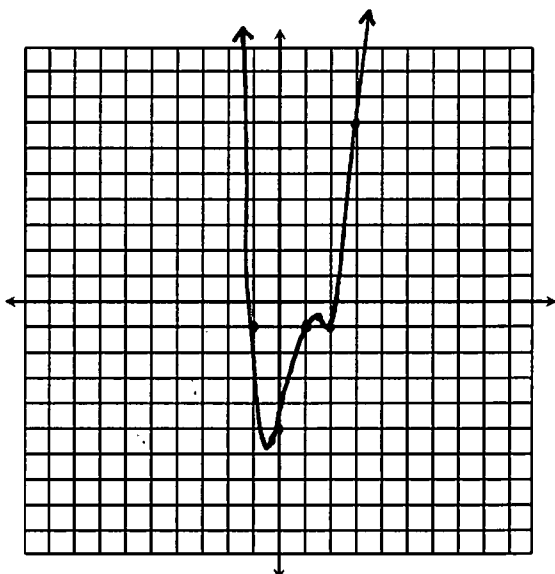
End Behavior: As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$

As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

Inc. Intervals: $(-\infty, 0), (2.33, \infty)$

Dec. Intervals: $(0, 2.33)$

7. $f(x) = x^4 - 4x^3 + 3x^2 + 4x - 5$



Domain: $\mathbb{R}$

Range: $\{y \mid y \geq -5.85\}$

Rel. Maximum(s): $(1.37, -0.65)$

Rel. Minimum(s): $(-0.37, -5.85), (2, -1)$

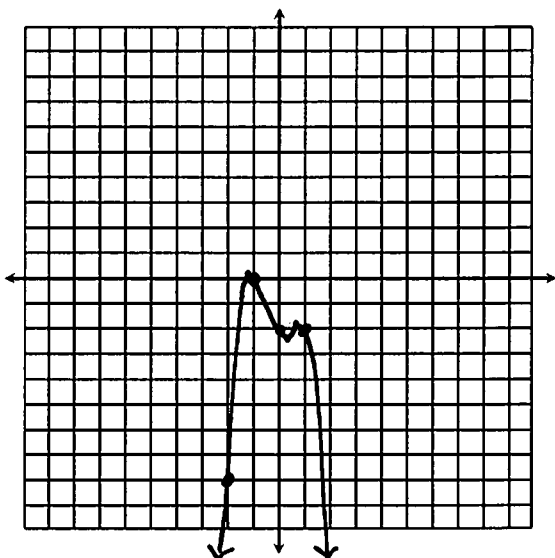
End Behavior: As $x \rightarrow -\infty$, $f(x) \rightarrow \infty$

As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

Inc. Intervals: $(-0.37, 1.37), (2, \infty)$

Dec. Intervals: $(-\infty, -0.37), (1.37, 2)$

8. $f(x) = -x^4 + 2x^2 - x - 2$



Domain: $\mathbb{R}$

Range: $\{y \mid y \leq 0.06\}$

Rel. Maximum(s): $(-1.11, 0.06), (0.84, -1.93)$

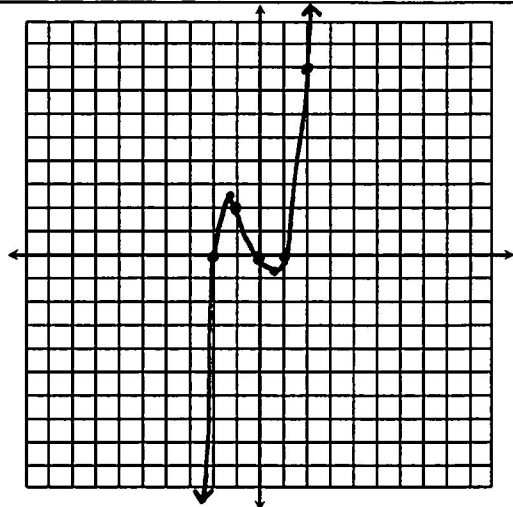
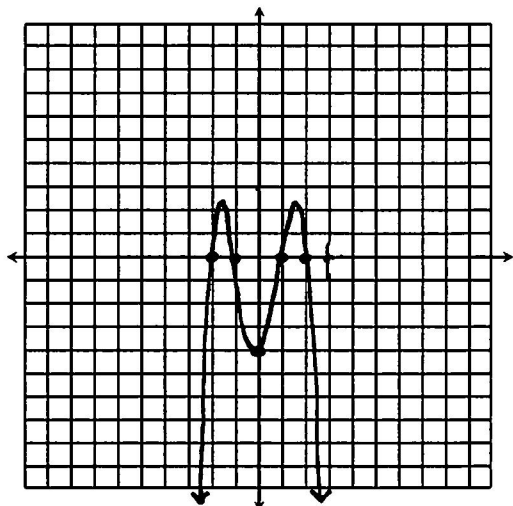
Rel. Minimum(s): $(0.27, -2.13)$

End Behavior: As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$

As $x \rightarrow \infty$, $f(x) \rightarrow -\infty$

Inc. Intervals: $(-\infty, -1.11), (0.27, 0.84)$

Dec. Intervals: $(-1.11, 0.27), (0.84, \infty)$

Name:		Date:	
Topic:		Class:	
Main Ideas/Questions		Notes/Examples	
ZEROS		- The point(s) at which the function intersects the <u>X - axis</u>. - A polynomial function of degree n can have at most <u>n</u> real zeros. - An even degree polynomial has an <u>even</u> number of zeros (or no zeros). - An odd degree polynomial has an <u>odd</u> number of zeros.	
FIND ZEROS <i>by Graphing</i>		Directions: Graph each function and identify its zeros. 1. $f(x) = x^3 + x^2 - 2x$ $(-2, 0)$ $(0, 0)$ $(1, 0)$ $x = \{-2, 0, 1\}$  2. $f(x) = -x^4 + 5x^2 - 4$ $(-2, 0)$ $(-1, 0)$ $(1, 0)$ $(2, 0)$ $x = \{-2, -1, 1, 2\}$ 	
LINEAR FACTORS		The linear factors of a polynomial reveal its zeros. Linear factors can be found by writing the polynomial in factored form. If $(x - a)$ is a factor of $f(x)$, then a is a zero of $f(x)$. 3. Write $f(x) = x^3 - 2x^2 - 15x$ in factored form and identify the zeros. $x(x^2 - 2x - 15)$ Factored Form: <u>$f(x) = x(x - 5)(x + 3)$</u> Zeros: <u>$x = \{-3, 0, 5\}$</u>	

MULTIPLICITY

The **multiplicity** of a zero is equal to the number of times its corresponding linear factor occurs.

4. Write $f(x) = x^3 - 6x^2 + 9x$ in factored form and identify the zeros and their multiplicity.

$$x(x^2 - 6x + 9)$$

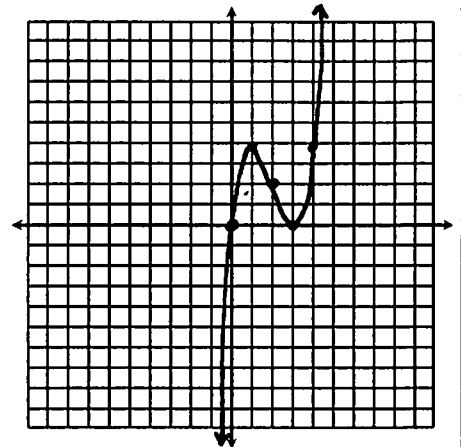
$$x(x-3)(x-3)$$

Factored Form: $f(x) = x(x-3)^2$

Zeros and Multiplicity: $x = 0$, mult. 1, $x = 3$, mult. 2

Effect of MULTIPLICITY ON A GRAPH

5. Graph $f(x) = x^3 - 6x^2 + 9x$



- If the **multiplicity** is **ODD**, then the graph crosses at that zero.
- If the **multiplicity** is **EVEN**, then the graph bounces at that zero.

EXAMPLES

Identify the zeros and state their multiplicities. Describe the effect on the graph.

6. $f(x) = (x+7)^2(2x+1)(x-4)^3$

Zero	Multiplicity	Effect
-7	2	bounces
$-\frac{1}{2}$	1	crosses
4	3	crosses

7. $f(x) = -2x^4 + 4x^3 + 16x^2$

$$-2x^2(x^2 - 2x - 8)$$

$$-2x^2(x-4)(x+2)$$

Zero	Multiplicity	Effect
0	2	bounces
4	1	crosses
-2	1	crosses

8. A polynomial function has a zero at -6 (multiplicity 2) and 2 (multiplicity 1). Write a polynomial function in standard form that could represent this function.

$$f(x) = (x+6)^2(x-2)$$

$$f(x) = (x+6)(x+6)(x-2)$$

$$f(x) = (x^2 + 12x + 36)(x-2)$$

$$f(x) = x^3 - 2x^2 + 12x^2 - 24x + 36x - 72$$

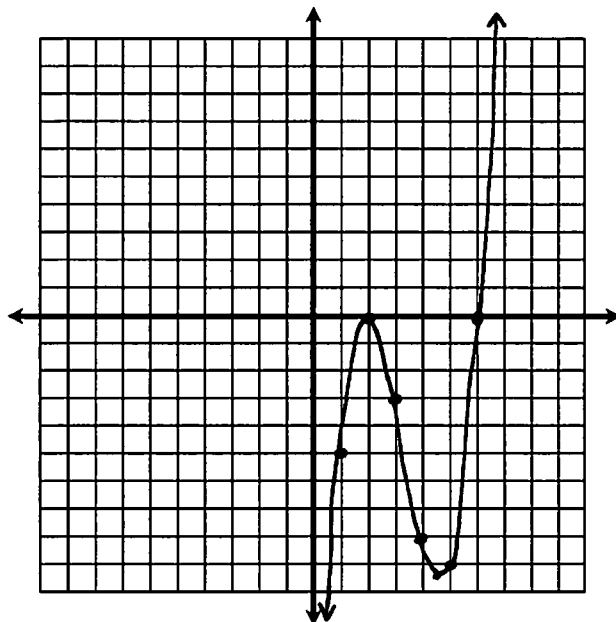
$$f(x) = x^3 + 10x^2 + 12x - 72$$

Analyzing Functions & Graphs

DIRECTIONS: Graph the function below. Then, write a written description of the graph that includes all key characteristics below.

CHARACTERISTICS: Type of polynomial function, domain, range, end behavior, turning points (including whether it's a relative maximum or relative minimum), zeros and their multiplicity, and the effect of the multiplicity on the graph.

$$f(x) = x^3 - 10x^2 + 28x - 24$$



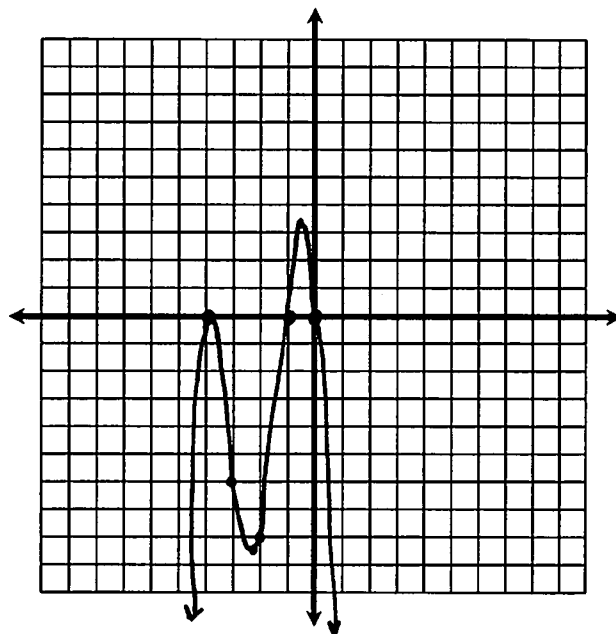
This polynomial has a degree of 3, making it a cubic function. Its domain and range are both all real numbers. The end behaviors follow the function in that as $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$ and as $x \rightarrow \infty$, $f(x) \rightarrow \infty$. There are two turning points: (2,0) is the relative maximum and (4.67, -9.48) is the relative minimum. One of the zeros is located at (2,0). This zero has a multiplicity of two, causing the graph to bounce at this point. The other zero is at (6,0). This zero has a multiplicity of one, causing the graph to cross at this point.

Analyzing Functions & Graphs

DIRECTIONS: Graph the function below. Then, write a written description of the graph that includes all key characteristics below.

CHARACTERISTICS: Type of polynomial function, domain, range, end behavior, turning points (including whether it's a relative maximum or relative minimum), zeros and their multiplicity, and the effect of the multiplicity on the graph.

$$f(x) = -x^4 - 9x^3 - 24x^2 - 16x$$



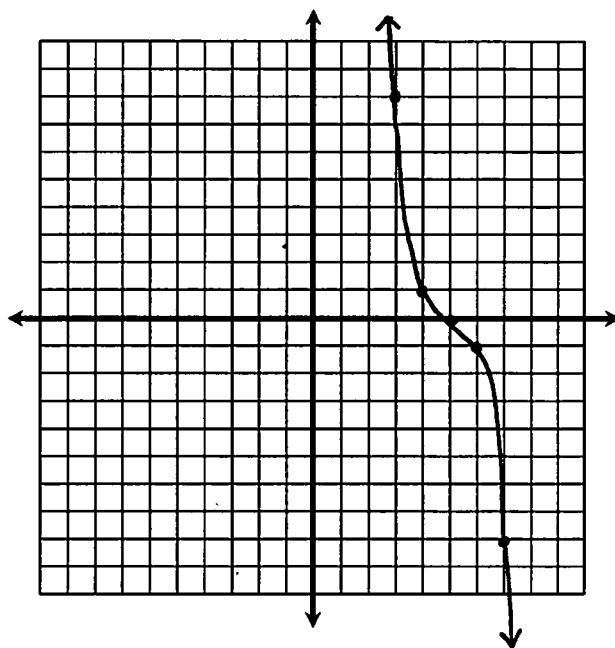
This polynomial has a degree of 4, making it a quartic function. Its domain is all real numbers and its range is $y \leq 3.12$. Since this is an even function, its end behaviors are the same; it approaches $-\infty$ on either side of the graph. There are three turning points: $(-2.32, -8.64)$ is the relative minimum, and $(-4, 0)$ and $(-0.43, 3.12)$ are the relative maximums. There are three zeros. One of the zeros is $(-4, 0)$ and has a multiplicity of 2 causing the graph to bounce at this point. The other zeros, $(-1, 0)$ and $(0, 0)$, each have a multiplicity of one which causes the graph to cross at that point.

Analyzing Functions & Graphs

DIRECTIONS: Graph the function below. Then, write a written description of the graph that includes all key characteristics below.

CHARACTERISTICS: Type of polynomial function, domain, range, end behavior, turning points (including whether it's a relative maximum or relative minimum), zeros and their multiplicity, and the effect of the multiplicity on the graph.

$$f(x) = -x^3 + 15x^2 - 75x + 125$$



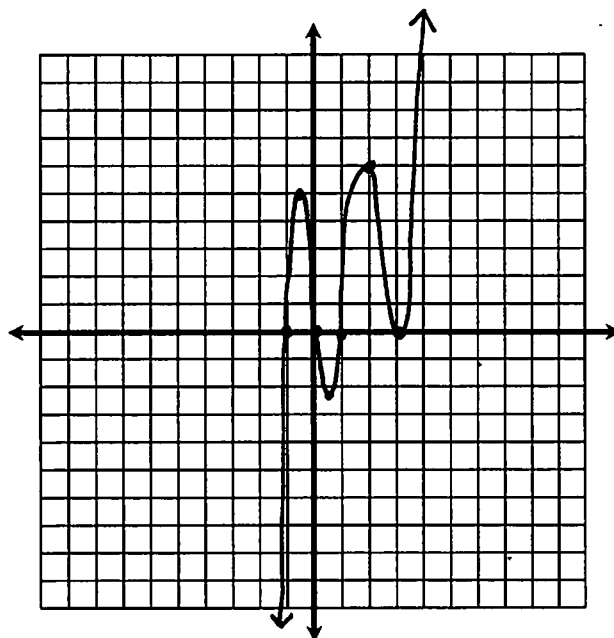
This polynomial is a cubic function, as its degree is three. The domain and range are both all real numbers. Its end behaviors are as follows: as $x \rightarrow -\infty$, $f(x) \rightarrow \infty$ and as $x \rightarrow \infty$, $f(x) \rightarrow -\infty$. There is one turning point at $(5, 0)$, which is also the only zero for the function. As a turning point, it does not serve as a relative minimum or a relative maximum. As a zero, this point has a multiplicity of three, causing it to cross the axis.

Analyzing Functions & Graphs

DIRECTIONS: Graph the function below. Then, write a written description of the graph that includes all key characteristics below.

CHARACTERISTICS: Type of polynomial function, domain, range, end behavior, turning points (including whether it's a relative maximum or relative minimum), zeros and their multiplicity, and the effect of the multiplicity on the graph.

$$f(x) = x^5 - 6x^4 + 8x^3 + 6x^2 - 9x$$



This polynomial of degree 5 is a quintic function. The domain and range are both all real numbers. Since the function is odd, the end behaviors are not the same; as $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$ and as $x \rightarrow \infty$, $f(x) \rightarrow \infty$. There are four turning points: $(-0.63, 5.61)$ and $(1.95, 6.03)$ are relative maximums and $(0.48, -2.35)$ and $(3, 0)$ are relative minimums. Three of the four zeros have a multiplicity of one. At $(-1, 0)$, $(0, 0)$, and $(1, 0)$, the graph crosses the x-axis. The last zero of $(3, 0)$ has a multiplicity of 2, which can be seen when the graph bounces at that point.

CHARACTERISTICS of a POLYNOMIAL FUNCTION

$$f(x) = x^3 + 6x^2 + 8x$$

TURNING POINTS

$(-3.15, 3.08)$ -
Relative Maximum

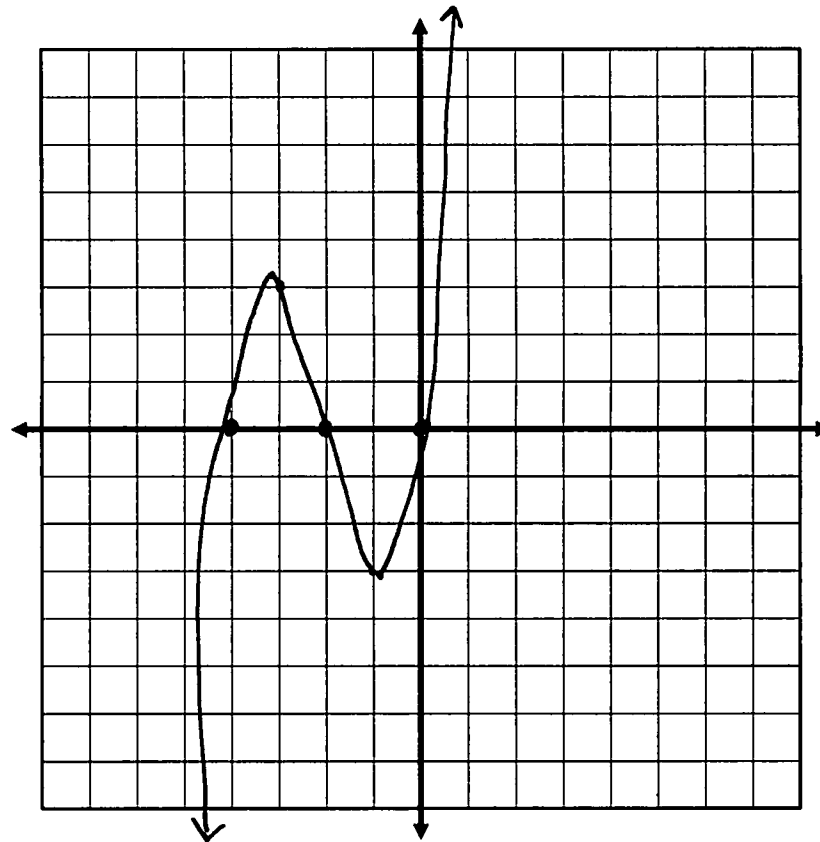
$(-0.85, -3.08)$ -
Relative Minimum

INCREASING INTERVALS

$(-\infty, -3.15)$,
 $(-0.85, \infty)$

DECREASING INTERVALS

$(-3.15, -0.85)$



PARENT FUNCTION

$$f(x) = x^3$$

END BEHAVIOR

As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$

As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

ZEROS

$$x = \{-4, -2, 0\}$$

DOMAIN

$\mathbb{R}$

RANGE

$\mathbb{R}$

CHARACTERISTICS of a POLYNOMIAL FUNCTION

$$f(x) = x^4 - 3x^2 - 2x$$

TURNING POINTS

$(-1, 0)$ and $(1.37, -4.85)$ -
Relative Minimum

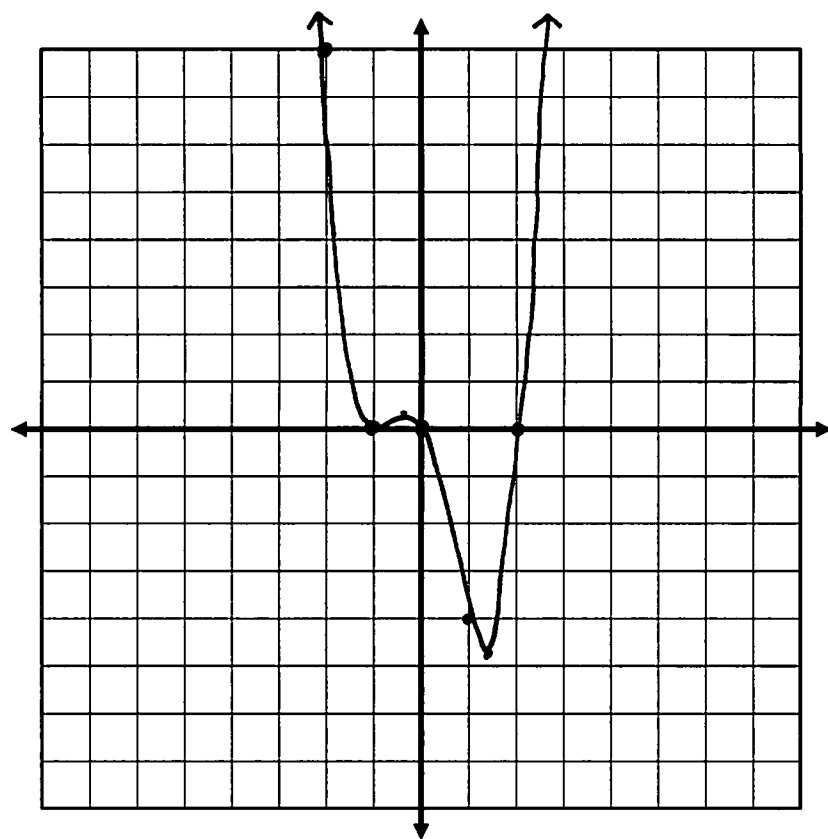
$(-0.37, 0.35)$ -
Relative Maximum

INCREASING INTERVALS

$(-1, -0.37)$,
 $(1.37, \infty)$

DECREASING INTERVALS

$(-\infty, -1)$,
 $(-0.37, 1.37)$



PARENT FUNCTION

$$f(x) = x^4$$

END BEHAVIOR

As $x \rightarrow -\infty$, $f(x) \rightarrow \infty$

As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

ZEROS

$$x = \{-1, 0, 2\}$$

DOMAIN

$$\mathbb{R}$$

RANGE

$$\{y \mid y \geq -4.85\}$$

Name: _____

Unit 5: Polynomial Functions

Date: _____ Per: _____

Homework 4: Zeros and Multiplicity

**** This is a 2-page document! ******Directions:** Identify the zeros, their multiplicity, and the effect of the multiplicity on the graph.

1. $f(x) = x^2(x-1)^4(x+5)$

Zero	Multiplicity	Effect
0	2	bounces
1	4	bounces
-5	1	crosses

2. $f(x) = -2x(3x+1)^2(x+7)^7$

Zero	Multiplicity	Effect
0	1	crosses
$-\frac{1}{3}$	2	bounces
-7	7	crosses

3. $f(x) = x^3 - 12x^2 + 35x$

$x(x^2 - 12x + 35)$

$x(x-7)(x-5)$

Zero	Multiplicity	Effect
0	1	crosses
7	1	crosses
5	1	crosses

4. $f(x) = 3x^3 - 21x^2 - 54x$

$3x(x^2 - 7x - 18)$

$3x(x-9)(x+2)$

Zero	Multiplicity	Effect
0	1	crosses
9	1	crosses
-2	1	crosses

5. $f(x) = x^4 - 16x^3 + 64x^2$

$x^2(x^2 - 16x + 64)$

$x^2(x-8)(x-8)$

$x^2(x-8)^2$

Zero	Multiplicity	Effect
0	2	bounces
8	2	bounces

6. $f(x) = -8x^3 - 20x^2$

$-4x^2(2x+5)$

Zero	Multiplicity	Effect
0	2	bounces
$-\frac{5}{2}$	1	crosses

7. A polynomial function has zeros at -1, 2, and 7 (all multiplicity 1). Write a function in standard form that could represent this function.

$f(x) = (x+1)(x-2)(x-7)$

$f(x) = (x+1)(x^2 - 9x + 14)$

$f(x) = x^3 - 9x^2 + 14x + x^2 - 9x + 14$

$f(x) = x^3 - 8x^2 + 5x + 14$

8. A polynomial function has zeros at $\frac{5}{2}$ (multiplicity 2), 3 (multiplicity 1), and 0 (multiplicity 4). Write a function in standard form that could represent this function.

$$f(x) = (x - \frac{5}{2})^2 (x - 3) x^4$$

$$f(x) = x^4 (x - 3) (2x - 5)^2$$

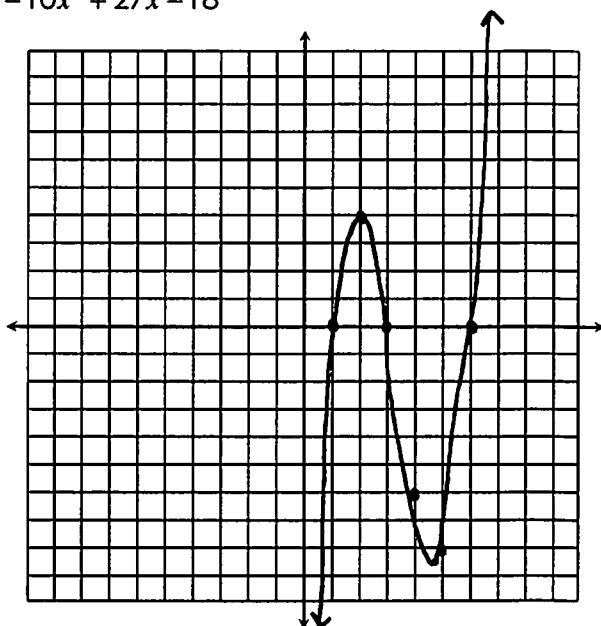
$$f(x) = (x^5 - 3x^4) (4x^2 - 20x + 25)$$

$$f(x) = 4x^7 - 20x^6 + 25x^5 - 12x^6 + 60x^5 - 75x^4$$

$$f(x) = 4x^7 - 32x^6 + 85x^5 - 75x^4$$

Directions: Graph each function and identify its key characteristics.

9. $f(x) = x^3 - 10x^2 + 27x - 18$



Domain: $\{\mathbb{R}\}$

Range: $\{\mathbb{R}\}$

Rel. Maximum(s): $(1.88, 4.06)$

Rel. Minimum(s): $(4.79, -8.21)$

End Behavior: As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$

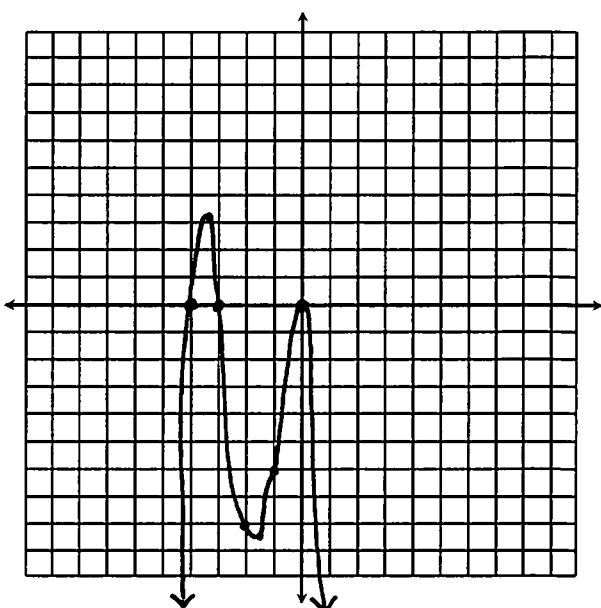
As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

Inc. Intervals: $(-\infty, 1.88)$, $(4.79, \infty)$

Dec. Intervals: $(1.88, 4.79)$

Zeros: $x = \{1, 3, 6\}$

10. $f(x) = -x^4 - 7x^3 - 12x^2$



Domain: $\{\mathbb{R}\}$

Range: $\{y \mid y \leq 3.12\}$

Rel. Maximum(s): $(-3.57, 3.12)$, $(0, 0)$

Rel. Minimum(s): $(-1.68, -8.64)$

End Behavior: As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$

As $x \rightarrow \infty$, $f(x) \rightarrow -\infty$

Inc. Intervals: $(-\infty, -3.57)$, $(-1.68, 0)$

Dec. Intervals: $(-3.57, -1.68)$, $(0, \infty)$

Zeros: $x = \{-4, -3, 0\}$

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples
SOLVING POLYNOMIAL EQUATIONS By Factoring	① MOVE ALL TERMS to one side and set the equation EQUAL TO 0.
	② FACTOR the polynomial completely!
	③ SET EACH FACTOR EQUAL TO 0 and SOLVE for each x-value.
	➤ For quadratic equations, solve by square roots, completing the square, or the quadratic formula. ➤ SIMPLIFY all irrational and complex solutions!

Directions: Solve each equation by factoring.

1. $x^3 + x = 0$

$$x(x^2 + 1) = 0$$

$x = 0$	$x^2 + 1 = 0$ $x^2 = -1$ $x = \pm \sqrt{-1}$ $x = \pm i$
---------	---

$x = \{ \pm i, 0 \}$

2. $4x^3 - 7x^2 = 3x^2$

$$4x^3 - 10x^2 = 0$$

$$2x^2(2x - 5) = 0$$

$2x^2 = 0$ $x^2 = 0$ $x = \pm \sqrt{0}$ $x = 0$	$2x - 5 = 0$ $x = \frac{5}{2}$
--	-----------------------------------

$x = \{ 0, \frac{5}{2} \}$

3. $x^4 - 64 = 0$

$$(x^2 + 8)(x^2 - 8) = 0$$

$x^2 + 8 = 0$ $x^2 = -8$ $x = \pm \sqrt{-8}$ $x = \pm 2i\sqrt{2}$	$x^2 - 8 = 0$ $x^2 = 8$ $x = \pm \sqrt{8}$ $x = \pm 2\sqrt{2}$
--	---

$x = \{ \pm 2\sqrt{2}, \pm 2i\sqrt{2} \}$

4. $5x^5 - 80x = 0$

$$5x(x^4 - 16) = 0$$

$$5x(x^2 + 4)(x^2 - 4) = 0$$

$$5x(x^2 + 4)(x + 2)(x - 2) = 0$$

$5x = 0$ $x = 0$	$x^2 + 4 = 0$ $x^2 = -4$ $x = \pm \sqrt{-4}$ $x = \pm 2i$	$x + 2 = 0$ $x = -2$	$x - 2 = 0$ $x = 2$
---------------------	--	-------------------------	------------------------

$x = \{ 0, \pm 2i, \pm 2 \}$

$$5. x^3 + 125 = 0$$

$$(x+5)(x^2 - 5x + 25) = 0$$

$$x+5=0$$

$$x = -5$$

$$x = \frac{5 \pm \sqrt{(-5)^2 - 4(1)(25)}}{2(1)}$$

$$x = \frac{5 \pm \sqrt{25 - 100}}{2}$$

$$x = \frac{5 \pm \sqrt{-75}}{2}$$

$$x = \frac{5 \pm 5i\sqrt{3}}{2}$$

$$x = \left\{ -5, \frac{5 \pm 5i\sqrt{3}}{2} \right\}$$

$$6. 8x^5 + 5x^2 = 4x^2$$

$$8x^5 + x^2 = 0$$

$$x^2(8x^3 + 1) = 0$$

$$x^2(2x+1)(4x^2 - 2x + 1) = 0$$

$$x^2 = 0$$

$$x = \pm \sqrt{0}$$

$$x = 0$$

$$2x+1=0$$

$$x = -\frac{1}{2}$$

$$x = \frac{2 \pm \sqrt{(-2)^2 - 4(4)(1)}}{2(4)}$$

$$x = \frac{2 \pm \sqrt{4 - 16}}{8}$$

$$x = \frac{2 \pm \sqrt{-12}}{8}$$

$$x = \frac{2 \pm 2i\sqrt{3}}{8}$$

$$x = \frac{1 \pm i\sqrt{3}}{4}$$

$$x = \left\{ 0, -\frac{1}{2}, \frac{1 \pm i\sqrt{3}}{4} \right\}$$

$$7. x^3 - 216 = 0$$

$$(x-6)(x^2 + 6x + 36) = 0$$

$$x-6=0$$

$$x = 6$$

$$x^2 + 6x = -36$$

$$x^2 + 6x + 9 = -36 + 9$$

$$(x+3)^2 = -27$$

$$x+3 = \pm \sqrt{-27}$$

$$x+3 = \pm 3i\sqrt{3}$$

$$x = -3 \pm 3i\sqrt{3}$$

$$x = \{6, -3 \pm 3i\sqrt{3}\}$$

$$8. 16x^3 = 54$$

$$16x^3 - 54 = 0$$

$$2(8x^3 - 27) = 0$$

$$2(2x-3)(4x^2 + 6x + 9) = 0$$

$$2 \neq 0$$

$$2x-3=0$$

$$x = \frac{3}{2}$$

$$x = \frac{-6 \pm \sqrt{6^2 - 4(4)(9)}}{2(4)}$$

$$x = \frac{-6 \pm \sqrt{36 - 144}}{8}$$

$$x = \frac{-6 \pm \sqrt{-108}}{8}$$

$$x = \frac{-6 \pm 6i\sqrt{3}}{8}$$

$$x = \frac{-3 \pm 3i\sqrt{3}}{4}$$

$$x = \left\{ \frac{3}{2}, \frac{-3 \pm 3i\sqrt{3}}{4} \right\}$$

$$9. x^4 + x^2 - 42 = 0$$

$$(x^2 + 7)(x^2 - 6) = 0$$

$$x^2 + 7 = 0$$

$$x^2 = -7$$

$$x = \pm \sqrt{-7}$$

$$x = \pm i\sqrt{7}$$

$$x^2 - 6 = 0$$

$$x^2 = 6$$

$$x = \pm \sqrt{6}$$

$$x = \{ \pm i\sqrt{7}, \pm \sqrt{6} \}$$

$$10. x^4 + 3 = 13 - 9x^4$$

$$10x^4 - 10 = 0$$

$$10(x^4 - 1) = 0$$

$$10(x^2 + 1)(x^2 - 1) = 0$$

$$10(x^2 + 1)(x+1)(x-1) = 0$$

$$10 \neq 0$$

$$x^2 + 1 = 0$$

$$x^2 = -1$$

$$x = \pm \sqrt{-1}$$

$$x = \pm i$$

$$x+1=0$$

$$x = -1$$

$$x-1=0$$

$$x = 1$$

$$x = \{ \pm i, \pm 1 \}$$

$$11. x^3 - 8x^2 + 16x = 0$$

$$x(x^2 - 8x + 16) = 0$$

$$x(x-4)(x-4) = 0$$

$x=0$	$x-4=0$	$x-4=0$
	$x=4$	$x=4$

$$x = \{0, 4\}$$

$$12. 2x^6 - 10x^4 - 48x^2 = 0$$

$$2x^2(x^4 - 5x^2 - 24) = 0$$

$$2x^2(x^2 - 8)(x^2 + 3) = 0$$

$2x^2=0$	$x^2-8=0$	$x^2+3=0$
$x^2=0$	$x^2=8$	$x^2=-3$
$x=\pm\sqrt{0}$	$x=\pm\sqrt{8}$	$x=\pm\sqrt{-3}$
$x=0$	$x=\pm 2\sqrt{2}$	$x=\pm i\sqrt{3}$

$$x = \{0, \pm 2\sqrt{2}, \pm i\sqrt{3}\}$$

$$13. 3x^4 + 2x^2 = 8$$

$$3x^4 + 2x^2 - 8 = 0$$

$$x^4 + 2x^2 - 24 = 0$$

$$(x^2 + \frac{6}{3})(x^2 - \frac{4}{3}) = 0$$

$$(x^2 + 2)(3x^2 - 4) = 0$$

$x^2+2=0$	$3x^2-4=0$
$x^2=-2$	$x^2=\frac{4}{3}$
$x=\pm\sqrt{-2}$	$x=\pm\sqrt{\frac{4}{3}}$
$x=\pm i\sqrt{2}$	$x=\pm\frac{2}{\sqrt{3}}$
	$x=\pm\frac{2\sqrt{3}}{3}$

$$x = \{\pm i\sqrt{2}, \pm \frac{2\sqrt{3}}{3}\}$$

$$15. x^3 + 2x^2 + 5x + 10 = 0$$

$$x^2(x+2) + 5(x+2) = 0$$

$$(x^2 + 5)(x+2) = 0$$

$x^2+5=0$	$x+2=0$
$x^2=-5$	$x=-2$
$x=\pm\sqrt{-5}$	
$x=\pm i\sqrt{5}$	

$$x = \{-2, \pm i\sqrt{5}\}$$

$$14. 5x^3 - 3x^2 + 32x = 2x + 18$$

$$5x^3 - 3x^2 + 30x - 18 = 0$$

$$x^2(5x-3) + 6(5x-3) = 0$$

$$(x^2 + 6)(5x - 3) = 0$$

$x^2+6=0$	$5x-3=0$
$x^2=-6$	$x=\frac{3}{5}$
$x=\pm\sqrt{-6}$	
$x=\pm i\sqrt{6}$	

$$x = \{\frac{3}{5}, \pm i\sqrt{6}\}$$

$$16. 2x^3 - 3x^2 - 32x + 48 = 0$$

$$x^2(2x-3) - 16(2x-3) = 0$$

$$(x^2 - 16)(2x - 3) = 0$$

$$(x+4)(x-4)(2x-3) = 0$$

$x+4=0$	$x-4=0$	$2x-3=0$
$x=-4$	$x=4$	$x=\frac{3}{2}$

$$x = \{\frac{3}{2}, \pm 4\}$$

Name: _____

Unit 5: Polynomial Functions

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Homework 5: Solving Polynomial Equations

**** This is a 2-page document! ******Directions:** Solve each equation by **factoring**. Simplify all irrational and complex solutions.

1. $12x^3 - 3x^2 = 0$

$$3x^2(4x-1) = 0$$

$3x^2 = 0$	$4x - 1 = 0$
$x^2 = 0$	$x = 1/4$
$x = \pm\sqrt{0}$	
$x = 0$	

$$x = \{0, 1/4\}$$

2. $x^4 - 64 = 0$

$$(x^2 + 8)(x^2 - 8) = 0$$

$x^2 + 8 = 0$	$x^2 - 8 = 0$
$x^2 = -8$	$x^2 = 8$
$x = \pm\sqrt{-8}$	$x = \pm\sqrt{8}$
$x = \pm 2i\sqrt{2}$	$x = \pm 2\sqrt{2}$

$$x = \{\pm 2i\sqrt{2}, \pm 2\sqrt{2}\}$$

3. $2x^4 = 24x^2$

$$2x^4 - 24x^2 = 0$$

$$2x^2(x^2 - 12) = 0$$

$2x^2 = 0$	$x^2 - 12 = 0$
$x^2 = 0$	$x^2 = 12$
$x = \pm\sqrt{0}$	$x = \pm\sqrt{12}$
$x = 0$	$x = \pm 2\sqrt{3}$

$$x = \{0, \pm 2\sqrt{3}\}$$

4. $x^3 + 512 = 0$

$$(x + 8)(x^2 - 8x + 64) = 0$$

$x + 8 = 0$	$x^2 - 8x = -64$
$x = -8$	$x^2 - 8x + 16 = -64 + 16$
	$(x - 4)^2 = -48$
	$x - 4 = \pm\sqrt{-48}$
	$x = 4 \pm 4i\sqrt{3}$

$$x = \{-8, 4 \pm 4i\sqrt{3}\}$$

5. $8x^3 - 125 = 0$

$$(2x - 5)(4x^2 + 10x + 25) = 0$$

$2x - 5 = 0$	$x = \frac{-10 \pm \sqrt{10^2 - 4(4)(25)}}{2(4)}$
$x = \frac{5}{2}$	$x = \frac{-10 \pm \sqrt{100 - 400}}{8}$
	$x = \frac{-10 \pm \sqrt{-300}}{8}$
	$x = \frac{-10 \pm 10i\sqrt{3}}{8}$
	$x = \frac{-5 \pm 5i\sqrt{3}}{4}$

$$x = \left\{ \frac{5}{2}, \frac{-5 \pm 5i\sqrt{3}}{4} \right\}$$

6. $81x^4 - 3x = 0$

$$81x^4 - 3x = 0$$

$$3x(27x^3 - 1) = 0$$

$$3x(3x - 1)(9x^2 + 3x + 1) = 0$$

$3x = 0$	$3x - 1 = 0$	$x = \frac{-3 \pm \sqrt{3^2 - 4(9)(1)}}{2(9)}$
$x = 0$	$x = \frac{1}{3}$	$x = \frac{-3 \pm \sqrt{-27}}{18}$
		$x = \frac{-3 \pm 3i\sqrt{3}}{18}$
		$x = \frac{-1 \pm i\sqrt{3}}{6}$

$$x = \left\{ 0, \frac{1}{3}, \frac{-1 \pm i\sqrt{3}}{6} \right\}$$

$$7. 2x^3 - 16x^2 - 40x = 0$$

$$2x(x^2 - 8x - 20) = 0$$

$$2x(x-10)(x+2) = 0$$

$2x = 0$ $x = 0$	$x - 10 = 0$ $x = 10$	$x + 2 = 0$ $x = -2$
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$$X = \{-2, 0, 10\}$$

$$8. x^4 - 16x^2 = x^2 + 18$$

$$x^4 - 17x^2 - 18 = 0$$

$$(x^2 - 18)(x^2 + 1) = 0$$

$x^2 - 18 = 0$ $x^2 = 18$ $x = \pm\sqrt{18}$ $x = \pm 3\sqrt{2}$	$x^2 + 1 = 0$ $x^2 = -1$ $x = \pm\sqrt{-1}$ $x = \pm i$
---	--

$$X = \{\pm 3\sqrt{2}, \pm i\}$$

$$9. 4x^4 + 35x^2 - 9 = 0$$

$$x^4 + 35x^2 - 36 = 0$$

$$(x^2 + \frac{36}{4})(x^2 - \frac{1}{4}) = 0$$

$(x^2 + 9)(4x^2 - 1) = 0$	
$x^2 + 9 = 0$ $x^2 = -9$ $x = \pm\sqrt{-9}$ $x = \pm 3i$	$4x^2 - 1 = 0$ $x^2 = \frac{1}{4}$ $x = \pm\sqrt{\frac{1}{4}}$ $x = \pm \frac{1}{2}$

$$X = \{\pm 3i, \pm \frac{1}{2}\}$$

$$10. x^3 + 3x^2 = 24x + 72$$

$$x^3 + 3x^2 - 24x - 72 = 0$$

$$x^2(x+3) - 24(x+3) = 0$$

$$(x^2 - 24)(x+3) = 0$$

$x^2 - 24 = 0$ $x^2 = 24$ $x = \pm\sqrt{24}$ $x = \pm 2\sqrt{6}$	$x + 3 = 0$ $x = -3$
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$$X = \{-3, \pm 2\sqrt{6}\}$$

$$11. 2x^3 - 5x^2 + 40x - 100 = 0$$

$$x^2(2x-5) + 20(2x-5) = 0$$

$$(x^2 + 20)(2x-5) = 0$$

$x^2 + 20 = 0$ $x^2 = -20$ $x = \pm\sqrt{-20}$ $x = \pm 2i\sqrt{5}$	$2x - 5 = 0$ $x = \frac{5}{2}$
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$$X = \{\frac{5}{2}, \pm 2i\sqrt{5}\}$$

12. The population of a species is modeled by the equation $p(t) = -t^4 + 56t^2 + 240$, where t is the number of years. Find the approximate number of years until the species is extinct.

$$0 = -t^4 + 56t^2 + 240$$

$$0 = -1(t^4 - 56t^2 - 240)$$

$$0 = -1(t^2 + 4)(t^2 - 60)$$

$0 \neq -1$	$t^2 + 4 = 0$ $t^2 = -4$ $t = \pm 2i$	$t^2 - 60 = 0$ $t^2 = 60$ $t = \pm 2\sqrt{15}$
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$$2\sqrt{15} \approx 7.75 \text{ years}$$

Name: _____

Algebra 2

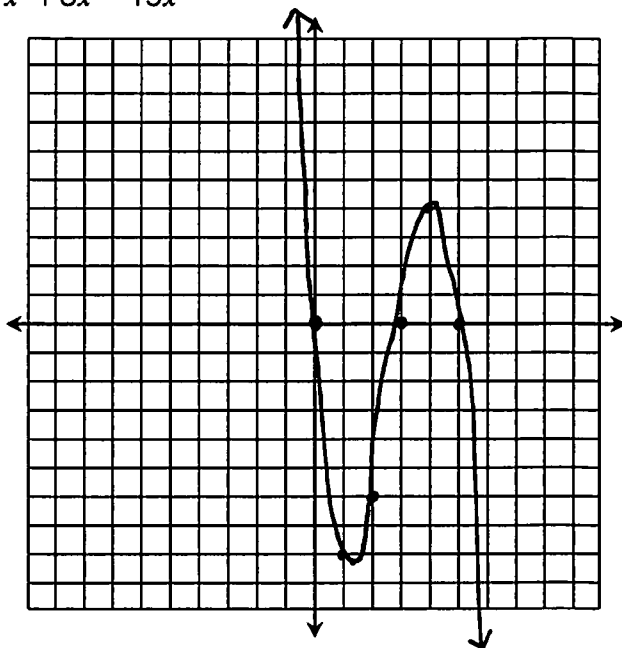
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Unit 5: Polynomial Functions

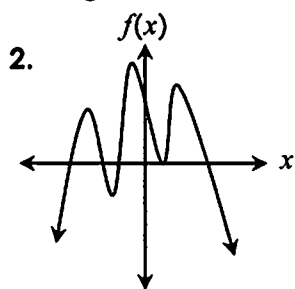
Quiz 5-2: Graphing Polynomial Functions & Solving Polynomial Equations

Graph the function below and identify its key characteristics.

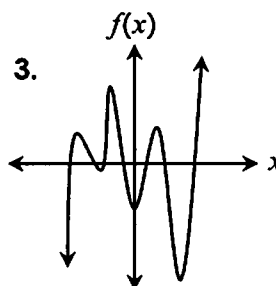
1. $f(x) = -x^3 + 8x^2 - 15x$

Domain: $\{ \mathbb{R} \}$ Range: $\{ \mathbb{R} \}$ Rel. Maximum(s): $(4.12, 4.06)$ Rel. Minimum(s): $(1.21, -8.21)$ End Behavior: As $x \rightarrow -\infty, f(x) \rightarrow -\infty$ As $x \rightarrow \infty, f(x) \rightarrow -\infty$ Inc. Intervals: $(1.21, 4.12)$ Dec. Intervals: $(-\infty, 1.21), (4.12, \infty)$ Zeros: $x = \{0, 3, 5\}$

Given the graph, determine whether the degree is even or odd and the leading coefficient is positive or negative.



Degree	Leading Coefficient
☒ Even	☐ Positive
☐ Odd	☒ Negative



Degree	Leading Coefficient
☐ Even	☒ Positive
☒ Odd	☐ Negative

Identify the zeros, their multiplicity, and their effect on the graph of the function.

4. $f(x) = -x(3x - 2)^2(x + 9)^5$

Zero	Multiplicity	Effect
0	1	crosses
$\frac{2}{3}$	2	bounces
-9	5	crosses

5. $f(x) = x^3 + 10x^2 + 25x$

$$f(x) = x(x^2 + 10x + 25)$$

$$f(x) = x(x+5)(x+5)$$

$$f(x) = x(x+5)^2$$

Zero	Multiplicity	Effect
0	1	crosses
-5	2	bounces

6. A polynomial function has a zero at 4 (multiplicity 3) and 0 (multiplicity 1). Write a function in standard form that could represent this function. Box your final answer.

$$f(x) = (x-4)^3 x$$

$$f(x) = x(x-4)(x-4)(x-4)$$

$$f(x) = (x^2 - 4x)(x^2 - 8x + 16)$$

$$f(x) = x^4 - 8x^3 + 16x^2 - 4x^3 + 32x^2 - 64x$$

$$f(x) = x^4 - 12x^3 + 48x^2 - 64x$$

Solve each equation by factoring. Simplify all irrational and complex solutions.

7. $2x^4 = 50$

$$2x^4 - 50 = 0$$

$$2(x^4 - 25) = 0$$

$$2(x^2 + 5)(x^2 - 5) = 0$$

$2 \neq 0$	$x^2 = -5$ $x = \pm \sqrt{-5}$ $x = \pm i\sqrt{5}$	$x^2 = 5$ $x = \pm \sqrt{5}$
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8. $x^3 + 216 = 0$

$$(x+6)(x^2 - 6x + 36) = 0$$

$x = -6$	$x^2 - 6x = -36$ $x^2 - 6x + 9 = -36 + 9$ $(x-3)^2 = -27$ $x-3 = \pm \sqrt{-27}$ $x = 3 \pm 3i\sqrt{3}$
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7. $x = \{\pm i\sqrt{5}, \pm \sqrt{5}\}$

8. $x = \{-6, 3 \pm 3i\sqrt{3}\}$

9. $x = \{-4, 0, 5\}$

10. $x = \{\pm i\sqrt{5}, \pm 2i\sqrt{2}\}$

11. $x = \{\pm 2, \pm i\sqrt{2}\}$

12. $x = \{-\frac{5}{4}, \pm 2\}$

9. $x^3 = x^2 + 20x$

$$x^3 - x^2 - 20x = 0$$

$$x(x^2 - x - 20) = 0$$

$$x(x-5)(x+4) = 0$$

$x = 0$	$x = 5$	$x = -4$
---------	---------	----------

10. $x^4 + 13x^2 + 40 = 0$

$$(x^2 + 5)(x^2 + 8) = 0$$

$x^2 = -5$ $x = \pm \sqrt{-5}$ $x = \pm i\sqrt{5}$	$x^2 = -8$ $x^2 = \pm \sqrt{-8}$ $x = \pm 2i\sqrt{2}$
--	---

11. $x^4 - x^2 = x^2 + 8$

$$x^4 - 2x^2 - 8 = 0$$

$$(x^2 - 4)(x^2 + 2) = 0$$

$x^2 = 4$ $x = \pm \sqrt{4}$ $x = \pm 2$	$x^2 = -2$ $x = \pm \sqrt{-2}$ $x = \pm i\sqrt{2}$
--	--

12. $4x^3 + 5x^2 - 16x - 20 = 0$

$$x^2(4x+5) - 4(4x+5) = 0$$

$$(x^2 - 4)(4x+5) = 0$$

$x^2 = 4$ $x = \pm \sqrt{4}$ $x = \pm 2$	$4x+5 = 0$ $x = -5/4$
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Name:		Date:
Topic:		Class:
Main Ideas/Questions	Notes/Examples	
DIVIDING by a Monomial	1. $\frac{6x^4 - 15x^3 + 3x^2}{3x^2}$ $2x^2 - 5x + 1$	2. $\frac{21a^5b^3 - 14a^3b^2 + 63a^2b}{7ab}$ $3a^4b^2 - 2a^2b + 9a$
	Use factoring to divide the polynomials below.	
DIVIDING by a Binomial	3. $\frac{w^2 - 7w + 10}{w - 5}$ $\frac{(w - 5)(w - 2)}{w - 5}$ $= w - 2$	4. $(4h^2 - 7h - 2) \div (h - 2)$ $\frac{4h^2 - 7h - 2}{h - 2}$ $= \frac{(4h + 1)(h - 2)}{h - 2}$ $= 4h + 1$
	5. $(x^3 + 9x^2 - 4x - 36) \div (x + 9)$ $= \frac{x^2(x + 9) - 4(x + 9)}{x + 9}$ $= \frac{(x^2 - 4)(x + 9)}{x + 9}$ $= \frac{(x + 2)(x - 2)(x + 9)}{x + 9}$ $= \frac{(x + 2)(x - 2)(x + 9)}{x + 9}$ $= (x + 2)(x - 2)$	6. $(2n^3 + 3n^2 - 16n - 24) \div (2n + 3)$ $= \frac{n^2(2n + 3) - 8(2n + 3)}{2n + 3}$ $= \frac{(n^2 - 8)(2n + 3)}{2n + 3}$ $= n^2 - 8$

Dividing Polynomials by
LONG DIVISION

When factoring is not possible, we can use long division to divide polynomials.

7. $(r^3 - 4r^2 - 26r - 15) \div (r + 3)$

$$\begin{array}{r} r^2 - 7r - 5 \\ r+3 \overline{) r^3 - 4r^2 - 26r - 15} \\ \underline{-(r^3 + 3r^2)} \\ -7r^2 - 26r - 15 \\ \underline{-(-7r^2 - 21r)} \\ -5r - 15 \\ \underline{-(-5r - 15)} \\ 0 \end{array}$$

$$r^2 - 7r - 5$$

8. $(k^3 - 8k^2 - 3k + 8) \div (k - 1)$

$$\begin{array}{r} k^2 - 7k - 10 \\ k-1 \overline{) k^3 - 8k^2 - 3k + 8} \\ \underline{-(k^3 - k^2)} \\ -7k^2 - 3k + 8 \\ \underline{-(-7k^2 + 7k)} \\ -10k + 8 \\ \underline{-(-10k + 10)} \\ -2 \end{array}$$

$$k^2 - 7k - 10 - \frac{2}{k-1}$$

9. $(6n^3 - 20n^2 - 42n - 47) \div (n - 5)$

$$\begin{array}{r} 6n^2 + 10n + 8 \\ n-5 \overline{) 6n^3 - 20n^2 - 42n - 47} \\ \underline{-(6n^3 - 30n^2)} \\ 10n^2 - 42n - 47 \\ \underline{-(10n^2 - 50n)} \\ 8n - 47 \\ \underline{-(8n - 40)} \\ -7 \end{array}$$

$$6n^2 + 10n + 8 - \frac{7}{n-5}$$

10. $(3a^3 - 7a^2 - 30a - 21) \div (3a + 2)$

$$\begin{array}{r} a^2 - 3a - 8 \\ 3a+2 \overline{) 3a^3 - 7a^2 - 30a - 21} \\ \underline{-(3a^3 + 2a^2)} \\ -9a^2 - 30a - 21 \\ \underline{-(-9a^2 - 6a)} \\ -24a - 21 \\ \underline{-(-24a - 16)} \\ -5 \end{array}$$

$$a^2 - 3a - 8 - \frac{5}{3a+2}$$

MISSING POWERS

NOTE: USE 0 as the coefficient when there is a missing power in the dividend.

11. $(k^3 - 9k - 2) \div (k - 3)$

$$\begin{array}{r} k^2 + 3k + 0 \\ k-3 \overline{) k^3 + 0k^2 - 9k - 2} \\ \underline{-(k^3 - 3k^2)} \\ 3k^2 - 9k \\ \underline{-(3k^2 - 9k)} \\ 0k - 2 \\ \underline{-(0k + 0)} \\ -2 \end{array}$$

$$k^2 + 3k - \frac{2}{k-3}$$

12. $(a^4 + 8a^3 - 3a - 16) \div (a + 8)$

$$\begin{array}{r} a^3 + 0a^2 + 0a - 3 \\ a+8 \overline{) a^4 + 8a^3 + 0a^2 - 3a - 16} \\ \underline{-(a^4 + 8a^3)} \\ 0a^3 + 0a^2 \\ \underline{-(0a^3 + 0a^2)} \\ 0a^2 - 3a \\ \underline{-(0a^2 + 0a)} \\ -3a - 16 \\ \underline{-(-3a - 24)} \\ 8 \end{array}$$

$$a^3 - 3 + \frac{8}{a+8}$$

13. $(3a^4 - 20a^2 - 6a + 14) \div (a + 2)$

$$\begin{array}{r} 3a^3 - 6a^2 - 8a + 10 \\ a+2 \overline{) 3a^4 + 0a^3 - 20a^2 - 6a + 14} \\ \underline{-(3a^4 + 6a^3)} \\ -6a^3 - 20a^2 \\ \underline{-(-6a^3 - 12a^2)} \\ -8a^2 - 6a \\ \underline{-(-8a^2 - 16a)} \\ 10a + 14 \\ \underline{-(10a + 20)} \\ -6 \end{array}$$

$$3a^3 - 6a^2 - 8a + 10 - \frac{6}{a+2}$$

14. $(36w^3 - 27w^2 - 6) \div (4w - 3)$

$$\begin{array}{r} 9w^2 + 0w + 0 \\ 4w-3 \overline{) 36w^3 - 27w^2 + 0w - 6} \\ \underline{-(36w^3 - 27w^2)} \\ 0w^2 + 0w \\ \underline{-(0w^2 + 0w)} \\ 0w - 6 \\ \underline{-(0w + 0)} \\ -6 \end{array}$$

$$9w^2 - \frac{6}{4w-3}$$

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Homework 6: Long Division with Polynomials

** This is a 2-page document! **

Find each quotient using long division.

1. $(k^3 - 10k^2 + k + 1) \div (k - 1)$

$$\begin{array}{r}
 k^2 - 9k - 8 \\
 k-1 \overline{) k^3 - 10k^2 + k + 1} \\
 \underline{-(k^3 - k^2)} \\
 -9k^2 + k \\
 \underline{-(-9k^2 + 9k)} \\
 -8k + 1 \\
 \underline{-(-8k + 8)} \\
 -7
 \end{array}$$

$$k^2 - 9k - 8 - \frac{7}{k-1}$$

2. $(x^4 + 4x^3 - 28x^2 - 45x + 26) \div (x + 7)$

$$\begin{array}{r}
 x^3 - 3x^2 - 7x + 4 \\
 x+7 \overline{) x^4 + 4x^3 - 28x^2 - 45x + 26} \\
 \underline{-(x^4 + 7x^3)} \\
 -3x^3 - 28x^2 \\
 \underline{-(-3x^3 - 21x^2)} \\
 -7x^2 - 45x \\
 \underline{-(-7x^2 - 49x)} \\
 4x + 26 \\
 \underline{-(4x + 28)} \\
 -2
 \end{array}$$

$$x^3 - 3x^2 - 7x + 4 - \frac{2}{x+7}$$

3. $(20c^3 + 22c^2 - 7c + 7) \div (5c - 2)$

$$\begin{array}{r}
 4c^2 + 6c + 1 \\
 5c-2 \overline{) 20c^3 + 22c^2 - 7c + 7} \\
 \underline{-(20c^3 - 8c^2)} \\
 30c^2 - 7c \\
 \underline{-(30c^2 - 12c)} \\
 5c + 7 \\
 \underline{-(5c - 2)} \\
 9
 \end{array}$$

$$4c^2 + 6c + 1 + \frac{9}{5c-2}$$

4. $(3n^4 + 6n^3 - 15n^2 + 32n - 25) \div (n + 4)$

$$\begin{array}{r}
 3n^3 - 6n^2 + 9n - 4 \\
 n+4 \overline{) 3n^4 + 6n^3 - 15n^2 + 32n - 25} \\
 \underline{-(3n^4 + 12n^3)} \\
 -6n^3 - 15n^2 \\
 \underline{-(-6n^3 - 24n^2)} \\
 9n^2 + 32n \\
 \underline{-(9n^2 + 36n)} \\
 -4n - 25 \\
 \underline{-(-4n - 16)} \\
 -9
 \end{array}$$

$$3n^3 - 6n^2 + 9n - 4 - \frac{9}{n+4}$$

5. $(p^3 - 17p - 13) \div (p + 4)$

$$\begin{array}{r}
 p^2 - 4p - 1 \\
 p+4 \overline{) p^3 + 0p^2 - 17p - 13} \\
 \underline{-(p^3 + 4p^2)} \\
 -4p^2 - 17p \\
 \underline{-(-4p^2 - 16p)} \\
 -p - 13 \\
 \underline{-(-p - 4)} \\
 -9
 \end{array}$$

$$p^2 - 4p - 1 - \frac{9}{p+4}$$

6. $(y^4 - y^2 + 8y + 5) \div (y + 1)$

$$\begin{array}{r}
 y^3 - y^2 + 0y + 18 \\
 y+1 \overline{) y^4 + 0y^3 - y^2 + 8y + 5} \\
 \underline{-(y^4 + y^3)} \\
 -y^3 - y^2 \\
 \underline{-(-y^3 - y^2)} \\
 0y^2 + 8y \\
 \underline{-(0y^2 + 0y)} \\
 8y + 5 \\
 \underline{-(8y + 8)} \\
 -3
 \end{array}$$

$$y^3 - y^2 + 8 - \frac{3}{y+1}$$

7. $(4v^3 - 12v^2 + 6) \div (v - 3)$

$$\begin{array}{r}
 4v^2 + 0v + 0 \\
 v-3 \overline{) 4v^3 - 12v^2 + 0v + 6} \\
 \underline{-(4v^3 - 12v^2)} \\
 0v^2 + 0v \\
 \underline{-(0v^2 + 0v)} \\
 0v + 6 \\
 \underline{-(0v + 0)} \\
 6
 \end{array}$$

$$4v^2 + \frac{6}{v-3}$$

8. $(3a^4 + 4a^3 + 15a + 10) \div (3a + 4)$

$$\begin{array}{r}
 a^3 + 0a^2 + 0a + 5 \\
 3a+4 \overline{) 3a^4 + 4a^3 + 0a^2 + 15a + 10} \\
 \underline{-(3a^4 + 4a^3)} \\
 0a^3 + 0a^2 \\
 \underline{-(0a^3 + 0a^2)} \\
 0a^2 + 15a \\
 \underline{-(0a^2 + 0a)} \\
 15a + 10 \\
 \underline{-(15a + 20)} \\
 -10
 \end{array}$$

$$a^3 + 5 - \frac{10}{3a+4}$$

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Main Ideas/Questions

Notes/Examples

Dividing Polynomials by SYNTHETIC DIVISION

Synthetic division is a shortcut method to divide polynomials.

This method only works when dividing by a binomial with a coefficient of 1.

1. $(x^2 + 3x - 43) \div (x + 8)$

$$\begin{array}{r|rrr} -8 & 1 & 3 & -43 \\ & \downarrow & -8 & 40 \\ \hline & 1 & -5 & (-3) \end{array}$$

$$x - 5 - \frac{3}{x+8}$$

2. $(2n^2 - 16n + 19) \div (n - 7)$

$$\begin{array}{r|rrr} 7 & 2 & -16 & 19 \\ & \downarrow & 14 & -14 \\ \hline & 2 & -2 & (5) \end{array}$$

$$2n - 2 + \frac{5}{n-7}$$

3. $(y^3 + 5y^2 - 7y + 2) \div (y - 2)$

$$\begin{array}{r|rrrr} 2 & 1 & 5 & -7 & 2 \\ & \downarrow & 2 & 14 & 14 \\ \hline & 1 & 7 & 7 & (16) \end{array}$$

$$y^2 + 7y + 7 + \frac{16}{y-2}$$

4. $(c^4 + 9c^3 + 13c^2 - 33c - 16) \div (c + 4)$

$$\begin{array}{r|rrrrr} -4 & 1 & 9 & 13 & -33 & -16 \\ & \downarrow & -4 & -20 & 28 & 20 \\ \hline & 1 & 5 & -7 & -5 & (4) \end{array}$$

$$c^3 + 5c^2 - 7c - 5 + \frac{4}{c+4}$$

MISSING POWERS

NOTE: USE 0 as the coefficient when there is a missing power in the dividend

5. $(k^3 - 17k + 32) \div (k + 5)$

$$\begin{array}{r|rrrr} -5 & 1 & 0 & -17 & 32 \\ & \downarrow & -5 & 25 & -40 \\ \hline & 1 & -5 & 8 & (-8) \end{array}$$

$$k^2 - 5k + 8 - \frac{8}{k+5}$$

6. $(8x^4 - 64x^3 + 3x - 28) \div (x - 8)$

$$\begin{array}{r|rrrrr} 8 & 8 & -64 & 0 & 3 & -28 \\ & \downarrow & 64 & 0 & 0 & 24 \\ \hline & 8 & 0 & 0 & 3 & (-4) \end{array}$$

$$8x^3 + 3 - \frac{4}{x-8}$$

LONG DIVISION vs. SYNTHETIC DIVISION

Find each quotient using long and synthetic division. Compare answers to check.

LONG DIVISION

7. $(n^3 - 3n^2 - 11n - 3) \div (n + 1)$

$$\begin{array}{r}
 n^2 - 4n - 7 \\
 n+1 \overline{) n^3 - 3n^2 - 11n - 3} \\
 \underline{-(n^3 + n^2)} \\
 -4n^2 - 11n \\
 \underline{-(-4n^2 - 4n)} \\
 -7n - 3 \\
 \underline{-(-7n - 7)} \\
 4
 \end{array}$$

$$n^2 - 4n - 7 + \frac{4}{n+1}$$

SYNTHETIC DIVISION

$$\begin{array}{r|rrrr}
 -1 & 1 & -3 & -11 & -3 \\
 & \downarrow & -1 & 4 & 7 \\
 \hline
 & 1 & -4 & -7 & 4
 \end{array}$$

$$n^2 - 4n - 7 + \frac{4}{n+1}$$

8. $(2k^4 + 10k^3 - 40k^2 + 61k - 24) \div (k + 8)$

$$\begin{array}{r}
 2k^3 - 6k^2 + 8k - 3 \\
 k+8 \overline{) 2k^4 + 10k^3 - 40k^2 + 61k - 24} \\
 \underline{-(2k^4 + 16k^3)} \\
 -6k^3 - 40k^2 \\
 \underline{-(-6k^3 - 48k^2)} \\
 8k^2 + 61k \\
 \underline{-(8k^2 + 64k)} \\
 -3k - 24 \\
 \underline{-(-3k - 24)} \\
 0
 \end{array}$$

$$2k^3 - 6k^2 + 8k - 3$$

$$\begin{array}{r|rrrrr}
 -8 & 2 & 10 & -40 & 61 & -24 \\
 & \downarrow & -16 & 48 & -64 & 24 \\
 \hline
 & 2 & -6 & 8 & -3 & 0
 \end{array}$$

$$2k^3 - 6k^2 + 8k - 3$$

9. $(w^4 - 4w^3 - 3w + 13) \div (w - 4)$

$$\begin{array}{r}
 w^3 + 0w^2 + 0w - 3 \\
 w-4 \overline{) w^4 - 4w^3 + 0w^2 - 3w + 13} \\
 \underline{-(w^4 - 4w^3)} \\
 0w^3 + 0w^2 \\
 \underline{-(0w^3 + 0w^2)} \\
 0w^2 - 3w \\
 \underline{-(0w^2 + 0w)} \\
 -3w + 13 \\
 \underline{-(-3w + 12)} \\
 1
 \end{array}$$

$$w^3 - 3 + \frac{1}{w-4}$$

$$\begin{array}{r|rrrrr}
 4 & 1 & -4 & 0 & -3 & 13 \\
 & \downarrow & 4 & 0 & 0 & -12 \\
 \hline
 & 1 & 0 & 0 & -3 & 1
 \end{array}$$

$$w^3 - 3 + \frac{1}{w-4}$$

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Homework 7: Synthetic Division with Polynomials

**** This is a 2-page document! ****

Find each quotient using synthetic division.

1. $(n^2 + 11n + 25) \div (n + 5)$

$$\begin{array}{r|rrrr} -5 & 1 & 11 & 25 & \\ & \downarrow & -5 & -30 & \\ \hline & 1 & 6 & -5 & \end{array}$$

$$n + 6 - \frac{5}{n+5}$$

2. $(3k^2 - k - 20) \div (k + 2)$

$$\begin{array}{r|rrrr} -2 & 3 & -1 & -20 & \\ & \downarrow & -6 & 14 & \\ \hline & 3 & -7 & -6 & \end{array}$$

$$3x - 7 - \frac{6}{k+2}$$

3. $(x^3 + 2x^2 - 22x - 45) \div (x + 5)$

$$\begin{array}{r|rrrrr} -5 & 1 & 2 & -22 & -45 & \\ & \downarrow & -5 & 15 & 35 & \\ \hline & 1 & -3 & -7 & -10 & \end{array}$$

$$x^2 - 3x - 7 - \frac{10}{x+5}$$

4. $(w^3 - 2w^2 - 2w + 1) \div (w - 1)$

$$\begin{array}{r|rrrrr} 1 & 1 & -2 & -2 & 1 & \\ & \downarrow & 1 & -1 & -3 & \\ \hline & 1 & -1 & -3 & -2 & \end{array}$$

$$x^2 - x - 3 - \frac{2}{w-1}$$

5. $(2c^3 + 13c^2 + 24c + 8) \div (c + 3)$

$$\begin{array}{r|rrrrr} -3 & 2 & 13 & 24 & 8 & \\ & \downarrow & -6 & -21 & -9 & \\ \hline & 2 & 7 & 3 & -1 & \end{array}$$

$$2c^2 + 7c + 3 - \frac{1}{c+3}$$

6. $(p^4 - 7p^2 - 32p - 15) \div (p - 4)$

$$\begin{array}{r|rrrrrr} 4 & 1 & 0 & -7 & -32 & -15 & \\ & \downarrow & 4 & 16 & 36 & 16 & \\ \hline & 1 & 4 & 9 & 4 & 1 & \end{array}$$

$$x^3 + 4x^2 + 9x + 4 + \frac{1}{p-4}$$

Find each quotient using long and synthetic division. Compare answers to check.

LONG DIVISION

SYNTHETIC DIVISION

7. $(m^3 - 12m^2 + 33m) \div (m - 5)$

$$\begin{array}{r}
 m^2 - 7m - 2 \\
 m-5 \overline{) m^3 - 12m^2 + 33m + 0} \\
 \underline{-(m^3 - 5m^2)} \\
 -7m^2 + 33m \\
 \underline{-(-7m^2 + 35m)} \\
 -2m + 0 \\
 \underline{-(-2m + 10)} \\
 -10
 \end{array}$$

$$m^2 - 7m - 2 - \frac{10}{m-5}$$

$$\begin{array}{r|rrrr}
 5 & 1 & -12 & 33 & 0 \\
 & \downarrow & 5 & -35 & -10 \\
 \hline
 & 1 & -7 & -2 & -10
 \end{array}$$

$$m^2 - 7m - 2 - \frac{10}{m-5}$$

8. $(2x^3 - 14x + 10) \div (x + 3)$

$$\begin{array}{r}
 2x^2 - 6x + 4 \\
 x+3 \overline{) 2x^3 + 0x^2 - 14x + 10} \\
 \underline{-(2x^3 + 6x^2)} \\
 -6x^2 - 14x \\
 \underline{-(-6x^2 - 18x)} \\
 4x + 10 \\
 \underline{-(4x + 12)} \\
 -2
 \end{array}$$

$$2x^2 - 6x + 4 - \frac{2}{x+3}$$

$$\begin{array}{r|rrrr}
 -3 & 2 & 0 & -14 & 10 \\
 & \downarrow & -6 & 18 & -12 \\
 \hline
 & 2 & -6 & 4 & -2
 \end{array}$$

$$2x^2 - 6x + 4 - \frac{2}{x+3}$$

9. $(x^4 - 2x^3 - 29x^2 - 43x + 8) \div (x - 7)$

$$\begin{array}{r}
 x^3 + 5x^2 + 6x - 1 \\
 x-7 \overline{) x^4 - 2x^3 - 29x^2 - 43x + 8} \\
 \underline{-(x^4 - 7x^3)} \\
 5x^3 - 29x^2 \\
 \underline{-(5x^3 - 35x^2)} \\
 6x^2 - 43x \\
 \underline{-(6x^2 - 42x)} \\
 -x + 8 \\
 \underline{-(-x + 7)} \\
 1
 \end{array}$$

$$x^3 + 5x^2 + 6x - 1 + \frac{1}{x-7}$$

$$\begin{array}{r|rrrrr}
 7 & 1 & -2 & -29 & -43 & 8 \\
 & \downarrow & 7 & 35 & 42 & -7 \\
 \hline
 & 1 & 5 & 6 & -1 & 1
 \end{array}$$

$$x^3 + 5x^2 + 6x - 1 + \frac{1}{x-7}$$

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Main Ideas/Questions	Notes/Examples
WARM-UP	1. Consider the polynomial function $f(x) = x^3 + 9x^2 + 22x - 1$ a) Find $f(-1)$ $f(-1) = (-1)^3 + 9(-1)^2 + 22(-1) - 1$ $= -1 + 9 - 22 - 1$ $= \boxed{-15}$
	b) Divide the function by $(x+1)$. $\begin{array}{r rrrr} -1 & 1 & 9 & 22 & -1 \\ & \downarrow & -1 & -8 & -14 \\ \hline & 1 & 8 & 14 & -15 \end{array}$ $\boxed{x^2 + 8x + 14 - \frac{15}{x+1}}$
	2. Consider the polynomial function $f(x) = 2x^4 - 30x^2 + 2x - 17$. a) Find $f(4)$ $f(4) = 2(4)^4 - 30(4)^2 + 2(4) - 17$ $= 2(256) - 30(16) + 8 - 17$ $= 512 - 480 + 8 - 17$ $= \boxed{23}$
	b) Divide the function by $(x-4)$. $\begin{array}{r rrrrr} 4 & 2 & 0 & -30 & 2 & -17 \\ & \downarrow & 8 & 32 & 8 & 40 \\ \hline & 2 & 8 & 2 & 10 & 23 \end{array}$ $\boxed{2x^3 + 8x^2 + 2x + 10 + \frac{23}{x-4}}$
REMAINDER <i>Theorem</i>	Remainder Theorem: If a polynomial function $f(x)$ is divided by <u>$x-c$</u>, then the <u>remainder</u> is <u>$f(c)$</u>. - Using synthetic division to evaluate a function is called <u>synthetic substitution</u>.
EXAMPLES	Use The Remainder Theorem (Synthetic Substitution) to find the given value. 3. $f(x) = x^3 - 13x^2 + 39x - 4$; $f(5)$ $\begin{array}{r rrrr} 5 & 1 & -13 & 39 & -4 \\ & \downarrow & 5 & -40 & -5 \\ \hline & 1 & -8 & -1 & -9 \end{array}$ $\boxed{f(5) = -9}$
	4. $f(x) = 5x^3 + 11x^2 - 22x - 28$; $f(-2)$ $\begin{array}{r rrrr} -2 & 5 & 11 & -22 & -28 \\ & \downarrow & -10 & -2 & 48 \\ \hline & 5 & 1 & -24 & 20 \end{array}$ $\boxed{f(-2) = 20}$

5. $f(x) = x^4 + 7x^3 - 12x^2 - 34x - 10; f(-8)$

$$\begin{array}{r|rrrrr} -8 & 1 & 7 & -12 & -34 & -10 \\ & \downarrow & & & & \\ & & -8 & 8 & 32 & 16 \\ \hline & 1 & -1 & -4 & -2 & \textcircled{6} \end{array}$$

$$f(-8) = 6$$

6. $f(x) = x^3 - 10x^2 - 4; f(10)$

$$\begin{array}{r|rrrr} 10 & 1 & -10 & 0 & -4 \\ & \downarrow & & & \\ & & 10 & 0 & 0 \\ \hline & 1 & 0 & 0 & \textcircled{-4} \end{array}$$

$$f(10) = -4$$

7. $f(x) = x^4 + x^3 + 7x - 1; f(-3)$

$$\begin{array}{r|rrrrr} -3 & 1 & 1 & 0 & 7 & -1 \\ & \downarrow & & & & \\ & & -3 & 6 & -18 & 33 \\ \hline & 1 & -2 & 6 & -11 & \textcircled{32} \end{array}$$

$$f(-3) = 32$$

8. $f(x) = 2x^4 - 18x^2 + 7; f(2)$

$$\begin{array}{r|rrrrr} 2 & 2 & 0 & -18 & 0 & 7 \\ & \downarrow & & & & \\ & & 4 & 8 & -20 & -40 \\ \hline & 2 & 4 & -10 & -20 & \textcircled{-33} \end{array}$$

$$f(2) = -33$$

APPLICATIONS

9. A bicycle manufacturer determines that the profit per bicycle when producing x bicycles per day is $P(x) = -x^4 + 7x^3 - 5x^2 + x$. Use the Remainder Theorem to find the profit per bicycle when four bicycles are produced each day.

$$\begin{array}{r|rrrrr} 4 & -1 & 7 & -5 & 1 & 0 \\ & \downarrow & & & & \\ & & -4 & 12 & 28 & 116 \\ \hline & -1 & 3 & 7 & 29 & 116 \end{array}$$

$$P(4) = 116$$

$$\$116$$

10. The value V (in dollars) of a signed baseball card can be modeled by the function $V(t) = 2t^3 - 48t^2 + 300t$ where t is the number of years since 1980. Use the Remainder Theorem to find the value of the card in 2025.

$$\begin{array}{r|rrrr} 45 & 2 & -48 & 300 & 0 \\ & \downarrow & & & \\ & & 90 & 1890 & 98550 \\ \hline & 2 & 42 & 2190 & 98550 \end{array}$$

$$V(45) = 98550$$

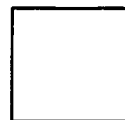
$$\$98,550$$

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Unit 5: Polynomial Functions

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Homework 8: Remainder Theorem



Evaluate each function using the Remainder Theorem (Synthetic Substitution).

1. $f(x) = x^3 - 11x^2 + 9x + 8; f(2)$

$$\begin{array}{r|rrrr} 2 & 1 & -11 & 9 & 8 \\ & \downarrow & & & \\ & & 2 & -18 & -18 \\ \hline & 1 & -9 & -9 & (-10) \end{array}$$

$$f(2) = -10$$

2. $f(x) = 4x^3 + 33x^2 + 25x - 68; f(-7)$

$$\begin{array}{r|rrrr} -7 & 4 & 33 & 25 & -68 \\ & \downarrow & & & \\ & & -28 & -35 & 70 \\ \hline & 4 & 5 & -10 & (2) \end{array}$$

$$f(-7) = 2$$

3. $f(x) = x^4 + 6x^3 + 6x^2 - 2x + 4; f(-1)$

$$\begin{array}{r|rrrrr} -1 & 1 & 6 & 6 & -2 & 4 \\ & \downarrow & & & & \\ & & -1 & -5 & -1 & 3 \\ \hline & 1 & 5 & 1 & -3 & (7) \end{array}$$

$$f(-1) = 7$$

4. $f(x) = 6x^4 - 26x^3 - 23x^2 + 17x - 4; f(5)$

$$\begin{array}{r|rrrrr} 5 & 6 & -26 & -23 & 17 & -4 \\ & \downarrow & & & & \\ & & 30 & 20 & -15 & 10 \\ \hline & 6 & 4 & -3 & 2 & (6) \end{array}$$

$$f(5) = 6$$

5. $f(x) = -x^3 - 8x^2 + 3; f(-4)$

$$\begin{array}{r|rrrr} -4 & -1 & -8 & 0 & 3 \\ & \downarrow & & & \\ & & 4 & 16 & -64 \\ \hline & -1 & -4 & 16 & (-61) \end{array}$$

$$f(-4) = -61$$

6. $f(x) = 2x^4 - 7x^3 - 2x + 19; f(3)$

$$\begin{array}{r|rrrrr} 3 & 2 & -7 & 0 & -2 & 19 \\ & \downarrow & & & & \\ & & 6 & -3 & -9 & -33 \\ \hline & 2 & -1 & -3 & -11 & (-14) \end{array}$$

$$f(3) = -14$$

7. $f(x) = x^4 - 7x^2 + 23; f(-2)$

$$\begin{array}{r|rrrrr} -2 & 1 & 0 & -7 & 0 & 23 \\ & \downarrow & & & & \\ & & -2 & 4 & 6 & -12 \\ \hline & 1 & -2 & -3 & 6 & (11) \end{array}$$

$$f(-2) = 11$$

8. The height h , in feet, of a rollercoaster x seconds into a ride can be modeled by the function $h(x) = -x^4 + 26x^3 - 212x^2 + 590x + 22$. Use the Remainder Theorem to find the height of the rollercoaster 4 seconds into the ride.

$$\begin{array}{r|rrrrr} 4 & -1 & 26 & -212 & 590 & 22 \\ & \downarrow & & & & \\ & & -4 & 88 & -496 & 376 \\ \hline & -1 & 22 & -124 & 94 & (398) \end{array}$$

$$P(4) = 398$$

$$398 \text{ ft}$$

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples	
<i>Function</i> OPERATIONS	Functions can added, subtracted, multiplied, and divided to create new functions.	
	Example: If $f(x) = 3x - 7$ and $g(x) = 13 - 2x$, find $f(x) + g(x)$: $(3x - 7) + (13 - 2x)$ $= \boxed{x + 6}$	
	This new function is denoted as $(f + g)(x)$	
OPERATIONS <i>Rules</i>	SUM	$(f + g)(x) = f(x) + g(x)$
	DIFFERENCE	$(f - g)(x) = f(x) - g(x)$
	PRODUCT	$(f \cdot g)(x) = f(x) \cdot g(x)$
	*QUOTIENT	$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$ * Must define domain
Functions: $f(x) = x^2 - 8x + 4$ $g(x) = 4x - 3$ $h(x) = x + 2$	Find each function using the functions given to the left. Indicate any restrictions in the domain.	
	1. $(f + g)(x)$	2. $(f - h)(x)$
	$(x^2 - 8x + 4) + (4x - 3)$ $= \boxed{x^2 - 4x + 1}$	$(x^2 - 8x + 4) - (x + 2)$ $= \boxed{x^2 - 9x + 2}$
	3. $(h \cdot g)(x)$	4. $\left(\frac{f}{h}\right)(x)$
	$(x + 2) \cdot (4x - 3)$ $= \boxed{4x^2 + 5x - 6}$	$= \boxed{\frac{x^2 - 8x + 4}{x + 2}, \{x x \neq -2\}}$
	5. $\left(\frac{h}{g}\right)(x)$	6. $(f \cdot g)(x)$
	$\frac{x + 2}{4x - 3}, \{x x \neq \frac{3}{4}\}$	$(x^2 - 8x + 4)(4x - 3)$ $= 4x^3 - 3x^2 - 32x^2 + 24x + 16x - 12$ $= \boxed{4x^3 - 35x^2 + 40x - 12}$

Functions: $f(x) = 2x^2 - x - 12$ $g(x) = x + 7$	Evaluate each function using the functions given to the left.	
	7. $(f + g)(-2)$ $f(-2) = 2(-2)^2 - (-2) - 12$ $= 8 + 2 - 12 = -2$ $g(-2) = -2 + 7 = 5$ $-2 + 5 = \boxed{3}$	8. $(f - g)(8)$ $f(8) = 2(8)^2 - 8 - 12$ $= 128 - 8 - 12 = 108$ $g(8) = 8 + 7 = 15$ $108 - 15 = \boxed{93}$
COMPOSITIONS of Functions	Another method to combine functions is called a composition. Given $f(x)$ and $g(x)$, the composite function $(f \circ g)(x)$ is defined as: $(f \circ g)(x) = f(g(x))$	
Functions: $f(x) = 5x - 3$ $g(x) = x - 1$ $h(x) = x^3 + 8$	Find each function value using the functions given to the left.	
	11. $(f \circ g)(x)$ $5(x - 1) - 3$ $= 5x - 5 - 3$ $= \boxed{5x - 8}$	12. $(g \circ h)(x)$ $(x^3 + 8) - 1$ $= \boxed{x^3 + 7}$
	13. $(h \circ g)(x)$ $(x - 1)^3 + 8$ $= (x - 1)(x - 1)(x - 1) + 8$ $= (x - 1)(x^2 - 2x + 1) + 8$ $= x^3 - 2x^2 + x - x^2 + 2x - 1 + 8$ $= \boxed{x^3 - 3x^2 + 3x + 7}$	14. $(g \circ f)(x)$ $(5x - 3) - 1$ $= \boxed{5x - 4}$
	Using the same set of functions, evaluate each function.	
	15. $(g \circ f)(2)$ $f(2) = 5(2) - 3$ $= 10 - 3$ $= 7$ $g(7) = 7 - 1 = \boxed{6}$	16. $(f \circ h)(-6)$ $h(-6) = (-6)^3 + 8$ $= -216 + 8$ $= -208$ $f(-208) = 5(-208) - 3$ $= \boxed{-1043}$

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Unit 5: Polynomial Functions

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Homework 9: Operations on Functions
& Compositions of Functions

Directions: Given $f(x) = 2x^2 - 9x + 2$, $g(x) = 1 - 6x$, and $h(x) = x^2 - 4$, find each function. Indicate any restrictions in the domain.

1. $(f - g)(x)$

$$(2x^2 - 9x + 2) - (1 - 6x)$$

$$= \boxed{2x^2 - 3x + 1}$$

2. $\left(\frac{f}{h}\right)(x)$

$$\frac{2x^2 - 9x + 2}{x^2 - 4}, \{x | x \neq -2, 2\}$$

3. $(g + f)(x)$

$$(1 - 6x) + (2x^2 - 9x + 2)$$

$$= \boxed{2x^2 - 15x + 3}$$

4. $(f \cdot h)(x)$

$$(2x^2 - 9x + 2)(x^2 - 4)$$

$$= 2x^4 - 8x^2 - 9x^3 + 36x + 2x^2 - 8$$

$$= \boxed{2x^4 - 9x^3 - 6x^2 + 36x - 8}$$

5. $\left(\frac{h}{g}\right)(x)$

$$\frac{x^2 - 4}{1 - 6x}, \{x | x \neq \frac{1}{6}\}$$

6. $(g \cdot h)(x)$

$$(1 - 6x)(x^2 - 4)$$

$$= x^2 - 4 - 6x^3 + 24x$$

$$= \boxed{-6x^3 + x^2 + 24x - 4}$$

7. $(f \circ g)(x)$

$$2(1 - 6x)^2 - 9(1 - 6x) + 2$$

$$= 2(1 - 12x + 36x^2) - 9 + 54x + 2$$

$$= 2 - 24x + 72x^2 - 9 + 54x$$

$$= \boxed{72x^2 + 30x - 5}$$

8. $(h \circ f)(x)$

$$(2x^2 - 9x + 2)^2 - 4$$

$$= (2x^2 - 9x + 2)(2x^2 - 9x + 2) - 4$$

$$= 4x^4 - 18x^3 + 4x^2 - 18x^3 + 81x^2 - 18x$$

$$+ 4x^2 - 18x + 4 - 4$$

$$= \boxed{4x^4 - 36x^3 + 89x^2 - 36x}$$

Directions: Using the functions above, find each function value.

9. $(h - g)(-7)$

$$h(-7) = (-7)^2 - 4$$

$$= 49 - 4 = 45$$

$$g(-7) = 1 - 6(-7)$$

$$= 1 + 42 = 43$$

$$45 - 43 = \boxed{2}$$

10. $(f \circ h)(2)$

$$h(2) = (2)^2 - 4$$

$$= 4 - 4 = 0$$

$$f(0) = 2(0)^2 - 9(0) + 2$$

$$= 0 - 0 + 2$$

$$= \boxed{2}$$

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Quiz 5-3: Dividing Polynomials, Remainder Theorem, Functions Operations

Find each quotient.

1. $(x^2 - 16x + 63) \div (x - 9)$

$$\begin{array}{r} (x-7)(x-9) \\ \hline x-9 \end{array}$$

2. $(y^2 - 2y - 29) \div (y - 6)$

$$\begin{array}{r} 6 \overline{) 1 \ -2 \ -29} \\ \underline{\downarrow \quad 6 \quad 24} \\ 1 \quad 4 \quad (-5) \end{array}$$

1. $\underline{x-7}$

2. $\underline{y+4 - \frac{5}{y-6}}$

3. $\underline{a^2 + 7a + 4 + \frac{16}{a+1}}$

4. $\underline{4v^2 + 5v - 1 + \frac{3}{3v+4}}$

5. $\underline{x^2 + 4x + \frac{9}{x-4}}$

6. $\underline{k^3 - 3k^2 + k - 7 + \frac{10}{k+7}}$

3. $(a^3 + 8a^2 + 11a + 20) \div (a + 1)$

$$\begin{array}{r} -1 \overline{) 1 \quad 8 \quad 11 \quad 20} \\ \underline{\downarrow \quad -1 \quad -7 \quad -4} \\ 1 \quad 7 \quad 4 \quad (16) \end{array}$$

4. $(12v^3 + 31v^2 + 17v - 1) \div (3v + 4)$

$$\begin{array}{r} 4v^2 + 5v - 1 \\ 3v+4 \overline{) 12v^3 + 31v^2 + 17v - 1} \\ \underline{-(12v^3 + 16v^2)} \\ 15v^2 + 17v \\ \underline{-(15v^2 + 20v)} \\ -3v - 1 \\ \underline{-(-3v - 4)} \\ (3) \end{array}$$

5. $(x^3 - 16x + 9) \div (x - 4)$

$$\begin{array}{r} 4 \overline{) 1 \quad 0 \quad -16 \quad 9} \\ \underline{\downarrow \quad 4 \quad 16 \quad 0} \\ 1 \quad 4 \quad 0 \quad (9) \end{array}$$

6. $(k^4 + 4k^3 - 20k^2 - 39) \div (k + 7)$

$$\begin{array}{r} -7 \overline{) 1 \quad 4 \quad -20 \quad 0 \quad -39} \\ \underline{\downarrow \quad -7 \quad 21 \quad -7 \quad 49} \\ 1 \quad -3 \quad 1 \quad -7 \quad (10) \end{array}$$

7. If $f(x) = 3x^4 - 8x^3 + 14x - 29$, use the Remainder Theorem to find $f(2)$.

$$\begin{array}{r|rrrrr} 2 & 3 & -8 & 0 & 14 & -29 \\ & \downarrow & 6 & -4 & -8 & 12 \\ \hline & 3 & -2 & -4 & 6 & -17 \end{array}$$

$$\begin{array}{l} 7. \quad -17 \\ 8. \quad 3278 \end{array}$$

8. The population P of an endangered species can be modeled by the function $P(t) = -t^3 + 32t^2 - 50t + 450$, where t is the years since 2010. Use the Remainder Theorem to find the population of the species in 2024.

$$\begin{array}{r|rrrr} 14 & -1 & 32 & -50 & 450 \\ & \downarrow & -14 & 252 & 2828 \\ \hline & -1 & 18 & 202 & 3278 \end{array}$$

Given $f(x) = x + 5$, $g(x) = x^2 - 9$, and $h(x) = x^2 - 10x + 7$, find each function. State any restrictions in the domain.

9. $(g - h)(x)$

$$\begin{aligned} (x^2 - 9) - (x^2 - 10x + 7) \\ = 10x - 16 \end{aligned}$$

10. $(f \cdot g)(x)$

$$\begin{aligned} (x + 5)(x^2 - 9) \\ = x^3 + 5x^2 - 9x - 45 \end{aligned}$$

9. $10x - 16$

10. $x^3 + 5x^2 - 9x - 45$

11. $\frac{x^2 - 10x + 7}{x + 5}, \{x | x \neq -5\}$

12. $x^2 + 10x + 16$

13. 44

14. 103

11. $\left(\frac{h}{f}\right)(x)$

$$\frac{x^2 - 10x + 7}{x + 5}$$

12. $(g \circ f)(x)$

$$\begin{aligned} (x + 5)^2 - 9 \\ = x^2 + 10x + 25 - 9 \\ = x^2 + 10x + 16 \end{aligned}$$

Find each function value.

13. $(h - f)(-3)$

$$\begin{aligned} h(-3) &= (-3)^2 - 10(-3) + 7 \\ &= 9 + 30 + 7 \\ &= 46 \end{aligned}$$

$$\begin{aligned} f(-3) &= -3 + 5 \\ &= 2 \end{aligned}$$

$$46 - 2 = 44$$

14. $(h \circ g)(5)$

$$\begin{aligned} g(5) &= 5^2 - 9 \\ &= 25 - 9 = 16 \end{aligned}$$

$$\begin{aligned} h(16) &= (16)^2 - 10(16) + 7 \\ &= 256 - 160 + 7 \\ &= 103 \end{aligned}$$

Name:

Date:

Topic:

Class:

Main Ideas/Questions

Notes/Examples

REVIEW:Linear &
Quadratic
Regression

1. The data below shows the average student loan debt accumulated by college graduates during certain years since 1996. Write an equation for the **line of best fit**, then estimate the average student loan debt for a student graduating in 2020.

$$\underline{x=24}$$

	Year	Loan Debt
0	1996	\$12,750
4	2000	\$17,350
8	2004	\$18,650
12	2008	\$23,200
13	2009	\$24,000

$$f(x) = 830.29x + 13045.89$$

$$f(24) = 830.29(24) + 13045.89$$

$$= \$32,972.85$$

2. Dan hit a golf ball from the top of a hill. The height of the ball after each second is given in the table below. Using a **quadratic model**, write an equation for the curve of best fit, then estimate the height of the ball after 5 seconds.

Time (s)	Height (ft)
0	60
1	83
2	95
3	81
4	59

$$f(x) = -8.29x^2 + 32.74x + 59.83$$

$$f(5) = -8.29(5)^2 + 32.74(5) + 59.83$$

$$= 16.28 \text{ ft}$$

NEW:Modeling with
Cubic & Quartic
Functions

3. The Millennium Force at Cedar Point is one of the world's most popular roller coasters. The table below shows the average number of people in line at each hour after the park opens. Use a **cubic model** to find an equation for the curve of best fit, then estimate the number of people in line 6 hours after the park opened.

Hour	People
0	0
1	120
2	324
3	375
4	394
5	312

$$f(x) = -5.20x^3 + 7.78x^2 + 154.47x - 8.17$$

$$f(6) = 75.53 \rightarrow 75 \text{ people}$$

4. On the first day back to school after winter break, 20 students were absent from school due to illness. The table below shows the number of students absent from school on certain days since returning from winter break due to illness. Use a **quartic model** to find an equation for the curve of best fit, then estimate the number of students absent 9 days after returning from break.

Day	# Absent
0	20
2	28
4	35
6	42
8	43

$$f(x) = -.02x^4 + .24x^3 - 1.05x^2 + 5.29x + 20$$

$$f(9) = 26.3 \rightarrow \boxed{26 \text{ students}}$$

5. The table below shows the average minimum wage during certain years since 1955. Use a **cubic model** to find an equation for the curve of best fit, then estimate the minimum wage in 2025.

$$x = 70$$

	Year	Min. Wage
0	1955	\$0.75
6	1961	\$1.15
12	1967	\$1.40
19	1974	\$2.00
23	1978	\$2.65
25	1980	\$3.10
35	1990	\$3.80

$$f(x) = -.00009x^3 + .01x^2 + .82$$

$$f(70) = \boxed{\$18.95}$$

6. The table below shows the opening value of a stock index on the first day of trading during certain years since 1988. Use a **quartic model** to find an equation for the curve of best fit, then estimate the value of the stock index in 2010.

$$x = 22$$

Year	Price
1988	\$1,800
1990	\$2,100
1993	\$5,500
1995	\$9,800
1999	\$7,000
2005	\$16,000

$$f(x) = 4.07x^4 - 123.8x^3 + 1079.55x^2 - 1912.70x + 1971.91$$

$$f(22) = \boxed{\$124,195.99}$$

Unit 5 Test Study Guide

(Polynomial Functions)

Name: _____

Date: _____ Per: _____

Topic 1: Classifying Polynomials & Polynomial Operations

Simplify each expression. Final answers should be written in standard form.

1. $(-4m^2n)^4 \cdot \frac{1}{6}m^{-10}n^{-4}$ $256m^8n^4 \cdot \frac{1}{6}m^{-10}n^{-4}$ $= \frac{128}{3m^2}$	2. $(8a^2 - 6 - 8a) + (1 - 6a - 7a^2)$ $a^2 - 14a - 5$	3. $(6x - 7x^2 + 7) - (5x^2 + 2x - 2x^3 - 1)$ $2x^3 - 12x^2 + 4x + 8$
4. $(y + 4)^3 - 2y(y - 1)$ $= (y + 4)(y + 4)(y + 4) - 2y^2 + 2y$ $= (y + 4)(y^2 + 8y + 16) - 2y^2 + 2y$ $= y^3 + 8y^2 + 16y + 4y^2 + 32y + 64 - 2y^2 + 2y$ $= y^3 + 10y^2 + 50y + 64$	5. $(3k - 6)(k^2 - k + 7)$ $= 3k^3 - 3k^2 + 21k - 6k^2 + 6k - 42$ $= 3k^3 - 9k^2 + 27k - 42$	6. $\frac{-8c^6d^4 + 56c^4d^2 - 24c^2d}{8c^2d}$ $= -c^4d^3 + 7c^2d - 3$

Topic 2: Factoring Polynomials

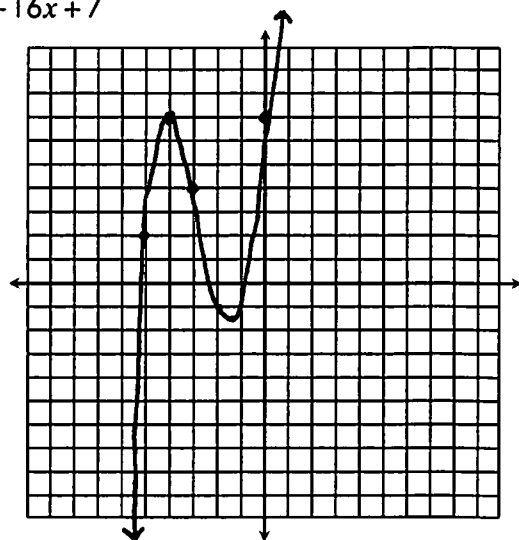
Differences of Squares $a^2 - b^2 =$ $(a + b)(a - b)$	Sum of Cubes $a^3 + b^3 =$ $(a + b)(a^2 - ab + b^2)$	Differences of Cubes $a^3 - b^3 =$ $(a - b)(a^2 + ab + b^2)$
7. $9x^3 + 21x^2$ $3x^2(3x + 7)$	8. $3n^4 - 147$ $3(n^4 - 49)$ $3(n^2 + 7)(n^2 - 7)$	
9. $64a^3 - 343b^3$ $(4a - 7b)(16a^2 + 28ab + 49b^2)$	10. $648w + 1029w^4$ $3w(216 + 343w^3)$ $3w(6 + 7w)(36 - 42w + 49w^2)$	
11. $32c^5d - 162cd^3$ $2cd(16c^4 - 81d^2)$ $2cd(4c^2 + 9d)(4c^2 - 9d)$	12. $216pq - p^7q$ $pq(216 - p^6)$ $pq(6 - p^2)(36 + 6p^2 + p^4)$	

13. $2c^5 - 2c^3 - 60c$ $2c(c^4 - c^2 - 30)$ $2c(c^2 - 6)(c^2 + 5)$	14. $9y^4 - 7y^2 - 16$ $y^4 - 7y^2 - 144$ $(y^2 - 16)(y^2 + 9)$ $(9y^2 - 16)(y^2 + 1)$ $(3y + 4)(3y - 4)(y^2 + 1)$
15. $n^3 + 2n^2 - 36n - 72$ $n^2(n+2) - 36(n+2)$ $(n^2 - 36)(n+2)$ $(n+6)(n-6)(n+2)$	16. $8x^3 - 10x^2 + 28x - 35$ $2x^2(4x-5) + 7(4x-5)$ $(2x^2 + 7)(4x-5)$

Topic 3: Graphing Polynomial Functions

Graph each function and identify its key characteristics.

17. $f(x) = x^3 + 8x^2 + 16x + 7$



Domain: $\mathbb{R}$

Range: $\mathbb{R}$

Rel. Maximum(s): $(-1.34, 7)$

Rel. Minimum(s): $(-4, -2.48)$

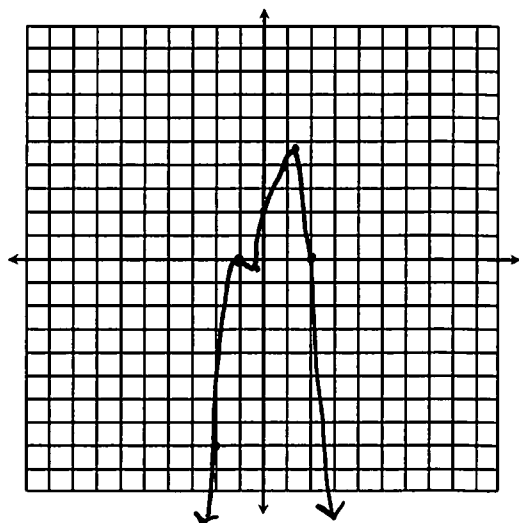
End Behavior: As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$

As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

Inc. Intervals: $(-\infty, -4)$, $(-1.34, \infty)$

Dec. Intervals: $(-4, -1.34)$

18. $f(x) = -x^4 + 3x^2 + 2x$



Domain: $\mathbb{R}$

Range: $\{y \mid y \leq 4.85\}$

Rel. Maximum(s): $(-1, 0)$, $(1.37, 4.85)$

Rel. Minimum(s): $(-0.36, -0.35)$

End Behavior: As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$

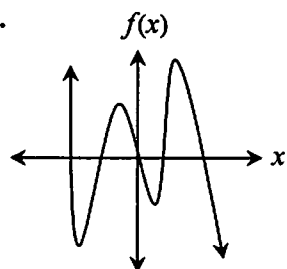
As $x \rightarrow \infty$, $f(x) \rightarrow -\infty$

Inc. Intervals: $(-\infty, -1)$, $(-0.36, 1.37)$

Dec. Intervals: $(-1, -0.36)$, $(1.37, \infty)$

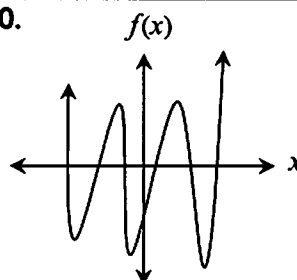
Determine the end behavior, whether the function is an even or odd degree, and the sign of the leading coefficient given each graph below.

19.



Odd degree,
Negative

20.



Even degree,
Positive

Identify the zeros (and their multiplicity) of each function below and the effect on the graph.

21. $f(x) = 3x^3(x-1)^2(x-8)$

Zero	Multiplicity	Effect
0	3	Crosses
1	2	bounces
8	1	crosses

22.. $f(x) = -x(5x+3)^3(x-2)^8$

Zero	Multiplicity	Effect
0	1	Crosses
$-3/5$	3	crosses
2	8	bounces

23. The graph of a polynomial function has zeros of 0 (multiplicity 2), 1 (multiplicity 2), and $5/2$ (multiplicity 2). Write a function in standard form that could represent this function.

$$f(x) = x^2(x-1)^2(2x-5)^2$$

$$f(x) = x^2(x^2-2x+1)(4x^2-20x+25)$$

$$f(x) = x^2(4x^4 - 20x^3 + 25x^2 - 8x^3 + 40x^2 - 50x + 4x^2 - 20x + 25)$$

$$f(x) = x^2(4x^4 - 28x^3 + 69x^2 - 70x + 25)$$

$$f(x) = 4x^6 - 28x^5 + 69x^4 - 70x^3 + 25x^2$$

Topic 4: Solving Polynomial Equations

Solve each equation. Simplify all irrational and complex solutions.

24. $2x^4 - 48x^2 = 0$

$$\begin{array}{l|l} 2x^2 & (x^2 - 24) = 0 \\ \hline 2x^2 = 0 & x^2 = 24 \\ x^2 = 0 & x = \pm \sqrt{24} \\ x = 0 & x = \pm 2\sqrt{6} \end{array}$$

$$x = \{0, \pm 2\sqrt{6}\}$$

25. $25x^3 = 64x$

$$25x^3 - 64x = 0$$

$$x(25x^2 - 64) = 0$$

$$x(5x+8)(5x-8) = 0$$

$$\begin{array}{l|l|l} x=0 & 5x+8=0 & 5x-8=0 \\ & x=-\frac{8}{5} & x=\frac{8}{5} \end{array}$$

$$x = \{0, \pm \frac{8}{5}\}$$

$$26. 108x^3 + 37 = 5$$

$$108x^3 + 32 = 0$$

$$4(27x^3 + 8) = 0$$

$$4 \overline{3x+2} \mid (9x^2 - 6x + 4) = 0$$

$$4 \neq 0 \quad 3x+2=0 \quad X = -\frac{2}{3}$$

$$X = \frac{6 \pm \sqrt{(-6)^2 - 4(9)(4)}}{2(9)}$$

$$X = \frac{6 \pm \sqrt{-108}}{18}$$

$$X = \frac{6 \pm 6i\sqrt{3}}{18}$$

$$X = \frac{1 \pm i\sqrt{3}}{3}$$

$$X = \left\{ -\frac{2}{3}, \frac{1 \pm i\sqrt{3}}{3} \right\}$$

$$27. 9x^5 - 72x^2 = 0$$

$$9x^2(x^3 - 8) = 0$$

$$9x^2 \mid (x-2)(x^2 + 2x + 4) = 0$$

$$9x^2=0 \quad X-2=0 \quad X = -2 \pm \frac{\sqrt{2^2 - 4(1)(4)}}{2(1)}$$

$$x^2=0 \quad X=2 \quad X = \frac{-2 \pm \sqrt{-12}}{2}$$

$$X=0 \quad X = \frac{-2 \pm 2i\sqrt{3}}{2}$$

$$X = -1 \pm i\sqrt{3}$$

$$X = \{0, 2, -1 \pm i\sqrt{3}\}$$

$$28. x^4 + 19x^2 - 20 = 0$$

$$(x^2 + 20)(x^2 - 1) = 0$$

$$x^2 + 20 = 0 \quad x^2 - 1 = 0$$

$$x^2 = -20 \quad x^2 = 1$$

$$x = \pm i\sqrt{20} \quad x = \pm 1$$

$$x = \pm 2i\sqrt{5}$$

$$X = \{ \pm 1, \pm 2i\sqrt{5} \}$$

$$29. x^5 = 18x^3 - 81x$$

$$x^5 - 18x^3 - 81x = 0$$

$$x(x^4 - 18x^2 - 81) = 0$$

$$x \mid (x^2 - 9)(x^2 - 9) = 0$$

$$x=0 \quad x^2 - 9 = 0 \quad x^2 - 9 = 0$$

$$x^2 = 9 \quad x^2 = 9$$

$$x = \pm 3 \quad x = \pm 3$$

$$X = \{0, \pm 3\}$$

$$30. 3x^4 - 14x^2 = 5$$

$$3x^4 - 14x^2 - 5 = 0$$

$$x^4 - 14x^2 - 15 = 0$$

$$(x^2 - 15)(x^2 + 1) = 0$$

$$(x^2 - 5)(3x^2 + 1) = 0$$

$$x^2 - 5 = 0 \quad 3x^2 + 1 = 0$$

$$x^2 = 5 \quad x^2 = -\frac{1}{3}$$

$$x = \pm \sqrt{5} \quad x = \pm \frac{i}{\sqrt{3}}$$

$$x = \pm \frac{i\sqrt{3}}{3}$$

$$X = \left\{ \pm \sqrt{5}, \pm \frac{i\sqrt{3}}{3} \right\}$$

$$31. 2x^3 + 7x^2 - 16x - 56 = 0$$

$$x^2(2x + 7) - 8(2x + 7) = 0$$

$$(x^2 - 8)(2x + 7) = 0$$

$$x^2 - 8 = 0 \quad 2x + 7 = 0$$

$$x^2 = 8 \quad x = -\frac{7}{2}$$

$$x = \pm 2\sqrt{2}$$

$$X = \left\{ -\frac{7}{2}, \pm 2\sqrt{2} \right\}$$

Topic 5: Dividing Polynomials & The Remainder Theorem

Find each quotient.

32. $(x^3 + x^2 - 71x + 9) \div (x + 9)$

$$\begin{array}{r} x^2 - 8x + 1 \\ x+9 \overline{) x^3 + x^2 - 71x + 9} \\ \underline{-(x^3 + 9x^2)} \\ -8x^2 - 71x \\ \underline{-(-8x^2 - 72x)} \\ x + 9 \\ \underline{-(x + 9)} \\ 0 \end{array}$$

$$\boxed{x^2 - 8x + 1}$$

33. $(8n^3 - 36n^2 - 15n - 16) \div (n - 5)$

$$\begin{array}{r} 8n^2 + 4n + 5 \\ 5 \overline{) 8 \quad -36 \quad -15 \quad -16} \\ \underline{\downarrow \quad 40 \quad 20 \quad 25} \\ 8 \quad 4 \quad 5 \quad \textcircled{9} \end{array}$$

$$\boxed{8n^2 + 4n + 5 + \frac{9}{n-5}}$$

34. $(12a^3 + 2a^2 - 6a - 30) \div (3a - 4)$

$$\begin{array}{r} 4a^2 + 6a + 6 \\ 3a-4 \overline{) 12a^3 + 2a^2 - 6a - 30} \\ \underline{-(12a^3 - 16a^2)} \\ 18a^2 - 6a \\ \underline{-(18a^2 - 24a)} \\ 18a - 30 \\ \underline{-(18a - 24)} \\ -6 \end{array}$$

$$\boxed{4a^2 + 6a + 6 - \frac{6}{3a-4}}$$

35. $(y^4 + 6y^3 - 4y - 31) \div (y + 6)$

$$\begin{array}{r} y^3 - 4 \\ -6 \overline{) 1 \quad 6 \quad 0 \quad -4 \quad -31} \\ \underline{\downarrow \quad -6 \quad 0 \quad 0 \quad 24} \\ 1 \quad 0 \quad 0 \quad -4 \quad \textcircled{-7} \end{array}$$

$$\boxed{y^3 - 4 - \frac{7}{y+6}}$$

36. Using the Remainder Theorem, find $f(-2)$ when $f(x) = 3x^3 - 28x^2 + 70$.

$$\begin{array}{r} -2 \overline{) 3 \quad -28 \quad 0 \quad 70} \\ \underline{\downarrow \quad -6 \quad 68 \quad -136} \\ 3 \quad -34 \quad 68 \quad \textcircled{-66} \end{array}$$

$$\boxed{f(-2) = -66}$$

37. The profit P of a small business (in thousands of dollars) since it was founded can be modeled by the function below, where t is the years since 1990. Use the Remainder Theorem to find the company's profit in 2017.

$$P(t) = 0.5t^4 - 3t^3 + t^2 + 25$$

$$\begin{array}{r} 27 \overline{) .5 \quad -3 \quad 1 \quad 0 \quad 25} \\ \underline{\downarrow \quad 13.5 \quad 283.5 \quad 7681.5 \quad 207400.5} \\ .5 \quad 10.5 \quad 284.5 \quad 7681.5 \quad 207425.5 \end{array}$$

$$\boxed{\$ 207,425,500}$$

Topic 6: Operations & Compositions of Functions

Given $f(x) = x^2 + 4x - 12$, $g(x) = 5x^2 - 2$, and $h(x) = x + 7$, find each function. Indicate any restrictions in the domain.

38. $(f - g)(x)$

$$(x^2 + 4x - 12) - (5x^2 - 2)$$

$$= \boxed{-4x^2 + 4x - 10}$$

39. $(h \cdot g)(x)$

$$(x + 7)(5x^2 - 2)$$

$$= \boxed{5x^3 + 35x^2 - 2x - 14}$$

40. $\left(\frac{h}{f}\right)(x)$

$$\frac{x + 7}{x^2 + 4x - 12}$$

$$= \boxed{\frac{x + 7}{(x + 6)(x - 2)} ; \{x | x \neq -6, 2\}}$$

41. $(g \circ h)(x)$

$$5(x + 7)^2 - 2$$

$$= 5(x^2 + 14x + 49) - 2$$

$$= \boxed{5x^2 + 70x + 243}$$

Use the same functions above, evaluate each function.

42. $(g + h)(-4)$

$$g(-4) = 5(-4)^2 - 2$$

$$= 5(16) - 2$$

$$= 78$$

$$h(-4) = -4 + 7$$

$$= 3$$

$$78 + 3 = \boxed{81}$$

43. $(h \circ f)(2)$

$$f(2) = (2)^2 + 4(2) - 12$$

$$= 4 + 8 - 12 = 0$$

$$h(0) = 0 + 7$$

$$= \boxed{7}$$

Topic 7: Regression

44. The population present in a bacteria culture over 5 days is given in the table below. Write a **cubic function** to represent the data.

Time (days)	0	1	2	3	4	5
Population	28	135	219	332	520	834

$$f(x) = 8.18x^3 - 35.41x^2 + 133.83x + 28.08$$

45. Use a **cubic function** to estimate the value of y when x is -8 . How does the estimate change when a quartic function is used instead?

Cubic:

x	-4	0	4	8	12
y	975	128	-9	-160	-893

$$f(x) = -1.68x^3 + 21.14x^2 - 97.48x + 136.91$$

$$f(-8) = \underline{3129.87}$$

Quartic:

$$f(x) = 0.03x^4 - 2.09x^3 + 21.78x^2 - 89.58x + 128$$

$$f(-8) = 3431.52$$

They differ by 301.65.

Name: _____

Date: _____ Per: _____

Unit 5 Test

Polynomial Functions

1. Simplify the expression given below.

$$\frac{(-6x^2y^3)^4}{8x^9y^7 \cdot 3xy^5} = \frac{1296x^8y^{12}}{24x^{10}y^{12}}$$

$$= 54x^{-2}$$

A. $-x^2y$ C. $-\frac{1}{x^2}$

B. $-54x^2y$ D. $\frac{54}{x^2}$

D

2. Simplify the expression below. Give your answer in standard form.

$$(18 - 2x^3 + 9x) - (4 - x^2 - 3x^3 + 11x)$$

$$x^3 + x^2 - 2x + 14$$

3. Simplify the expression below. Give your answer in standard form.

$$(2w - 5)^3$$

$$(2w - 5)(2w - 5)(2w - 5)$$

$$= (2w - 5)(4w^2 - 20w + 25)$$

$$= 8w^3 - 40w^2 + 50w - 20w^2 + 100w - 125$$

$$8w^3 - 60w^2 + 150w - 125$$

4. Simplify the expression below. Give your answer in standard form.

$$\frac{-18a^7b^3 + 9a^4b^2 - 63a^2b}{9ab}$$

$$-2a^6b^2 + a^3b - 7a$$

Factor each polynomial completely.

5. $50x^5 - 18x$

$$2x(25x^4 - 9)$$

$$2x(5x^2 + 3)(5x^2 - 3)$$

$$2x(5x^2 + 3)(5x^2 - 3)$$

6. $x^4 + 216x$

$$x(x^3 + 216)$$

$$x(x + 6)(x^2 - 6x + 36)$$

$$x(x + 6)(x^2 - 6x + 36)$$

7. $14x^3 - 49x^2 + 21x$

$$7x(2x^2 - 7x + 3)$$

$$7x(2x - 1)(x - 3)$$

$$7x(2x - 1)(x - 3)$$

8. $x^3 - 5x^2 - x + 5$

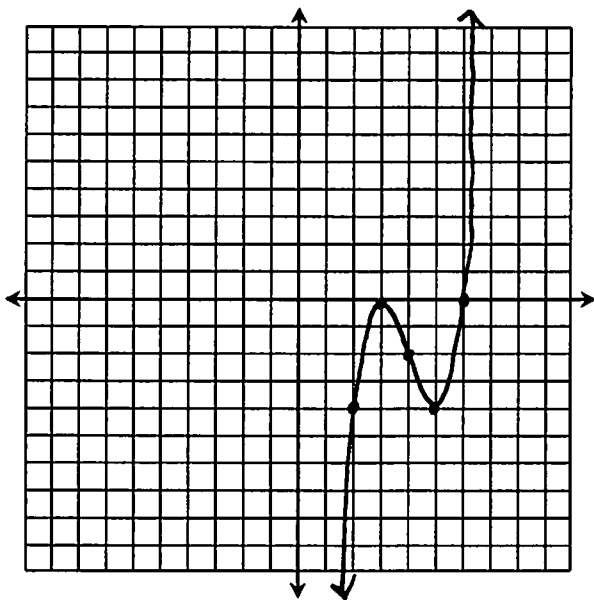
$$x^2(x - 5) - 1(x - 5)$$

$$(x^2 - 1)(x - 5)$$

$$(x + 1)(x - 1)(x - 5)$$

$$(x + 1)(x - 1)(x - 5)$$

9. Graph the function and identify all key characteristics: $f(x) = x^3 - 12x^2 + 45x - 54$



Domain: $\{\mathbb{R}\}$

Range: $\{\mathbb{R}\}$

Rel. Maximum(s): $(3, 0)$

Rel. Minimum(s): $(5, -4)$

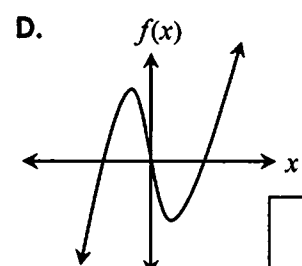
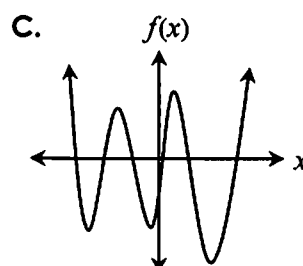
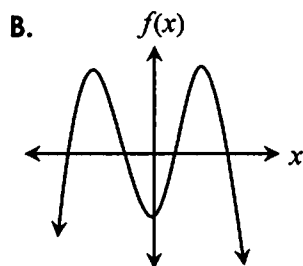
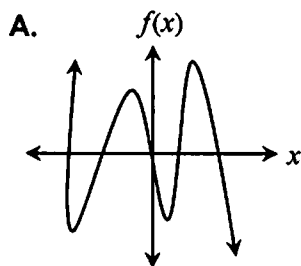
Inc. Intervals: $(-\infty, 3), (5, \infty)$

Dec. Intervals: $(3, 5)$

Zeros: $x = \{3, 6\}$

End Behavior: As $x \rightarrow -\infty, f(x) \rightarrow -\infty$
As $x \rightarrow \infty, f(x) \rightarrow \infty$

10. If an even-degree polynomial function has a positive leading coefficient, which graph could represent this function?



C

11. Which of the following describes the end behavior of the function below? Check all that apply.

$$f(x) = -x^9 + bx^4 + c$$

odd, negative

☐ As $x \rightarrow -\infty, f(x) \rightarrow -\infty$

☒ As $x \rightarrow \infty, f(x) \rightarrow -\infty$

☒ As $x \rightarrow -\infty, f(x) \rightarrow \infty$

☐ As $x \rightarrow \infty, f(x) \rightarrow \infty$

12. Identify the zeros, their multiplicity, and the effect (bounces or intersects) on the graph of the function given below.

$$f(x) = 2x(x + 4)^4(3x - 2)^3$$

Zero	Multiplicity	Effect
0	1	Crosses
-4	4	bounces
$\frac{2}{3}$	3	Crosses

13. The graph of a polynomial function has zeros of 3 (multiplicity 2), and -7 (multiplicity 1). Write a function in standard form that could represent this function.

$$\begin{aligned} f(x) &= (x-3)^2(x+7) \\ &= (x^2-6x+9)(x+7) \\ &= x^3+7x^2-6x^2-42x+9x+63 \\ &= x^3+x^2-33x+63 \end{aligned}$$

$$f(x) = x^3 + x^2 - 33x + 63$$

Solve each equation by factoring. Simplify all irrational and complex solutions.

14. $8x^3 - 98x = 0$

$$2x(4x^2 - 49) = 0$$

$$\begin{array}{c|c|c} 2x(2x+7)(2x-7) = 0 \\ \hline 2x=0 & 2x+7=0 & 2x-7=0 \\ \hline x=0 & x=-\frac{7}{2} & x=\frac{7}{2} \end{array}$$

$$x = \left\{ 0, \pm \frac{7}{2} \right\}$$

15. $80x^4 = 10x$

$$80x^4 - 10x = 0$$

$$10x(8x^3 - 1) = 0$$

$$\begin{array}{c|c|c} 10x(2x-1)(4x^2+2x+1) = 0 \\ \hline 10x=0 & 2x-1=0 & x = \frac{-2 \pm \sqrt{2^2 - 4(4)(1)}}{2(4)} \\ \hline x=0 & x=\frac{1}{2} & x = \frac{-2 \pm \sqrt{-12}}{8} \\ & & x = \frac{-2 \pm 2i\sqrt{3}}{8} \\ & & x = \frac{-1 \pm i\sqrt{3}}{4} \end{array}$$

$$x = \left\{ 0, \frac{1}{2}, \frac{-1 \pm i\sqrt{3}}{4} \right\}$$

16. $x^4 - 5x^2 = 3x^2 + 48$

$$x^4 - 8x^2 - 48 = 0$$

$$\begin{array}{c|c} (x^2-12)(x^2+4) = 0 \\ \hline x^2-12=0 & x^2+4=0 \\ \hline x^2=12 & x^2=-4 \\ \hline x=\pm\sqrt{12} & x=\pm\sqrt{-4} \\ \hline x=\pm 2\sqrt{3} & x=\pm 2i \end{array}$$

$$x = \{ \pm 2\sqrt{3}, \pm 2i \}$$

17. $3x^3 + 4x^2 = 12x + 16$

$$3x^3 + 4x^2 - 12x - 16 = 0$$

$$x^2(3x+4) - 4(3x+4) = 0$$

$$(x^2-4)(3x+4) = 0$$

$$\begin{array}{c|c|c} (x+2)(x-2)(3x+4) = 0 \\ \hline x+2=0 & x-2=0 & 3x+4=0 \\ \hline x=-2 & x=2 & x=-\frac{4}{3} \end{array}$$

$$x = \left\{ \pm 2, -\frac{4}{3} \right\}$$

18. Which of the following represents the zeros of the function $f(x) = x^3 + 512$? Check all that apply.

$$\begin{array}{c|c} (x+8)(x^2-8x+64) = 0 \\ \hline x+8=0 & x^2-8x = -64 \\ \hline x=-8 & x^2-8x+16 = -64+16 \\ & (x-4)^2 = -48 \\ & x-4 = \pm\sqrt{-48} \\ & x-4 = \pm 4i\sqrt{3} \\ & x = 4 \pm 4i\sqrt{3} \end{array}$$

☐ 8	☒ -8
☐ $4 + 4\sqrt{3}$	☐ $4 - 4\sqrt{3}$
☒ $4 + 4i\sqrt{3}$	☒ $4 - 4i\sqrt{3}$

Find the quotient.

19. $(x^3 + 3x^2 - 15x + 18) \div (x + 6)$

$$\begin{array}{r|rrrr} -6 & 1 & 3 & -15 & 18 \\ & \downarrow & -6 & 18 & -18 \\ \hline & 1 & -3 & 3 & 0 \end{array}$$

$$x^2 - 3x + 3$$

20. $(2a^3 - 10a^2 + 17a - 7) \div (a - 1)$

$$\begin{array}{r|rrrr} 1 & 2 & -10 & 17 & -7 \\ & \downarrow & 2 & -8 & 9 \\ \hline & 2 & -8 & 9 & 2 \end{array}$$

$$2a^2 - 8a + 9 + \frac{2}{a-1}$$

21. $(16w^3 + 58w^2 + 39w - 17) \div (2w + 5)$

$$\begin{array}{r} 8w^2 + 9w - 3 \\ 2w+5 \overline{) 16w^3 + 58w^2 + 39w - 17} \\ \underline{-(16w^3 + 40w^2)} \\ 18w^2 + 39w \\ \underline{-(18w^2 + 45w)} \\ -6w - 17 \\ \underline{-(-6w - 15)} \\ -2 \end{array}$$

$$8w^2 + 9w - 3 - \frac{2}{2w+5}$$

22. $(y^4 - 64y^2 + 2y - 9) \div (y - 8)$

$$\begin{array}{r|rrrrr} 8 & 1 & 0 & -64 & 2 & -9 \\ & \downarrow & 8 & 64 & 0 & 16 \\ \hline & 1 & 8 & 0 & 2 & 7 \end{array}$$

$$y^3 + 8y^2 + 2 + \frac{7}{y-8}$$

23. Kyle ran for 30 minutes. His elevation f (in feet) x minutes into his run can be represented by the function $f(x) = 0.04x^3 - 2x^2 + 24x + 150$. Using the Remainder theorem, find his elevation halfway through his run.

$$\begin{array}{r|rrrr} f(15) \rightarrow 15 & 0.04 & -2 & 24 & 150 \\ & \downarrow & .6 & -21 & 45 \\ \hline & 0.04 & -1.4 & 3 & 195 \end{array}$$

$$195 \text{ ft}$$

Use $f(x) = x^2 - x$, $g(x) = 9 - 2x$, and $h(x) = x^2 + 2x - 63$ for questions 24-29.

24. Find $(f + h)(x)$

$$(x^2 - x) + (x^2 + 2x - 63)$$

$$2x^2 + x - 63$$

25. Find $(g \cdot f)(x)$

$$(9 - 2x)(x^2 - x) \\ = 9x^2 - 2x^3 - 9x + 2x^2$$

$$-2x^3 + 11x^2 - 9x$$

26. The domain of the function $\left(\frac{g}{h}\right)(x)$ is all real numbers except -

$$\frac{9 - 2x}{x^2 + 2x - 63} \\ = \frac{9 - 2x}{(x + 9)(x - 7)}$$

- A. -9 and 7
- B. -7 and 9
- C. -9, 0 and 7
- D. -7, 0 and 9

A

27. Find $(f \circ g)(x)$

$$(9 - 2x)^2 - (9 - 2x) \\ 81 - 36x + 4x^2 - 9 + 2x$$

$$4x^2 - 34x + 72$$

28. Evaluate $(h - g)(4)$

$$h(4) = 4^2 + 2(4) - 63 \\ = 16 + 8 - 63 = -39$$

$$g(4) = 9 - 2(4) \\ = 9 - 8 = 1$$

$$-39 - 1 = -40$$

-40

29. Evaluate $(g \circ h)(-1)$

$$h(-1) = (-1)^2 + 2(-1) - 63 \\ = 1 - 2 - 63 \\ = -64$$

$$g(-64) = 9 - 2(-64) \\ = 9 + 128 \\ = 137$$

137

30. The data below shows the attendance at a water park over five days. Write a **quartic** function to model the data, then estimate the attendance at the park on the 8th day.

Day	Attendance
1	605
2	921
3	977
4	1,037
5	1,221

Equation:

$$f(x) = -6x^4 + 104x^3 - 604x^2 + 1490x - 379$$

Attendance:

1,557

CREDITS

I use clipart and
fonts in my products by:



Art with Jenny K



Many thanks to these
talented artists!